Section 5.1 Orthogonal Subspaces

Definition 5.1.1

Let $\boldsymbol{u} = (u_1, u_2, \cdots, u_n)^T$ and $\boldsymbol{v} = (v_1, v_2, \cdots, v_n)^T$ be vectors in $\boldsymbol{R}^n$. The number

$$\boldsymbol{u}^T\boldsymbol{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n$$

is called the **dot product** of $\boldsymbol{u}$ and $\boldsymbol{v}$, and often it is written as $\boldsymbol{u} \cdot \boldsymbol{v}$.

Theorem 5.1.2

Suppose $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{w}$ are vectors in $\boldsymbol{R}^n$ and let c be a scalar in $\boldsymbol{R}$. Then

(i) $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$

(ii) $c(\boldsymbol{u} \cdot \boldsymbol{v}) = (c\boldsymbol{u}) \cdot \boldsymbol{v} = \boldsymbol{u} \cdot (c\boldsymbol{v})$

(iii) $\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \cdot \boldsymbol{v} + \boldsymbol{u} \cdot \boldsymbol{w}$

The proofs are easy.

Definition 5.1.3

The **length** (or **norm**) of $\boldsymbol{v} = (v_1, v_2, \cdots, v_n)^T$ in $\boldsymbol{R}^n$ is the nonnegative scalar $\|\boldsymbol{v}\|$ defined by

$$\|\boldsymbol{v}\| = \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

The distance between $\boldsymbol{u}$ and $\boldsymbol{v}$ is defined to be $\|\boldsymbol{u} - \boldsymbol{v}\|$.

A vector whose length is 1 is called a **unit vector**. If we *divide* a nonzero vector $\boldsymbol{v}$ by its length—that is, multiply by $\frac{1}{\|\boldsymbol{v}\|}$—we obtain a unit vector $\frac{1}{\|\boldsymbol{v}\|}\boldsymbol{v}$ because its length is 1. The process of creating unit vector from $\boldsymbol{v}$ is sometimes called **normalizing** $\boldsymbol{v}$, and we say that $\frac{1}{\|\boldsymbol{v}\|}\boldsymbol{v}$ is *in the same direction* as $\boldsymbol{v}$.

Theorem 5.1.4

(i) $\|\boldsymbol{v}\| \geq 0$, and $\boldsymbol{v} \cdot \boldsymbol{v} = 0$ if and only if $\boldsymbol{v} = \boldsymbol{0}$

(ii) $\|c\boldsymbol{v}\| = |c|\|\boldsymbol{v}\|$ for every vector $\boldsymbol{v} = (v_1, v_2, \cdots, v_n)^T$ in $\boldsymbol{R}^n$ and every scalar c in $\boldsymbol{R}$.

(iii) $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$ for every vector $\boldsymbol{u} = (u_1, u_2, \cdots, u_n)^T$ and $\boldsymbol{v} = (v_1, v_2, \cdots, v_n)^T$ in $\boldsymbol{R}^n$.

Proof of (i) and (ii) are easy. Before giving the proof of (iii), let us proof the following theorem first.

Theorem 5.1.5

Let $\boldsymbol{u}$ and $\boldsymbol{v}$ be vectors in $\boldsymbol{R}^n$. Then

$$|\boldsymbol{u} \cdot \boldsymbol{v}| \leq \|\boldsymbol{u}\|\|\boldsymbol{v}\|$$

with equality holding if and only if one of the vectors is $\boldsymbol{0}$ or one vector is a multiple of the other.

Proof

$0 \leq \|\boldsymbol{u} + t\boldsymbol{v}\|^2 = (\boldsymbol{u} + t\boldsymbol{v}) \cdot (\boldsymbol{u} + t\boldsymbol{v}) = \boldsymbol{u} \cdot \boldsymbol{u} + 2t\boldsymbol{u} \cdot \boldsymbol{v} + t^2 \boldsymbol{v} \cdot \boldsymbol{v} = \|\boldsymbol{u}\|^2 + 2t\boldsymbol{u} \cdot \boldsymbol{v} + t^2 \|\boldsymbol{v}\|^2$ for any variable t

$\Delta = (2\boldsymbol{u} \cdot \boldsymbol{v})^2 - 4\|\boldsymbol{u}\|^2 \|\boldsymbol{v}\|^2 \leq 0$

$|\boldsymbol{u} \cdot \boldsymbol{v}| \leq \|\boldsymbol{u}\|\|\boldsymbol{v}\|$

$|\boldsymbol{u} \cdot \boldsymbol{v}| = \|\boldsymbol{u}\|\|\boldsymbol{v}\|$

$\Leftrightarrow \quad \Delta = 0$

$\Leftrightarrow$ $\begin{cases} \boldsymbol{v} = \boldsymbol{0} \\ \boldsymbol{v} \neq \boldsymbol{0} \text{ and } \|\boldsymbol{u} + t\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + 2t\boldsymbol{u} \cdot \boldsymbol{v} + t^2 \|\boldsymbol{v}\|^2 = 0 \quad \text{has a repeated root } t = -\dfrac{\boldsymbol{u} \cdot \boldsymbol{v}}{\boldsymbol{v} \cdot \boldsymbol{v}} \end{cases}$

$\Leftrightarrow$ $\begin{cases} \boldsymbol{v} = \boldsymbol{0} \\ \boldsymbol{v} \neq \boldsymbol{0} \text{ and } \boldsymbol{u} + t\boldsymbol{v} = \boldsymbol{0} \text{ for some } t = -\dfrac{\boldsymbol{u} \cdot \boldsymbol{v}}{\boldsymbol{v} \cdot \boldsymbol{v}} \end{cases}$

Proof of (iii) in Theorem 5.1.4.

$\|\boldsymbol{u} + \boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + 2\boldsymbol{u} \cdot \boldsymbol{v} + \|\boldsymbol{v}\|^2 \leq \|\boldsymbol{u}\|^2 + 2\|\boldsymbol{u}\|\|\boldsymbol{v}\| + \|\boldsymbol{v}\|^2 = (\|\boldsymbol{u}\| + \|\boldsymbol{v}\|)^2 \quad \Rightarrow \quad \|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$

Example 5.1.6

Let $\boldsymbol{u}=(u_1,u_2,u_3)$ and $\boldsymbol{v}=(v_1,v_2,v_3)$ in $\boldsymbol{R}^3$. Suppose θ is the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$.

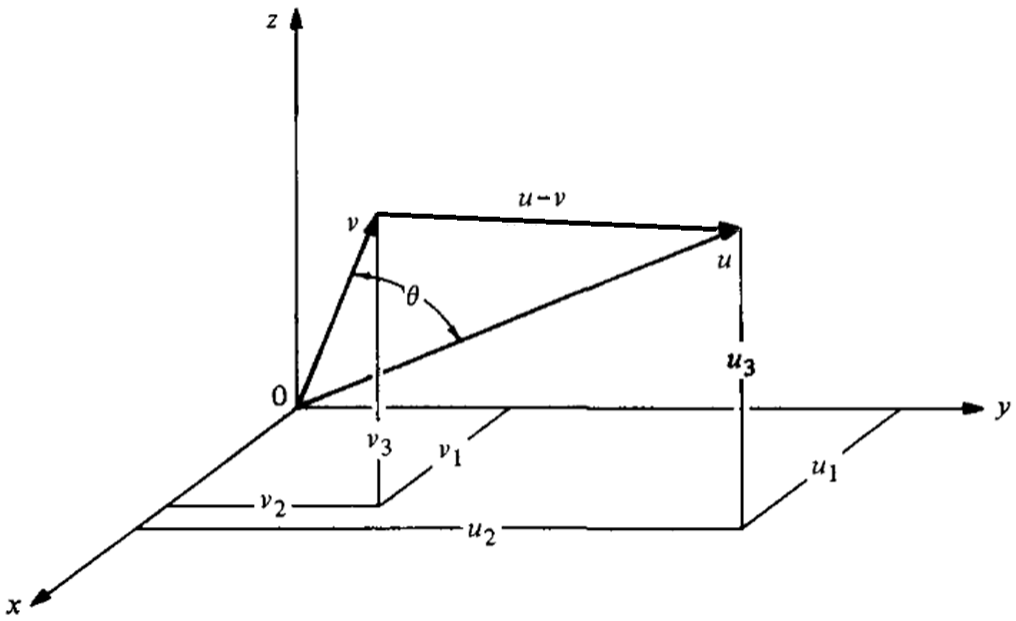
Then $\boldsymbol{u}\cdot\boldsymbol{v}=\|\boldsymbol{u}\|\|\boldsymbol{v}\|\cos\theta$ where $0\le\theta\le\pi$.

Proof

Consider a triangle formed by $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{u}-\boldsymbol{v}$. By the Law of Cosines, we have

$$\|\boldsymbol{u}-\boldsymbol{v}\|^2=\|\boldsymbol{u}\|^2+\|\boldsymbol{v}\|^2-2\|\boldsymbol{u}\|\|\boldsymbol{v}\|\cos\theta.$$

$$\begin{aligned}\|\boldsymbol{u}\|\|\boldsymbol{v}\|\cos\theta&=\frac{1}{2}\left(\|\boldsymbol{u}\|^2+\|\boldsymbol{v}\|^2-\|\boldsymbol{u}-\boldsymbol{v}\|^2\right)\\&=\frac{1}{2}\left(\boldsymbol{u}\cdot\boldsymbol{u}+\boldsymbol{v}\cdot\boldsymbol{v}-(\boldsymbol{u}-\boldsymbol{v})\cdot(\boldsymbol{u}-\boldsymbol{v})\right)\\&=\frac{1}{2}\left(\boldsymbol{u}\cdot\boldsymbol{u}+\boldsymbol{v}\cdot\boldsymbol{v}-(\boldsymbol{u}\cdot\boldsymbol{u}-2\boldsymbol{u}\cdot\boldsymbol{v}+\boldsymbol{v}\cdot\boldsymbol{v})\right)\\&=\boldsymbol{u}\cdot\boldsymbol{v}\end{aligned}$$



Definition 5.1.7

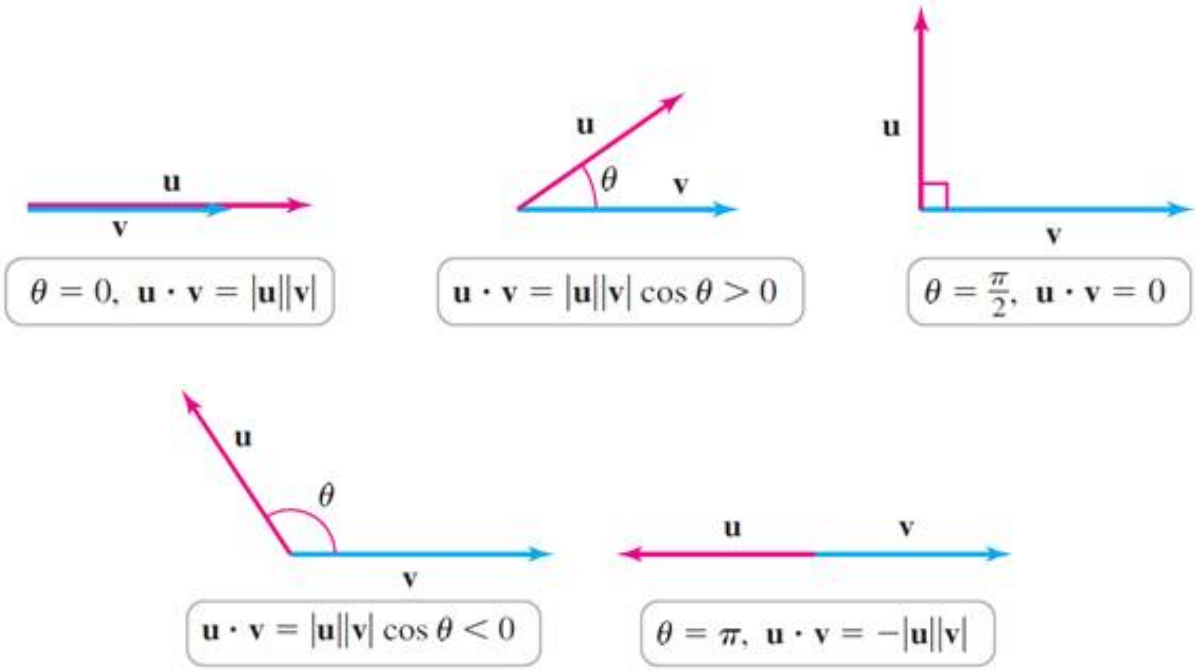
The angle θ between two nonzero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\boldsymbol{R}^n$ is defined by

$$\cos\theta=\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{\|\boldsymbol{u}\|\|\boldsymbol{v}\|}$$

where $0\le\theta\le\pi$. If $\boldsymbol{u}=\boldsymbol{0}$ or $\boldsymbol{v}=\boldsymbol{0}$, then θ is undefined.

$\boldsymbol{u}$ and $\boldsymbol{v}$ are parallel $(\theta = 0$ or $\theta = \pi)$ if and only if $\boldsymbol{u} \cdot \boldsymbol{v} = \pm \|\boldsymbol{u}\| \|\boldsymbol{v}\|$.

$\boldsymbol{u}$ and $\boldsymbol{v}$ are perpendicular $(\theta = \pi/2)$ or either one of the vectors is the zero vector if and only if $\boldsymbol{u} \cdot \boldsymbol{v} = 0$.

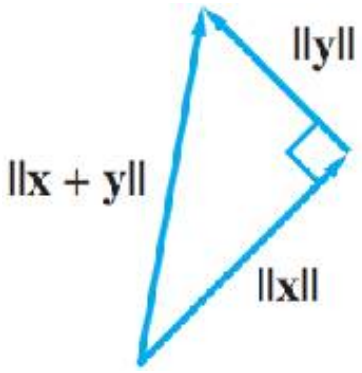


Definition 5.1.8

Two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\boldsymbol{R}^n$ are said to be **orthogonal** (to each other) if $\boldsymbol{u} \cdot \boldsymbol{v} = 0$.

Theorem 5.1.9

Two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are orthogonal if and only if $\|\boldsymbol{u} + \boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2$.



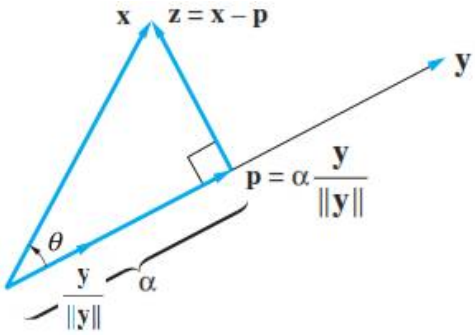
$$\|\mathbf{x}+\mathbf{y}\|^2 = (\mathbf{x}+\mathbf{y})^T(\mathbf{x}+\mathbf{y}) = \|\mathbf{x}\|^2 + 2\mathbf{x}^T\mathbf{y} + \|\mathbf{y}\|^2$$

$$\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \quad \Longrightarrow \quad \mathbf{x}^T\mathbf{y} = 0.$$

Definition 5.1.10

Scalar projection of **x** onto **y**: $\alpha = \dfrac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{y}\|}$

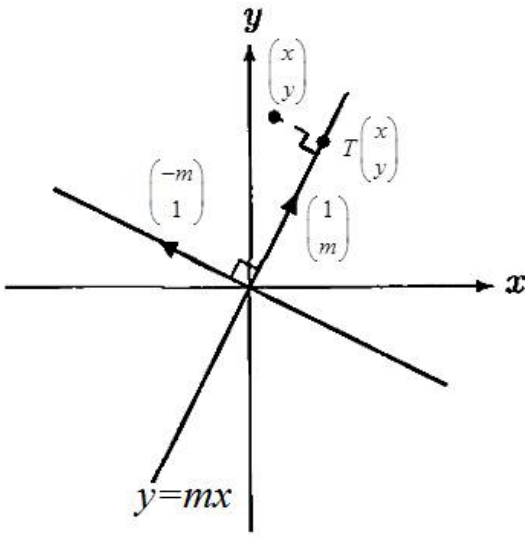
Vector projection of **x** onto **y**: $\mathbf{p} = \alpha \dfrac{1}{\|\mathbf{y}\|}\mathbf{y} = \dfrac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y}$



$$\begin{aligned}\alpha &= \|\mathbf{x}\|\cos\theta \\ &= \frac{\|\mathbf{x}\|\,\|\mathbf{y}\|\cos\theta}{\|\mathbf{y}\|} \\ &= \frac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{y}\|}\end{aligned}$$

Example 5.1.11

Find the vector projection of $\begin{pmatrix} x \\ y \end{pmatrix}$ onto $\begin{pmatrix} 1 \\ m \end{pmatrix}$.



Solution

$$\boldsymbol{p} = \frac{(x,y)\begin{pmatrix}1\\m\end{pmatrix}}{(1,m)\begin{pmatrix}1\\m\end{pmatrix}}\begin{pmatrix}1\\m\end{pmatrix} = \frac{x+my}{1+m^2}\begin{pmatrix}1\\m\end{pmatrix} = \begin{pmatrix}\frac{1}{1+m^2} & \frac{m}{1+m^2}\\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2}\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$$

which is the same as $T\begin{pmatrix} x \\ y \end{pmatrix}$ in Example 4.2.9

Example 5.1.12

Find the distance from the point $(2,0,0)^T$ to the plane $x+2y+2z=0$.

Solution

The vector $\mathbf{N} = (1,2,2)^T$ is normal to the plane and the plane passes through the origin. Let $\boldsymbol{v} = (2,0,0)^T$. The distance d from $(2,0,0)^T$ to the plane is simply the absolute value of the scalar projection of $\boldsymbol{v}$ onto $\mathbf{N}$. Thus,

$$d = \frac{\left|\boldsymbol{v}^T\mathbf{N}\right|}{\|\mathbf{N}\|} = \frac{2}{3}.$$

Definition 5.2.1

Two subspaces U and V of $\boldsymbol{R}^n$ are said to be **orthogonal** if $\boldsymbol{u}\cdot\boldsymbol{v}=0$ for every $\boldsymbol{u}\in U$ and every $\boldsymbol{v}\in V$. If U and V are orthogonal, we write $U\perp V$.

Definition 5.2.2

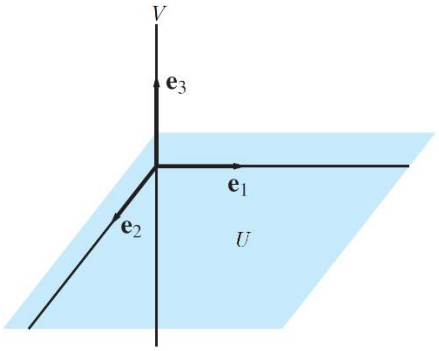
Let V be a subspace of $\boldsymbol{R}^n$. The set of all vectors in $\boldsymbol{R}^n$ that are orthogonal to every vector in V will be denoted $V^\perp$. Thus,

$$V^\perp=\{\boldsymbol{u}\in\boldsymbol{R}^n:\boldsymbol{u}\cdot\boldsymbol{v}=0 \text{ for every } \boldsymbol{v}\in V\}.$$

The set $V^\perp$ is called the **orthogonal complement** of V.

Example 5.2.3

Let $X=span\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}\right\}$ and $Y=span\left\{\begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$ be subspace of $\boldsymbol{R}^3$. For any $\begin{pmatrix}x_1\\0\\0\end{pmatrix}\in X$ and $\begin{pmatrix}0\\0\\y_3\end{pmatrix}\in Y$, $(x_1,0,0)\begin{pmatrix}0\\0\\y_3\end{pmatrix}=0$. Thus $X\perp Y$. Furthermore, $X^\perp=span\left\{\begin{pmatrix}0\\1\\0\end{pmatrix},\begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$.



Theorem 5.2.4

(a) If U and V are orthogonal subspaces of $\boldsymbol{R}^n$, then $U\cap V=\{\boldsymbol{0}\}$.

(b) If U is a subspace of $\boldsymbol{R}^n$, then $U^\perp$ is also a subspace of $\boldsymbol{R}^n$.

Proof

(a) Let $\boldsymbol{u} \in U \cap V$. Then $\|\boldsymbol{u}\|^2 = \overset{\in U \ \in V}{\boldsymbol{u} \cdot \boldsymbol{u}} = 0$. Hence $\boldsymbol{u} = \boldsymbol{0}$.

(b) $\boldsymbol{0} \in U^{\perp}$ because $\boldsymbol{0} \cdot \boldsymbol{u} = 0$ for every $\boldsymbol{u} \in U$.

Let a, b be any scalar and $\boldsymbol{v}_1, \boldsymbol{v}_2 \in U^{\perp}$. Then $(a\boldsymbol{v}_1 + b\boldsymbol{v}_2) \cdot \boldsymbol{u} = a(\boldsymbol{v}_1 \cdot \boldsymbol{u}) + b(\boldsymbol{v}_2 \cdot \boldsymbol{u}) = 0$ for every $\boldsymbol{u} \in U$.

Let $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_m$ be column vectors in $\boldsymbol{R}^n$ and $V = span\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_m\}$. Then $V^{\perp} = N(A)$

where $A = \begin{pmatrix} -\boldsymbol{v}_1^T - \\ -\boldsymbol{v}_2^T - \\ \vdots \\ -\boldsymbol{v}_m^T - \end{pmatrix}$, an $m \times n$ matrix.

Theorem 5.2.5 (Fundamental Subspaces Theorem)

If A is an $m \times n$ matrix, then $N(A) = \text{Row}(A)^{\perp} \subset \boldsymbol{R}^n$ and $N(A^T) = \text{Col}(A)^{\perp} \subset \boldsymbol{R}^m$.

Proof.

$\boldsymbol{v} \in N(A) \quad \Leftrightarrow \quad A\boldsymbol{v} = \boldsymbol{0}$

$\Leftrightarrow \quad \boldsymbol{v}$ is orthogonal to every row of A

$\Leftrightarrow \quad \boldsymbol{v} \in \text{Row}(A)^{\perp}$.

Furthermore, $N(A^T) = \text{Row}(A^T)^{\perp} = \text{Col}(A)^{\perp} \subset \boldsymbol{R}^m$.

Example 5.2.6

Consider $\boldsymbol{v}_1^T = (1,1,2), \boldsymbol{v}_2^T = (0,1,1), \boldsymbol{v}_3^T = (1,3,4)$ and $V = span\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$. Let

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{pmatrix}.$$

Find the bases for $N(A), \text{Col}(A^T), N(A^T), \text{Col}(A)$ and verify Theorem 5.2.5.

Solution

The reduced row echelon form of A is

$$\begin{pmatrix}1&1&2\\0&1&1\\1&3&4\end{pmatrix}\xrightarrow{R_3-R_1\to R_3}\begin{pmatrix}1&1&2\\0&1&1\\0&2&2\end{pmatrix}\xrightarrow{R_3-2R_2\to R_3}\begin{pmatrix}1&1&2\\0&1&1\\0&0&0\end{pmatrix}\xrightarrow{R_1-R_2\to R_1}\begin{pmatrix}1&0&1\\0&1&1\\0&0&0\end{pmatrix}$$

Since (1, 0, 1) and (0, 1, 1) form a basis of Row(A), $\left\{\begin{pmatrix}1\\0\\1\end{pmatrix},\begin{pmatrix}0\\1\\1\end{pmatrix}\right\}$ forms a basis of $\text{Col}\left(A^T\right)$.

Let $\boldsymbol{x}\in N(A)$. We have $x_1+x_3=0$ and $x_2+x_3=0$. Thus, $x_1=x_2=-x_3$.

$N(A)=\left\{\begin{pmatrix}\alpha\\\alpha\\-\alpha\end{pmatrix}\middle|\alpha\in R\right\}$ and $\begin{pmatrix}1\\1\\-1\end{pmatrix}$ forms a basis of $N(A)$.

Clearly, $\text{Col}(A^T)=N(A)^{\perp}$.

The reduced row echelon form of A^T is

$$\begin{pmatrix}1&0&1\\1&1&3\\2&1&4\end{pmatrix}\xrightarrow{R_2-R_1\to R_2}\begin{pmatrix}1&0&1\\0&1&2\\2&1&4\end{pmatrix}\xrightarrow{R_3-2R_1\to R_3}\begin{pmatrix}1&0&1\\0&1&2\\0&1&2\end{pmatrix}\xrightarrow{R_3-R_2\to R_3}\begin{pmatrix}1&0&1\\0&1&2\\0&0&0\end{pmatrix}$$

Since (1, 0, 1) and (0, 1, 2) form a basis of $\text{Row}\left(A^T\right)$, $\left\{\begin{pmatrix}1\\0\\1\end{pmatrix},\begin{pmatrix}0\\1\\2\end{pmatrix}\right\}$ forms a basis of Col(A).

Let $\boldsymbol{x}\in N(A^T)$. We have $x_1+x_3=0$ and $x_2+2x_3=0$. Thus, $x_1=-x_3, x_2=-2x_3$.

$N(A^T)=\left\{\begin{pmatrix}\alpha\\2\alpha\\-\alpha\end{pmatrix}\middle|\alpha\in R\right\}$ and $\begin{pmatrix}1\\2\\-1\end{pmatrix}$ forms a basis for $N(A^T)$.

Clearly, $N(A^T)=\text{Col}(A)^{\perp}$.

Definition 5.2.7

Let U and V are subspaces of a vector space W. We define the **sum** of U and V to be the set

$$\{\boldsymbol{u}+\boldsymbol{v}:\ \text{for some } \boldsymbol{u}\in U,\ \boldsymbol{v}\in V\}$$

which we denoted by $U + V$.

Theorem 5.2.8

Let U and V are subspaces of a vector space W. Then $U + V$ is a subspace of W.

Proof.

Choosing any 2 vectors $\boldsymbol{u}_1+\boldsymbol{v}_1, \boldsymbol{u}_2+\boldsymbol{v}_2 \in U+V$ where $\boldsymbol{u}_1, \boldsymbol{u}_2 \in U$ and $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$. Then

$$\boldsymbol{u}_1+\boldsymbol{v}_1+\boldsymbol{u}_2+\boldsymbol{v}_2 = (\underbrace{\boldsymbol{u}_1+\boldsymbol{u}_2}_{\in U})+(\underbrace{\boldsymbol{v}_1+\boldsymbol{v}_2}_{\in V}) \in U+V.$$

Choosing any scalar a in $\boldsymbol{R}$ and any vector $\boldsymbol{u}, \boldsymbol{v} \in U+V$ where $\boldsymbol{u} \in U$ and $\boldsymbol{v} \in V$. Then

$$a(\boldsymbol{u}+\boldsymbol{v}) = \underbrace{a\boldsymbol{u}_1}_{\in U}+\underbrace{a\boldsymbol{v}_1}_{\in V} \in U+V.$$

Theorem 5.2.9

Let U and V be two subspaces of a finite dimensional vector space W. Then

$$\dim(U+V) = \dim U + \dim V - \dim(U \cap V).$$

Outline of Proof

Let $\boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_k$ be a basis of $U \cap V$. Extend this basis to $\boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_k, \boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_i$ and form a basis of U. Similarly, $\boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_k, \boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_j$ form a basis of V. Then it can be shown that $\boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_k, \boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_i, \boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_j$ form a basis of $U + V$.

Example 5.2.10

Let U be x-y plane and V be x-z plane in $\boldsymbol{R}^3$. Then $U+V = \boldsymbol{R}^3$ and $U \cap V$ is x-axis.

Definition 5.2.11

If U and V are subspaces of a vector space W and each $\boldsymbol{w} \in W$ can be written uniquely as a sum $\boldsymbol{u} + \boldsymbol{v}$, where $\boldsymbol{u} \in U$ and $\boldsymbol{v} \in V$, then we say that W is a **direct sum** of U and V, and we write $W = U \oplus V$.

Theorem 5.2.12

Let U and V be subspaces of a vector space. Then the sum $U + V$ is a direct sum if and only if $U \cap V = \{\mathbf{0}\}$.

Proof

$\Rightarrow$) Let $\boldsymbol{w} \in U \cap V$. $\boldsymbol{w} = \underset{\in U}{\mathbf{0}} + \underset{\in V}{\boldsymbol{w}} = \underset{\in U}{\boldsymbol{w}} + \underset{\in V}{\mathbf{0}}$. Because of unique representation $U + V$, $\boldsymbol{w} = \mathbf{0}$.

$\Leftarrow$) Let $\boldsymbol{w} = \boldsymbol{u}_1 + \boldsymbol{v}_1 = \boldsymbol{u}_2 + \boldsymbol{v}_2$ where $\boldsymbol{u}_1, \boldsymbol{u}_2 \in U$ and $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$. Then

$$\boldsymbol{u}_1 - \boldsymbol{u}_2 = \boldsymbol{v}_2 - \boldsymbol{v}_1 \in U \cap V = \{\mathbf{0}\}$$

Hence $\boldsymbol{u}_1 = \boldsymbol{u}_2$ and $\boldsymbol{v}_1 = \boldsymbol{v}_2$. $\boldsymbol{w} \in W$ can be written uniquely as a sum $\boldsymbol{u} + \boldsymbol{v}$, where $\boldsymbol{u} \in U$ and $\boldsymbol{v} \in V$.

$U + V$ in Example 5.2.10 is not a direct sum because $U \cap V$ is x-axis but not $\{\mathbf{0}\}$. On the other hand, any vector in $\boldsymbol{R}^3$ can be expressed sum of a vector in U and a vector in V but in difference ways. Say for instance

$$(1,1,1) = \underset{\in U}{(1,1,0)} + \underset{\in V}{(0,0,1)} = \underset{\in U}{(1,0,0)} + \underset{\in V}{(0,1,1)}.$$

Theorem 5.2.13

If V is a subspace of $\boldsymbol{R}^n$, then $\dim V + \dim V^{\perp} = n$. Hence $V \oplus V^{\perp} = \boldsymbol{R}^n$.

Proof

If $V = \{\mathbf{0}\}$, then $V^{\perp} = \boldsymbol{R}^n$ and $\dim V + \dim V^{\perp} = 0 + n = n$.

Suppose $V \neq \{\mathbf{0}\}$. Let $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_k\}$ be column basis for V where $k = \dim V$ and let

$A = \begin{pmatrix} -\boldsymbol{v}_1^T - \\ -\boldsymbol{v}_2^T - \\ \\ -\boldsymbol{v}_k^T - \end{pmatrix}$, a $k \times n$ matrix. We have $V^\perp = N(A)$. By Rank–Nullity Theorem,

$$\dim V + \dim V^\perp = k + \dim N(A) = k + n - k = n.$$

Let $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_{n-k}\}$ be column basis for $V^\perp$. We need to show $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_k, \boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_{n-k}\}$ is a basis for $\boldsymbol{R}^n$. Suppose $a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_k\boldsymbol{v}_k + b_1\boldsymbol{u}_1 + b_2\boldsymbol{u}_2 + \cdots + b_{n-k}\boldsymbol{u}_{n-k} = \mathbf{0}$. Then

$$a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_k\boldsymbol{v}_k = -(b_1\boldsymbol{u}_1 + b_2\boldsymbol{u}_2 + \cdots + b_{n-k}\boldsymbol{u}_{n-k}) \in V \cap V^\perp = \{\mathbf{0}\}.$$

We have $a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_k\boldsymbol{v}_k = \mathbf{0}$ and $b_1\boldsymbol{u}_1 + b_2\boldsymbol{u}_2 + \cdots + b_{n-k}\boldsymbol{u}_{n-k} = \mathbf{0}$. Because of linear independent,

$$a_1 = a_2 = \cdots = a_k = b_1 = b_2 = \cdots = b_{n-k} = 0.$$

Since $V \oplus V^\perp \subset \boldsymbol{R}^n$ and $\dim(V \oplus V^\perp) = \dim V + \dim V^\perp = n = \dim \boldsymbol{R}^n$, we have

$$V \oplus V^\perp = \boldsymbol{R}^n.$$

Corollary 5.2.14

Let A be an $m \times n$ matrix, then $N(A) \oplus \mathrm{Row}(A) = \boldsymbol{R}^n$ and $N(A^T) \oplus \mathrm{Col}(A) = \boldsymbol{R}^m$.

Theorem 5.2.15

If V is a subspace of $\boldsymbol{R}^n$, then $(V^\perp)^\perp = V$.

Proof

On the one hand, if $\boldsymbol{v} \in V$, then $\boldsymbol{v}$ is orthogonal to each $\boldsymbol{u}$ in $V^\perp$. Therefore, $\boldsymbol{v} \in (V^\perp)^\perp$ and hence $V \subset (V^\perp)^\perp$. On the other hand,

$$\dim V + \dim V^\perp = \dim \boldsymbol{R}^n = n$$

$$\dim V^{\perp} + \dim (V^{\perp})^{\perp} = \dim \boldsymbol{R}^n = n$$
$$\dim V = \dim (V^{\perp})^{\perp}$$

Since $V \subset (V^{\perp})^{\perp}$, therefore $(V^{\perp})^{\perp} = V$.

Theorem 5.2.16

For any $m \times n$ matrix A, we have

$$N(A^T A) = N(A), \quad N(AA^T) = N(A^T), \qquad \text{and} \qquad \text{Col}(A^T A) = \text{Col}(A^T), \quad \text{Col}(AA^T) = \text{Col}(A).$$

Proof

Clearly, $N(A) \subset N(A^T A)$. Let $\boldsymbol{x} \in N(A^T A)$. Then $A^T A\boldsymbol{x} = \boldsymbol{0}$. Hence

$$\left\| A\boldsymbol{x} \right\|^2 = \boldsymbol{x}^T A^T A\boldsymbol{x} = 0.$$

It implies $A\boldsymbol{x} = \boldsymbol{0}$ and $\boldsymbol{x} \in N(A)$. By Theorem 5.2.5,

$$\text{Col}(A^T) = N(A)^{\perp} = N(A^T A)^{\perp} = N\left((A^T A)^T \right)^{\perp} = \text{Col}(A^T A).$$

Applying A^T into A, we have

$$N(AA^T) = N\left((A^T)^T A^T \right) = N(A^T) \quad \text{and} \quad \text{Col}(AA^T) = \text{Col}\left((A^T)^T A^T \right) = \text{Col}(A).$$

Note that $A^T A$ and AA^T are always square matrices for any matrix A.

Corollary 5.2.17

Let A be an $m \times n$ matrix. Then

(a) $A^T A$ is nonsingular if and only if rank(A) = n,

(b) AA^T is nonsingular if and only if rank(A) = m.

Proof

$A^T A$ is nonsingular if and only if $N(A^T A) = N(A) = \{\boldsymbol{0}\}$ if and only if $\text{Row}(A) = \boldsymbol{R}^n$ if and only if rank(A) = n. The proof of (b) is similar to that of (a).

Section 5.3 Least Squares Problems

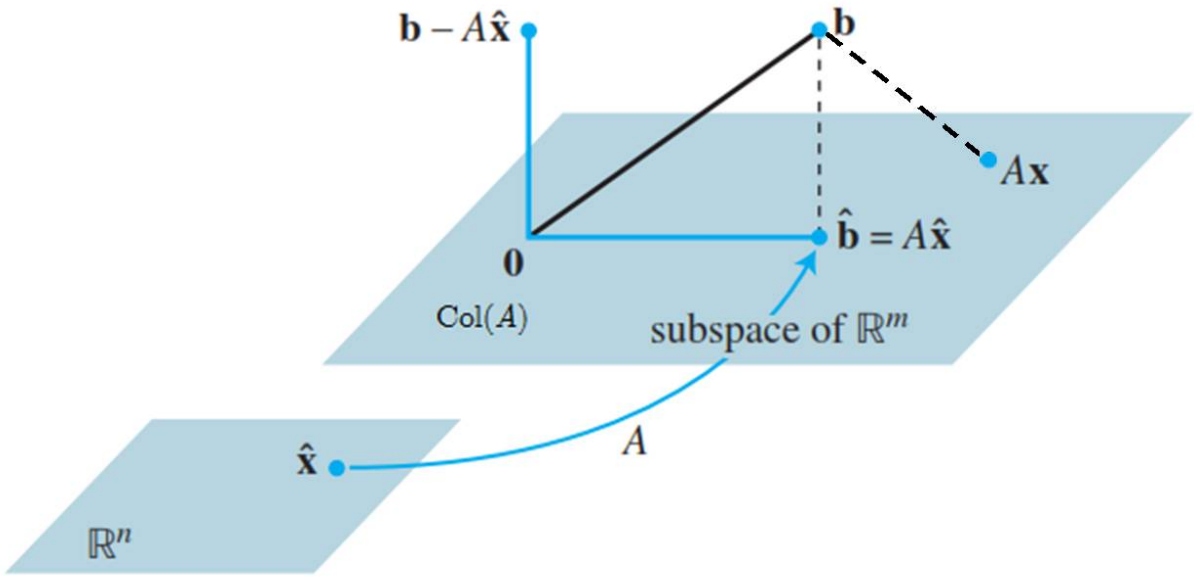
A least squares problem can generally be formulated as an overdetermined linear system of equations (more equations than unknowns). Such systems are usually inconsistent. Thus given an $m \times n$ system $A\boldsymbol{x} = \boldsymbol{b}$ with $m > n$, we cannot expect in general to find a vector $\boldsymbol{x} \in \boldsymbol{R}^n$ for which $A\boldsymbol{x} = \boldsymbol{b}$. Instead, we can look for a vector $\hat{\boldsymbol{x}} \in \boldsymbol{R}^n$ for which $A\hat{\boldsymbol{x}}$ is closest to $\boldsymbol{b}$.

Definition 5.3.1

If A is an $m \times n$ matrix and $\boldsymbol{b}$ is in $\boldsymbol{R}^m$, a **least-squares solution** of $A\boldsymbol{x} = \boldsymbol{b}$ is an $\hat{\boldsymbol{x}}$ in $\boldsymbol{R}^n$ such that

$$\|\boldsymbol{b} - A\hat{\boldsymbol{x}}\| \le \|\boldsymbol{b} - A\boldsymbol{x}\|$$

for all $\boldsymbol{x}$ in $\boldsymbol{R}^n$.



The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

Theorem 5.3.2

Let A be an $m \times n$ matrix with $m > n$. For each $\boldsymbol{b} \in \boldsymbol{R}^m$, there is a unique vector $\boldsymbol{p} = A(\hat{\boldsymbol{x}}) \in \text{Col}(A)$ such that

$$\|\boldsymbol{b}-\boldsymbol{v}\| > \|\boldsymbol{b}-A\hat{\boldsymbol{x}}\|$$

for any $\boldsymbol{v} \neq A\hat{\boldsymbol{x}}$ in Col(A). Furthermore, $\boldsymbol{b}-A\hat{\boldsymbol{x}} \in \text{Col}(A)^{\perp} = N(A^T)$.

Proof

Since $\boldsymbol{R}^m = \text{Col}(A) \oplus \text{Col}(A)^{\perp}$, for each $\boldsymbol{b} \in \boldsymbol{R}^m$, there is a unique vector $\boldsymbol{p} = A(\hat{\boldsymbol{x}}) \in \text{Col}(A)$ and $\boldsymbol{b}-A\hat{\boldsymbol{x}} \in \text{Col}(A)^{\perp} = N(A^T)$. For any $\boldsymbol{v} \neq A\hat{\boldsymbol{x}}$ in Col(A), we have

$$\|\boldsymbol{b}-\boldsymbol{v}\|^2 = \|\boldsymbol{b}-A\hat{\boldsymbol{x}}\|^2 + \|A\hat{\boldsymbol{x}}-\boldsymbol{v}\|^2 > \|\boldsymbol{b}-A\hat{\boldsymbol{x}}\|^2$$

That is $\|\boldsymbol{b}-\boldsymbol{v}\| > \|\boldsymbol{b}-A\hat{\boldsymbol{x}}\|$

Theorem 5.3.3

If A is an $m \times n$ matrix of rank n, $\hat{\boldsymbol{x}} = \left(A^T A\right)^{-1} A^T \boldsymbol{b}$ is the unique solution of the system $A\boldsymbol{x} = \boldsymbol{b}$.

Outline of the proof

Since rank of $A = n$, $A^T A$ is invertible. By Theorem 5.3.2, we have

$$A^T(\boldsymbol{b}-A\hat{\boldsymbol{x}}) = \boldsymbol{0}$$
$$A^T A\hat{\boldsymbol{x}} = A^T\boldsymbol{b}$$
$$\hat{\boldsymbol{x}} = \left(A^T A\right)^{-1} A^T\boldsymbol{b}.$$

Example 5.3.4

Find a least-squares solution of the inconsistent system $A\boldsymbol{x} = \boldsymbol{b}$ for

$$A = \begin{pmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 2 \\ 0 \\ 11 \end{pmatrix}.$$

Determine the least-squares error in the least squares solution of $A\boldsymbol{x} = \boldsymbol{b}$.

Solution

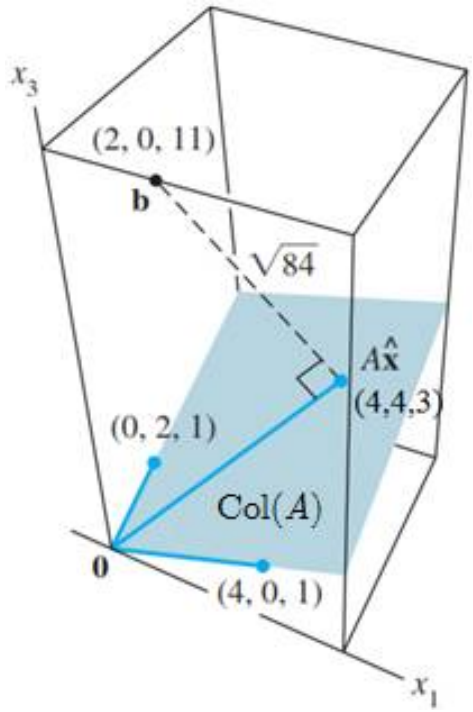
We have

$$A^TA=\begin{pmatrix}4&0&1\\0&2&1\end{pmatrix}\begin{pmatrix}4&0\\0&2\\1&1\end{pmatrix}=\begin{pmatrix}17&1\\1&5\end{pmatrix},\quad A^T\boldsymbol{b}=\begin{pmatrix}4&0&1\\0&2&1\end{pmatrix}\begin{pmatrix}2\\0\\11\end{pmatrix}=\begin{pmatrix}19\\11\end{pmatrix}.$$

$$\begin{aligned}\hat{\boldsymbol{x}}&=\left(A^TA\right)^{-1}A^T\boldsymbol{b}\\&=\begin{pmatrix}17&1\\1&5\end{pmatrix}^{-1}\begin{pmatrix}19\\11\end{pmatrix}\\&=\frac{1}{84}\begin{pmatrix}5&-1\\-1&17\end{pmatrix}\begin{pmatrix}19\\11\end{pmatrix}\\&=\begin{pmatrix}1\\2\end{pmatrix}\end{aligned}$$

Clearly, $A\hat{\boldsymbol{x}}=\begin{pmatrix}4&0\\0&2\\1&1\end{pmatrix}\begin{pmatrix}1\\2\end{pmatrix}=\begin{pmatrix}4\\4\\3\end{pmatrix}$. The least-squares error is

$$\left\|\boldsymbol{b}-A\hat{\boldsymbol{x}}\right\|=\left\|\begin{pmatrix}2\\0\\11\end{pmatrix}-\begin{pmatrix}4\\4\\3\end{pmatrix}\right\|=\left\|\begin{pmatrix}-2\\-4\\8\end{pmatrix}\right\|=\sqrt{84}.$$



Given a table of data

x	x_1	x_2	$\cdots$	x_m
y	y_1	y_2	$\cdots$	y_m

Consider a line $y = \beta_0 + \beta_1 x$. We call y_j the j-th *observed value* of y and $\beta_0 + \beta_1 x_j$ the j-th **predicted value**. The difference between an observed value and a predicted value is called a **residual**.

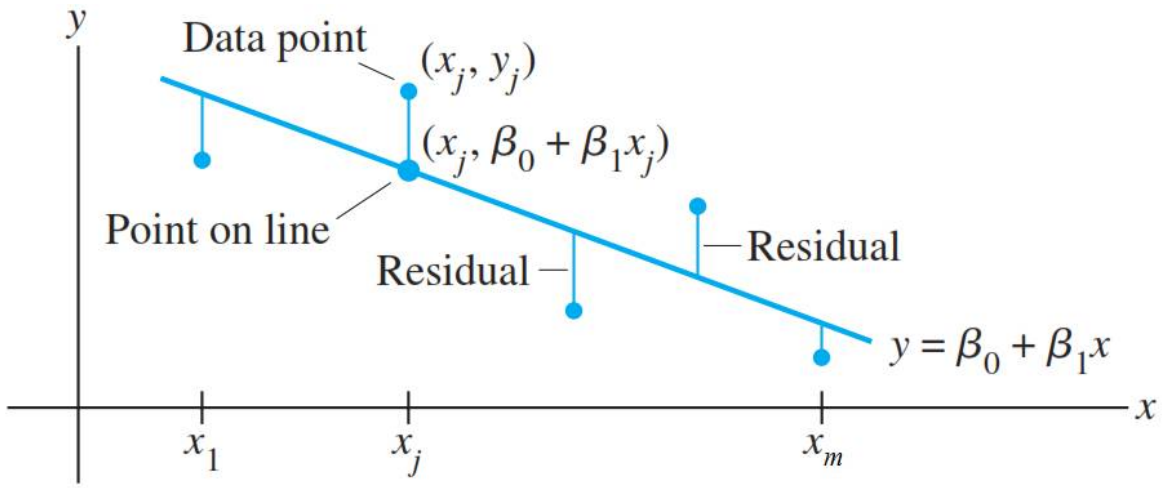


FIGURE 1 Fitting a line to experimental data.

We wish to find a linear function $y = \beta_0 + \beta_1 x$ that minimizes the sum of squares of the residuals. That means we want to find a least-squares solution of the inconsistent system of linear equations

$$\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}.$$

Let $A = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{pmatrix}$, $\boldsymbol{x} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$, $\boldsymbol{b} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$. Clearly,

$$A^TA=\begin{pmatrix}1 & 1 & \cdots & 1\\ x_1 & x_2 & \cdots & x_m\end{pmatrix}\begin{pmatrix}1 & x_1\\ 1 & x_2\\ \vdots & \vdots\\ 1 & x_m\end{pmatrix}=\begin{pmatrix}m & \sum_{j=1}^{m}x_j\\ \sum_{j=1}^{m}x_j & \sum_{j=1}^{m}x_j^2\end{pmatrix},\quad A^T\boldsymbol{b}=\begin{pmatrix}1 & 1 & \cdots & 1\\ x_1 & x_2 & \cdots & x_m\end{pmatrix}\begin{pmatrix}y_1\\ y_2\\ \vdots\\ y_m\end{pmatrix}=\begin{pmatrix}\sum_{j=1}^{m}y_j\\ \sum_{j=1}^{m}x_jy_j\end{pmatrix}.$$

$$\begin{aligned}\hat{\boldsymbol{x}}&=\left(A^TA\right)^{-1}A^T\boldsymbol{b}\\ &=\begin{pmatrix}m & \sum_{j=1}^{m}x_j\\ \sum_{j=1}^{m}x_j & \sum_{j=1}^{m}x_j^2\end{pmatrix}^{-1}\begin{pmatrix}\sum_{j=1}^{m}y_j\\ \sum_{j=1}^{m}x_jy_j\end{pmatrix}.\end{aligned}$$

Example 5.3.5

Given the data

x	0	3	6
y	1	4	5

Find the best least squares fit by a linear function. Determine the least-squares error. Find the predicted value of $x = 2$.

Solution

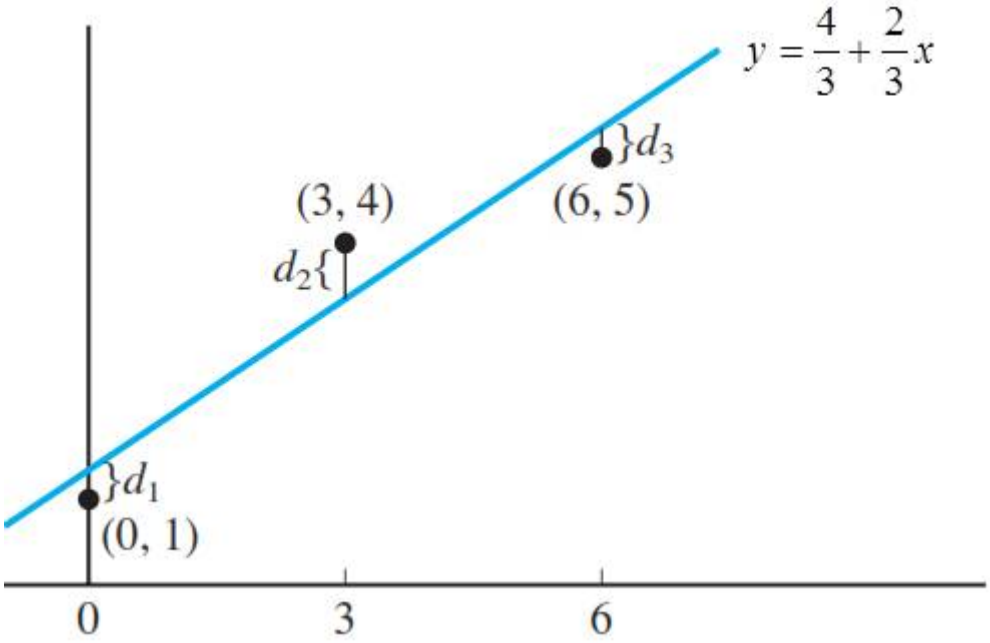
Since $m=3\quad \sum_{j=1}^{3}x_j=9\quad \sum_{j=1}^{3}x_j^2=3^2+6^2=45\quad \sum_{j=1}^{3}y_j=10\quad \sum_{j=1}^{3}x_jy_j=12+30=42,$

$$\begin{aligned}\hat{\boldsymbol{x}}&=\left(A^TA\right)^{-1}A^T\boldsymbol{b}\\ &=\begin{pmatrix}3 & 9\\ 9 & 45\end{pmatrix}^{-1}\begin{pmatrix}10\\ 42\end{pmatrix}\\ &=\frac{1}{54}\begin{pmatrix}72\\ 36\end{pmatrix}\\ &=\frac{1}{3}\begin{pmatrix}4\\ 2\end{pmatrix}\end{aligned}$$

The best least squares fit is $y=\frac{4}{3}+\frac{2}{3}x.$

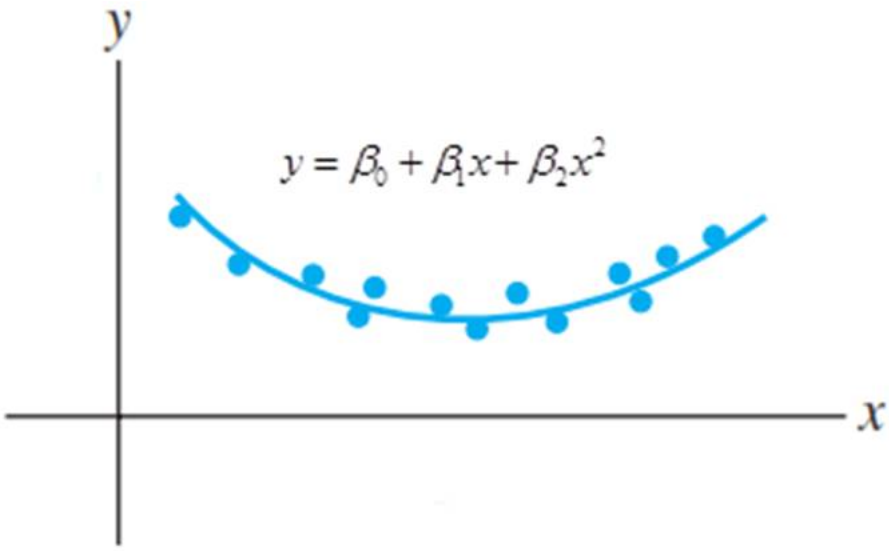
The least-squares error is $\left\|\boldsymbol{b}-A\hat{\boldsymbol{x}}\right\|=\left\|\begin{pmatrix}1\\ 4\\ 5\end{pmatrix}-\frac{1}{3}\begin{pmatrix}1 & 0\\ 1 & 3\\ 1 & 6\end{pmatrix}\begin{pmatrix}4\\ 2\end{pmatrix}\right\|=\left\|\begin{pmatrix}-\frac{1}{3}\\ \frac{2}{3}\\ -\frac{1}{3}\end{pmatrix}\right\|=\sqrt{\frac{2}{3}}.$

The predicted value of $x = 2$ is $y = \frac{4}{3} + \frac{2}{3} \cdot 2 = \frac{8}{3}$.



Similarly, we can find the best quadratic least squares fit $y = \beta_0 + \beta_1 x + \beta_2 x^2$ by considering the inconsistent system of linear equations

$$\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_m & x_m^2 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$$



Example 5.3.6

Find the best quadratic least squares fit to the data

x	0	1	2	3
y	3	2	4	4

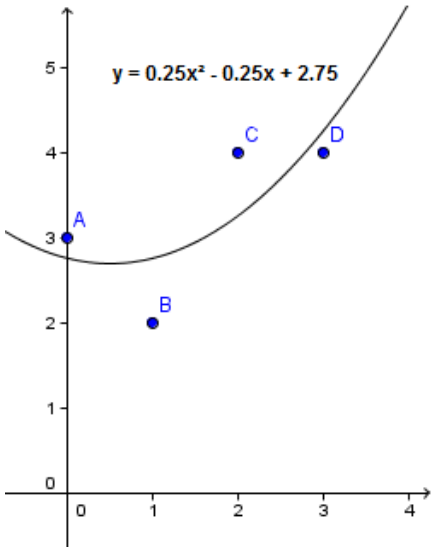
Determine the least-squares error.

Solution

Let $A=\begin{pmatrix}1 & x_1 & x_1^2\\ 1 & x_2 & x_2^2\\ 1 & x_3 & x_3^2\\ 1 & x_4 & x_4^2\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 1 & 1 & 1\\ 1 & 2 & 4\\ 1 & 3 & 9\end{pmatrix}$, $\boldsymbol{x}=\begin{pmatrix}\beta_0\\ \beta_1\\ \beta_2\end{pmatrix}$, $\boldsymbol{b}=\begin{pmatrix}y_1\\ y_2\\ y_3\\ y_4\end{pmatrix}=\begin{pmatrix}3\\ 2\\ 4\\ 4\end{pmatrix}$.

$$A^TA=\begin{pmatrix}1 & 1 & 1 & 1\\ 0 & 1 & 2 & 3\\ 0 & 1 & 4 & 9\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 1 & 1 & 1\\ 1 & 2 & 4\\ 1 & 3 & 9\end{pmatrix}=\begin{pmatrix}4 & 6 & 14\\ 6 & 14 & 36\\ 14 & 36 & 98\end{pmatrix},\quad A^T\boldsymbol{b}=\begin{pmatrix}1 & 1 & 1 & 1\\ 0 & 1 & 2 & 3\\ 0 & 1 & 4 & 9\end{pmatrix}\begin{pmatrix}3\\ 2\\ 4\\ 4\end{pmatrix}=\begin{pmatrix}13\\ 22\\ 54\end{pmatrix}.$$

$$\begin{aligned}\hat{x}&=\left(A^TA\right)^{-1}A^Tb\\ &=\begin{pmatrix}4 & 6 & 14\\ 6 & 14 & 36\\ 14 & 36 & 98\end{pmatrix}^{-1}\begin{pmatrix}13\\ 22\\ 54\end{pmatrix}\\ &=\frac{1}{20}\begin{pmatrix}19 & -21 & 5\\ -21 & 49 & -15\\ 5 & -15 & 5\end{pmatrix}\begin{pmatrix}13\\ 22\\ 54\end{pmatrix}\\ &=\frac{1}{20}\begin{pmatrix}55\\ -5\\ 5\end{pmatrix}=\frac{1}{4}\begin{pmatrix}11\\ -1\\ 1\end{pmatrix}\end{aligned}$$

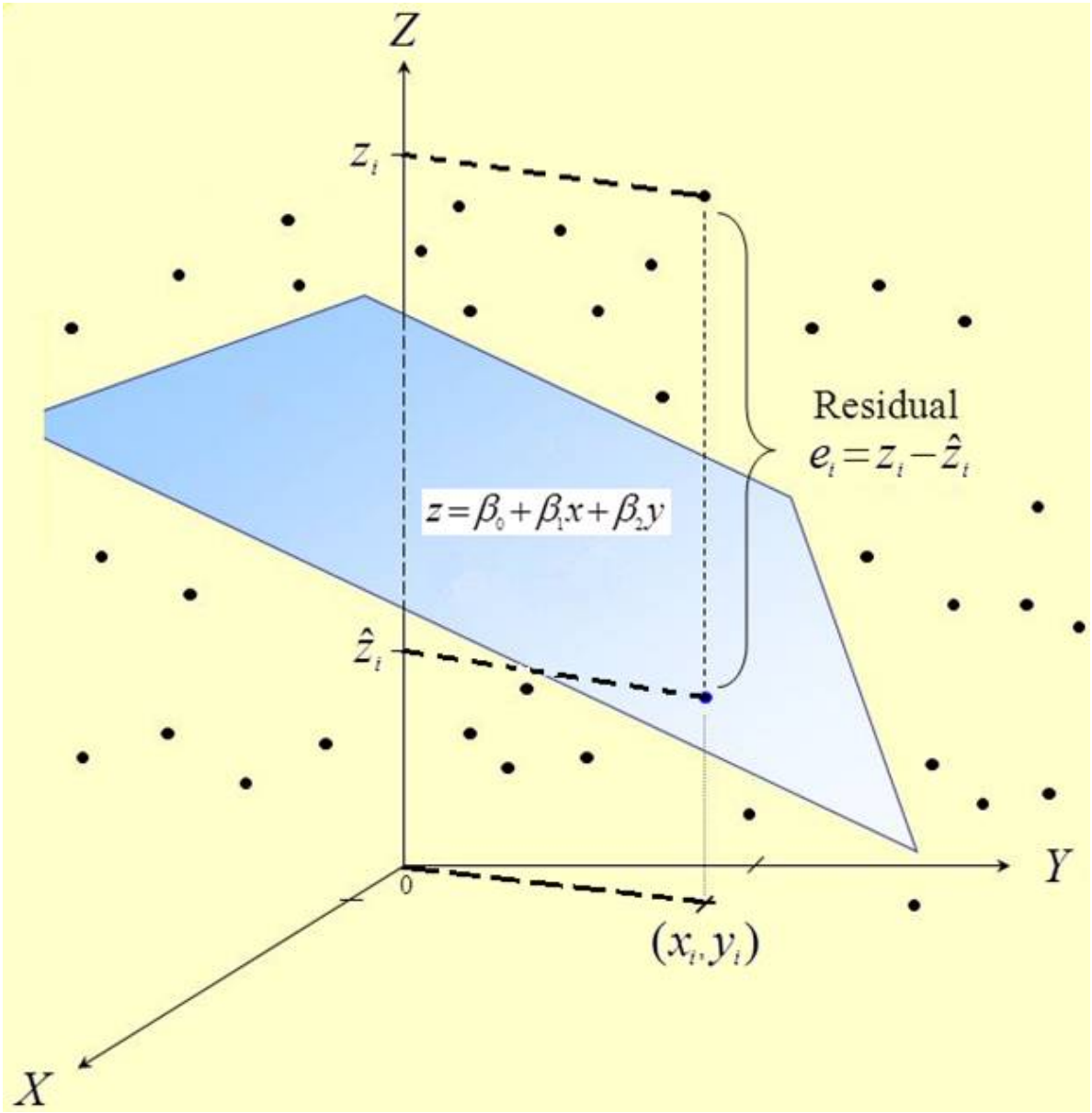


The best quadratic least squares fit is $y=2.75-0.25x+0.25x^2$.

The least-squares error is $\left\|\boldsymbol{b}-A\hat{\boldsymbol{x}}\right\|=\left\|\begin{pmatrix}3\\ 2\\ 4\\ 4\end{pmatrix}-\frac{1}{4}\begin{pmatrix}1 & 0 & 0\\ 1 & 1 & 1\\ 1 & 2 & 4\\ 1 & 3 & 9\end{pmatrix}\begin{pmatrix}11\\ -1\\ 1\end{pmatrix}\right\|=\left\|\begin{pmatrix}0.25\\ -0.75\\ 0.75\\ -0.25\end{pmatrix}\right\|=\sqrt{\frac{5}{4}}$.

We can also find the best fitted plane $z = \beta_0 + \beta_1 x + \beta_2 y$ by considering the inconsistent system of linear equations

$$\begin{pmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_m & y_m \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_m \end{pmatrix}$$



Example 5.3.7

Find the best fitted plane to the data

(x, y)	$(0,0)$	$(1,0)$	$(0,1)$	$(1,1)$
z	2	3	5	7

Determine the least-squares error.

Solution

Let $A=\begin{pmatrix}1 & x_1 & y_1\\ 1 & x_2 & y_2\\ 1 & x_3 & y_3\\ 1 & x_4 & y_4\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 0 & 1\\ 1 & 1 & 1\end{pmatrix}$, $\boldsymbol{x}=\begin{pmatrix}\beta_0\\ \beta_1\\ \beta_2\end{pmatrix}$, $\boldsymbol{b}=\begin{pmatrix}z_1\\ z_2\\ z_3\\ z_4\end{pmatrix}=\begin{pmatrix}2\\ 3\\ 5\\ 7\end{pmatrix}$.

$$A^TA=\begin{pmatrix}1 & 1 & 1 & 1\\ 0 & 1 & 0 & 1\\ 0 & 0 & 1 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 0 & 1\\ 1 & 1 & 1\end{pmatrix}=\begin{pmatrix}4 & 2 & 2\\ 2 & 2 & 1\\ 2 & 1 & 2\end{pmatrix},\quad A^T\boldsymbol{b}=\begin{pmatrix}1 & 1 & 1 & 1\\ 0 & 1 & 0 & 1\\ 0 & 0 & 1 & 1\end{pmatrix}\begin{pmatrix}2\\ 3\\ 5\\ 7\end{pmatrix}=\begin{pmatrix}17\\ 10\\ 12\end{pmatrix}.$$

$$\begin{aligned}\hat{\boldsymbol{x}}&=\left(A^TA\right)^{-1}A^T\boldsymbol{b}\\ &=\begin{pmatrix}4 & 2 & 2\\ 2 & 2 & 1\\ 2 & 1 & 2\end{pmatrix}^{-1}\begin{pmatrix}17\\ 10\\ 12\end{pmatrix}\\ &=\frac{1}{4}\begin{pmatrix}3 & -2 & -2\\ -2 & 4 & 0\\ -2 & 0 & 4\end{pmatrix}\begin{pmatrix}17\\ 10\\ 12\end{pmatrix}\\ &=\frac{1}{4}\begin{pmatrix}7\\ 6\\ 14\end{pmatrix}\end{aligned}$$

The best fitted plane is $z=1.75+1.5x+3.5y$

The least-squares error is $\left\|\boldsymbol{b}-A\hat{\boldsymbol{x}}\right\|=\left\|\begin{pmatrix}2\\ 3\\ 5\\ 7\end{pmatrix}-\frac{1}{4}\begin{pmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 0 & 1\\ 1 & 1 & 1\end{pmatrix}\begin{pmatrix}7\\ 6\\ 14\end{pmatrix}\right\|=\left\|\begin{pmatrix}0.25\\ -0.25\\ -0.25\\ 0.25\end{pmatrix}\right\|=\frac{1}{2}.$

Section 5.4 Inner product space

Definition 5.4.1

An **inner product** on a vector space V is an operation on V that assigns, to each pair vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ in V, a real number $\langle \boldsymbol{x}, \boldsymbol{y} \rangle$ satisfying the following conditions:

I. $\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$ with equality if and only if $\boldsymbol{x} = \boldsymbol{0}$.

II. $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle$ for all $\boldsymbol{x}$ and $\boldsymbol{y}$ in V.

III. $\langle \alpha \boldsymbol{x} + \beta \boldsymbol{y}, \boldsymbol{z} \rangle = \alpha \langle \boldsymbol{x}, \boldsymbol{z} \rangle + \beta \langle \boldsymbol{y}, \boldsymbol{z} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y}$ and $\boldsymbol{z}$ in V and all scalars α and β.

A vector space V with an inner product is called an **inner product space**.

Definition 5.4.2

The **standard inner product** for $\boldsymbol{R}^n$ is the dot product $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^T \boldsymbol{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$.

Given a vector $\boldsymbol{w} = (w_1, \cdots, w_n)$ with positive entries, we could also define an inner product on $\boldsymbol{R}^n$ by

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = w_1 x_1 y_1 + w_2 x_2 y_2 + \cdots + w_n x_n y_n.$$

The entries w_i are referred to as **weights**.

Definition 5.4.3

We may define an inner product on $C[a, b]$, vector space of continuous functions on $[a, b]$, by

$$\langle f, g \rangle = \int_a^b f(x) g(x) dx$$

If $w(x) > 0$ and continuous on $[a, b]$, then

$$\langle f, g \rangle = \int_a^b w(x) f(x) g(x) dx$$

also defines an inner product on $C[a, b]$. The function $w(x)$ is called a **weight function**.

Definition 5.4.4

Two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ in an inner product space V are said to be **orthogonal** if $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = 0$. Two subspaces W_1 and W_2 of V are said to be **orthogonal** if $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = 0$ for every $\boldsymbol{u} \in W_1$ and $\boldsymbol{v} \in W_2$. If W_1 and W_2 are orthogonal, we write $W_1 \perp W_2$.

Theorem 5.4.5

For the vector space $C[-L, L]$, define an inner product by

$$\langle f, g \rangle = \frac{1}{L} \int_{-L}^{L} f(x) g(x) dx.$$

Let $W_1 = \{ f \in C[-L, L] : f(-x) = -f(x) \text{ for all } x \in [-L, L] \}$ be subspace of odd functions and $W_2 = \{ g \in C[-L, L] : g(-x) = g(x) \text{ for all } x \in [-L, L] \}$ be subspace of even functions, then $W_1 \perp W_2$. That means $\langle f, g \rangle = 0$ for any $f \in W_1$ and $g \in W_2$.

Proof

Let $f \in W_1$ and $g \in W_2$. We have

$$\begin{aligned}
\langle f, g \rangle &= \frac{1}{L} \int_{-L}^{L} f(x) g(x) dx \quad [x = -y] \\
&= \frac{1}{L} \int_{-y=L}^{-y=-L} f(-y) g(-y) d(-y) \\
&= \frac{1}{L} \int_{y=L}^{y=-L} -f(y) g(y) (-dy) \\
&= \frac{-1}{L} \int_{-L}^{L} f(y) g(y) dy \\
&= -\langle f, g \rangle
\end{aligned}$$

Hence $\langle f, g \rangle = 0$.

Definition 5.4.6

If $\boldsymbol{v}$ is a vector in an inner product space V, the **length** or **norm**, of $\boldsymbol{v}$ is given by

$$\| \boldsymbol{v} \| = \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}.$$

Theorem 5.4.7 (Pythagorean Law)

If $\boldsymbol{u}$ and $\boldsymbol{v}$ are orthogonal vectors in an inner product space V, then $\|\boldsymbol{u}+\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2$.

Definition 5.4.8

Let $\boldsymbol{v} \neq \mathbf{0}$. Define scalar projection of $\boldsymbol{u}$ onto $\boldsymbol{v}$ by $\alpha = \dfrac{\langle \boldsymbol{u}, \boldsymbol{v} \rangle}{\|\boldsymbol{v}\|}$ and vector projection of $\boldsymbol{u}$ onto $\boldsymbol{v}$ by $\mathbf{p} = \dfrac{\alpha}{\|\boldsymbol{v}\|}\boldsymbol{v} = \dfrac{\langle \boldsymbol{u}, \boldsymbol{v} \rangle}{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}\boldsymbol{v}$

Remark 5.4.9

If $\boldsymbol{v} \neq \mathbf{0}$ and $\mathbf{p}$ is the vector projection of $\boldsymbol{u}$ onto $\boldsymbol{v}$, then

(i) $\boldsymbol{u} - \mathbf{p}$ and $\mathbf{p}$ are orthogonal.

(ii) $\mathbf{u} = \mathbf{p}$ if and only if $\mathbf{u}$ is a scalar multiple of $\mathbf{v}$.

Theorem 5.4.10

If $\boldsymbol{u}$ and $\boldsymbol{v}$ are any two vectors in an inner product space V, then

$$|\langle \boldsymbol{u}, \boldsymbol{v} \rangle| \leq \|\boldsymbol{u}\|\|\boldsymbol{v}\|$$

Equality holds if and only if $\boldsymbol{u}$ and $\boldsymbol{v}$ are linearly dependent.

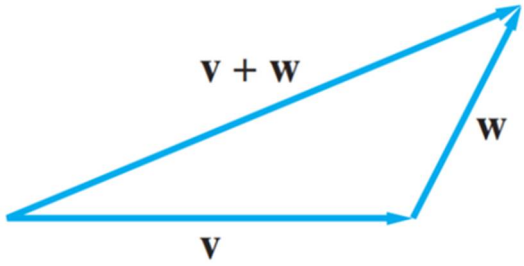
Definition 5.4.11

A vector space V is said to be a **normed linear space** if, for each vector $\boldsymbol{v} \in V$, there is associated a real number $\|\boldsymbol{v}\|$, called the norm of $\boldsymbol{v}$, satisfying

(i) $\|\boldsymbol{v}\| \geq 0$ with equality if and only if $\boldsymbol{v} = 0$.

(ii) $\|\alpha \boldsymbol{v}\| = |\alpha| \|\boldsymbol{v}\|$ for any scalar α.

(iii) $\|\boldsymbol{v} + \boldsymbol{w}\| \le \|\boldsymbol{v}\| + \|\boldsymbol{w}\|$ for all $\boldsymbol{v}, \boldsymbol{w} \in V$.



The 3rd condition is called triangle inequality.

Definition 5.4.12

Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be vectors in a normed space. The **distance** between $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined to be the number $\|\boldsymbol{y} - \boldsymbol{x}\|$.

Theorem 5.4.13

If V is an inner product space, then

$$\|\boldsymbol{v}\| = \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle} \text{ for all } \boldsymbol{v} \in V$$

defines a norm on V.

Example 5.4.14

For the vector space $C[-L, L]$, define an inner product as in Theorem 5.4.5. Define a norm

$$\|f\| = \sqrt{\frac{1}{L}\int_{-L}^{L} f(x)^2 \, dx}.$$

Then for any positive integer m and n,

$$\left\| \frac{1}{\sqrt{2}} \right\| = \sqrt{\frac{1}{L}\int_{-L}^{L} \frac{1}{2} dx} = \sqrt{\frac{2L}{2L}} = 1$$

$$\left\| \cos\frac{n\pi x}{L} \right\| = \sqrt{\frac{1}{L}\int_{-L}^{L} \cos^2\frac{n\pi x}{L} dx} = \sqrt{\frac{1}{2L}\int_{-L}^{L} \left(1 + \cos\frac{2n\pi x}{L}\right) dx} = \sqrt{\frac{2L}{2L} + \frac{1}{4n\pi}\left[\sin\frac{2n\pi x}{L}\right]_{-L}^{L}} = 1$$

$$\left\| \sin\frac{n\pi x}{L} \right\| = \sqrt{\frac{1}{L}\int_{-L}^{L} \sin^2\frac{n\pi x}{L} dx} = \sqrt{\frac{1}{2L}\int_{-L}^{L} \left(1 - \cos\frac{2n\pi x}{L}\right) dx} = \sqrt{\frac{2L}{2L} - \frac{1}{4n\pi}\left[\sin\frac{2n\pi x}{L}\right]_{-L}^{L}} = 1$$

Section 5.5 Orthonormal Sets

Definition 5.5.1

Let $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ be nonzero vectors in an inner product space V. If $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$ whenever $i \neq j$, then $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$ is said to be an **orthogonal set** of vectors.

Example 5.5.2

The set $\{(1,2,2)^T, (2,1,-2)^T, (2,-2,1)^T\}$ is an orthogonal set in $\boldsymbol{R}^3$ since

$$\langle (1,2,2)^T, (2,1,-2)^T \rangle = \langle (1,2,2)^T, (2,-2,1)^T \rangle = \langle (2,1,-2)^T, (2,-2,1)^T \rangle = 0$$

Theorem 5.5.3

If $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$ is an orthogonal set of nonzero vectors in an inner product space V, then $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ are linearly independent.

Proof

Suppose $a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_n\boldsymbol{v}_n = \boldsymbol{0}$. Since $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$ for $i \neq j$, we have

$$0 = \langle \boldsymbol{v}_i, a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_n\boldsymbol{v}_n \rangle = a_1 \langle \boldsymbol{v}_i, \boldsymbol{v}_1 \rangle \boldsymbol{v}_1 + \cdots + a_i \langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle + \cdots + a_n \langle \boldsymbol{v}_i, \boldsymbol{v}_n \rangle = a_i \langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle$$

Since each $\boldsymbol{v}_i$ is nonzero, $\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle = \|\boldsymbol{v}_i\|^2 \neq 0$. Therefore $a_i = 0$.

Definition 5.5.4

An **orthonormal set** of vectors is an orthogonal set of unit vectors. In other words,

$\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n\}$ is orthonormal if and only if $\langle \boldsymbol{u}_i, \boldsymbol{u}_j \rangle = \delta_{ij}$ where $\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$

Given any orthogonal set of nonzero vectors $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$, it is possible to form an orthonormal set by defining

$$\boldsymbol{u}_i = \left(\frac{1}{\|\boldsymbol{v}_i\|}\right)\boldsymbol{v}_i \quad \text{for } i = 1, 2, \cdots, n$$

Example 5.5.5

In Example 5.5.2, $\{(1,2,2)^T, (2,1,-2)^T, (2,-2,1)^T\}$ is an orthogonal set. To form an orthonormal set, let

$$\boldsymbol{u}_1 = \left(\frac{1}{\|\boldsymbol{v}_1\|}\right)\boldsymbol{v}_1 = \frac{1}{3}(1,2,2)^T, \quad \boldsymbol{u}_2 = \left(\frac{1}{\|\boldsymbol{v}_2\|}\right)\boldsymbol{v}_2 = \frac{1}{3}(2,1,-2)^T, \quad \boldsymbol{u}_3 = \left(\frac{1}{\|\boldsymbol{v}_3\|}\right)\boldsymbol{v}_3 = \frac{1}{3}(2,-2,1)^T$$

$\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$ forms an orthonormal set.

Theorem 5.5.6 (Fourier Series)

For the vector space $C[-L, L]$, define an inner product as in Theorem 5.4.5. Show that $\left\{\frac{1}{\sqrt{2}}, \cos\frac{\pi x}{L}, \sin\frac{\pi x}{L}, \cdots, \cos\frac{n\pi x}{L}, \sin\frac{n\pi x}{L}, \cdots\right\}$ is an orthonormal set of nonzero vectors.

Proof

$$\left\langle \frac{1}{\sqrt{2}}, \cos\frac{m\pi x}{L} \right\rangle = \frac{1}{\sqrt{2}L}\int_{-L}^{L} \cos\frac{m\pi x}{L}\,dx = \frac{1}{m\sqrt{2}\pi}\left[\sin\frac{m\pi x}{L}\right]_{-L}^{L} = 0$$

$$\left\langle \frac{1}{\sqrt{2}}, \sin\frac{m\pi x}{L} \right\rangle = \frac{1}{\sqrt{2}L}\int_{-L}^{L} \sin\frac{m\pi x}{L}\,dx = \frac{-1}{m\sqrt{2}\pi}\left[\cos\frac{m\pi x}{L}\right]_{-L}^{L} = 0$$

$$\left\langle \sin\frac{m\pi x}{L}, \cos\frac{n\pi x}{L} \right\rangle = 0 \quad \text{by Theorem 5.4.5}$$

For any distinct positive integers m and n,

$$\begin{aligned}\left\langle \cos\frac{m\pi x}{L}, \cos\frac{n\pi x}{L} \right\rangle &= \frac{1}{L}\int_{-L}^{L} \cos\frac{m\pi x}{L}\cos\frac{n\pi x}{L}\,dx \\ &= \frac{1}{2L}\int_{-L}^{L} \cos\frac{(m-n)\pi x}{L} + \cos\frac{(m+n)\pi x}{L}\,dx \\ &= \frac{1}{2(m-n)\pi}\left[\sin\frac{(m-n)\pi x}{L}\right]_{-L}^{L} + \frac{1}{2(m+n)\pi}\left[\sin\frac{(m+n)\pi x}{L}\right]_{-L}^{L} \\ &= 0\end{aligned}$$

$$\left\langle \sin\frac{m\pi x}{L}, \sin\frac{n\pi x}{L} \right\rangle = \frac{1}{L}\int_{-L}^{L} \sin\frac{m\pi x}{L}\sin\frac{n\pi x}{L}\,dx$$
$$= \frac{1}{2L}\int_{-L}^{L} \cos\frac{(m-n)\pi x}{L} - \cos\frac{(m+n)\pi x}{L}\,dx$$
$$= \frac{1}{2(m-n)\pi}\left[\sin\frac{(m-n)\pi x}{L}\right]_{-L}^{L} - \frac{1}{2(m+n)\pi}\left[\sin\frac{(m+n)\pi x}{L}\right]_{-L}^{L}$$
$$= 0$$

The above result together with Example 5.4.14 imply $\left\{\frac{1}{\sqrt{2}}, \cos\frac{\pi x}{L}, \sin\frac{\pi x}{L}, \cdots, \cos\frac{n\pi x}{L}, \sin\frac{n\pi x}{L}, \cdots\right\}$ is an orthonormal set.

Theorem 5.5.7

Let $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n\}$ be orthonormal basis for an inner product space V. If $\boldsymbol{u} = \sum_{i=1}^{n} a_i \boldsymbol{u}_i$ and $\boldsymbol{v} = \sum_{i=1}^{n} b_i \boldsymbol{u}_i$, then $a_i = \langle \boldsymbol{u}, \boldsymbol{u}_i \rangle, b_i = \langle \boldsymbol{v}, \boldsymbol{u}_i \rangle, \langle \boldsymbol{u}, \boldsymbol{v} \rangle = \sum_{i=1}^{n} a_i b_i$ and $\|\boldsymbol{u}\|^2 = \sum_{i=1}^{n} a_i^2$.

Proof

Suppose $\boldsymbol{u} = a_1\boldsymbol{u}_1 + a_2\boldsymbol{u}_2 + \cdots + a_n\boldsymbol{u}_n$. We have

$$\langle \boldsymbol{u}, \boldsymbol{u}_i \rangle = \langle a_1\boldsymbol{u}_1 + a_2\boldsymbol{u}_2 + \cdots + a_n\boldsymbol{u}_n, \boldsymbol{u}_i \rangle = a_1 \langle \boldsymbol{u}_1, \boldsymbol{u}_i \rangle + \cdots + a_i \langle \boldsymbol{u}_i, \boldsymbol{u}_i \rangle + \cdots + a_n \langle \boldsymbol{u}_n, \boldsymbol{u}_n \rangle = a_i.$$

Similarly, $b_i = \langle \boldsymbol{v}, \boldsymbol{u}_i \rangle$. Furthermore,

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \left\langle \sum_{i=1}^{n} a_i \boldsymbol{u}_i, \sum_{j=1}^{n} b_j \boldsymbol{u}_j \right\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i b_j \langle \boldsymbol{u}_i, \boldsymbol{u}_j \rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i b_j \delta_{ij} = \sum_{i=1}^{n} a_i b_i$$

In particular for $\boldsymbol{u} = \boldsymbol{v}$, we have $\|\boldsymbol{u}\|^2 = \sum_{i=1}^{n} a_i^2$.

Example 5.5.8

Express $(1,1,3)^T$ and $(1,4,3)^T$ as linear combination of $\boldsymbol{u}_1 = \frac{1}{3}(1,2,2)^T, \boldsymbol{u}_2 = \frac{1}{3}(2,1,-2)^T$ and $\boldsymbol{u}_3 = \frac{1}{3}(2,-2,1)^T$.

Solution

Since $(1,1,3)^T, (1,4,3)^T \in \boldsymbol{R}^3 = span\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$, we have $(1,1,3)^T = a_1\boldsymbol{u}_1 + a_2\boldsymbol{u}_2 + a_3\boldsymbol{u}_3$ and $(1,4,3)^T = b_1\boldsymbol{u}_1 + b_2\boldsymbol{u}_2 + b_3\boldsymbol{u}_3$. Then

$$a_1 = \langle (1,1,3)^T, \boldsymbol{u}_1 \rangle = 3 \quad a_2 = \langle (1,1,3)^T, \boldsymbol{u}_2 \rangle = -1 \quad a_3 = \langle (1,1,3)^T, \boldsymbol{u}_3 \rangle = 1$$
$$b_1 = \langle (1,4,3)^T, \boldsymbol{u}_1 \rangle = 5 \quad b_2 = \langle (1,4,3)^T, \boldsymbol{u}_2 \rangle = 0 \quad b_3 = \langle (1,4,3)^T, \boldsymbol{u}_3 \rangle = -1$$

Hence $(1,1,3)^T = 3\boldsymbol{u}_1 - \boldsymbol{u}_2 + \boldsymbol{u}_3$ and $(1,4,3)^T = 5\boldsymbol{u}_1 + 0\boldsymbol{u}_2 - \boldsymbol{u}_3$.

Furthermore, $\|(1,4,3)\|^2 = 1^2 + 4^2 + 3^2 = 26 = 5^2 + 0^2 + (-1)^2 = b_1^2 + b_2^2 + b_3^2$ and $(1,1,3)\cdot(1,4,3)$ $= 14 = 3(5) - 1(0) + 1(-1) = a_1b_1 + a_2b_2 + a_3b_3$.

Theorem 5.5.9 (Fourier Series)

In $C[-L,L]$, define an inner product as in Theorem 5.3.5. By Theorem 5.5.6, the set $\left\{\frac{1}{\sqrt{2}}, \cos\frac{\pi x}{L}, \sin\frac{\pi x}{L}, \cdots, \cos\frac{n\pi x}{L}, \sin\frac{n\pi x}{L}, \cdots\right\}$ is an orthonormal set of nonzero vectors. Let f be a function on $[-L,L]$. Suppose

$$f(x) = \frac{a_0}{2} + a_1\cos\frac{\pi x}{L} + b_1\sin\frac{\pi x}{L} + \cdots + a_n\cos\frac{n\pi x}{L} + b_n\sin\frac{n\pi x}{L} + \cdots$$

Then

$$a_0 = \frac{1}{L}\int_{-L}^{L}\frac{a_0}{2}dx = \langle f, 1\rangle = \frac{1}{L}\int_{-L}^{L} f(x)dx$$
$$a_n = \left\langle f, \cos\frac{n\pi x}{L}\right\rangle = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}dx \quad \text{for } n = 1,2,\cdots$$
$$b_n = \left\langle f, \sin\frac{n\pi x}{L}\right\rangle = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}dx \quad \text{for } n = 1,2,\cdots$$

The coefficient a_n and b_n are called the Fourier coefficients of f.

Example 5.5.10

The set

$$S = \left\{ \tfrac{1}{\sqrt{2}}, \cos x, \cos 2x, \cos 3x, \cos 4x \right\}$$

is an orthonormal set of vectors in $C[-\pi, \pi]$ with inner product defined by Definition 5.4.3.

(a) Use trigonometric identities to write the functions $\sin^2 x$ and $\cos^4 x$ as a linear combination of elements of *S*.

(b) Use part (a) and Theorem 5.5.7 to find the value of $\int_{-\pi}^{\pi} \sin^2 x \cos^4 x dx$

Solution

(a) $\cos^4 x = \left(\dfrac{1+\cos 2x}{2} \right)^2$

$$= \frac{1 + 2\cos 2x + \cos^2 2x}{4}$$

$$= \frac{1 + 2\cos 2x + \frac{1}{2}(1+\cos 4x)}{4}$$

$$= \frac{3}{8} + \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$$

$$= \frac{3}{4\sqrt{2}}\left(\frac{1}{\sqrt{2}} \right) + \frac{1}{2}(\cos 2x) + \frac{1}{8}(\cos 4x)$$

$$\sin^2 x = \left(\frac{1-\cos 2x}{2} \right) = \frac{1}{2} - \frac{1}{2}\cos 2x = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}} \right) - \frac{1}{2}(\cos 2x)$$

(b) $\dfrac{1}{\pi}\int_{-\pi}^{\pi} \sin^2 x \cos^4 x dx = \left\langle \sin^2 x, \cos^4 x \right\rangle = \left(\dfrac{3}{4\sqrt{2}} \right)\left(\dfrac{1}{\sqrt{2}} \right) + \left(\dfrac{1}{2} \right)\left(-\dfrac{1}{2} \right) = \dfrac{1}{8}$

$$\int_{-\pi}^{\pi} \sin^2 x \cos^4 x dx = \frac{\pi}{8}$$

Theorem 5.5.10

If $f \in C[-L, L]$ is an odd function then $a_n = 0$ for $n = 0,1,2,\cdots$. We have

$$f(x) = b_1 \sin \frac{\pi x}{L} + \cdots + b_n \sin \frac{n\pi x}{L} + \cdots$$

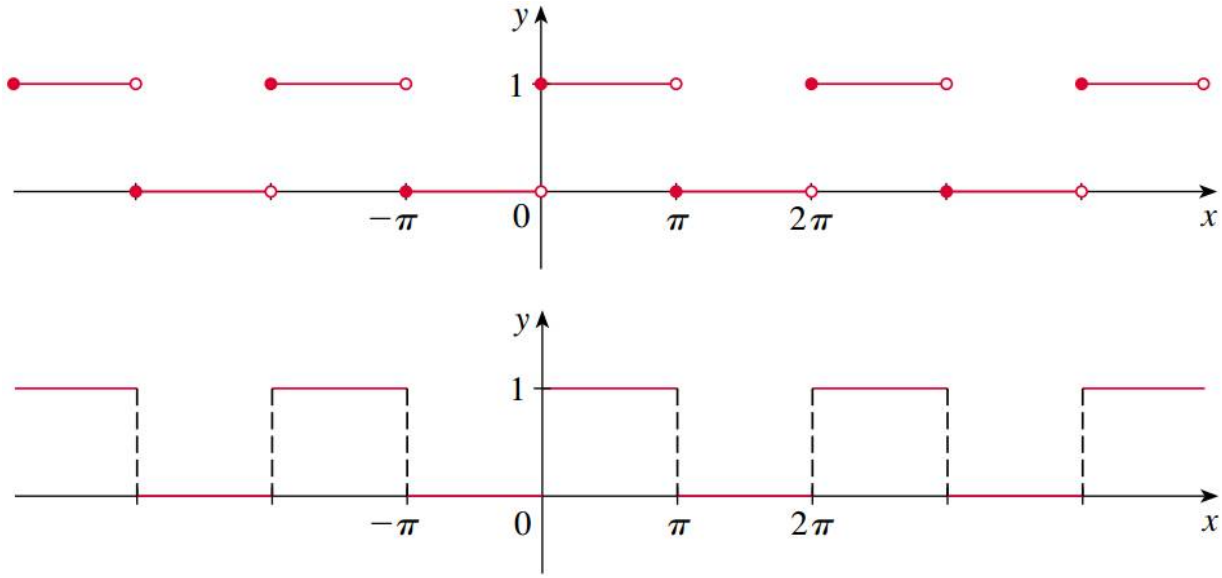
If $f \in C[-L, L]$ is an even function then $b_n = 0$ for $n = 1,2,\cdots$. We have

$$f(x) = \frac{a_0}{2} + a_1 \cos \frac{\pi x}{L} + \cdots + a_n \cos \frac{n\pi x}{L} + \cdots$$

Example 5.5.11

Find the Fourier coefficients and Fourier series of the square-wave function f defined by

$$f(x)=\begin{cases}0 & \text{if } -\pi \le x<0\\ 1 & \text{if } 0\le x<\pi\end{cases} \quad \text{and} \quad f(x+2\pi)=f(x)$$



Solution

$$a_0=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)dx=\frac{1}{\pi}\int_{-\pi}^{0}f(x)dx+\frac{1}{\pi}\int_{0}^{\pi}f(x)dx=\frac{1}{\pi}\int_{-\pi}^{0}0dx+\frac{1}{\pi}\int_{0}^{\pi}1dx=1.$$

For $n\ge 1$,

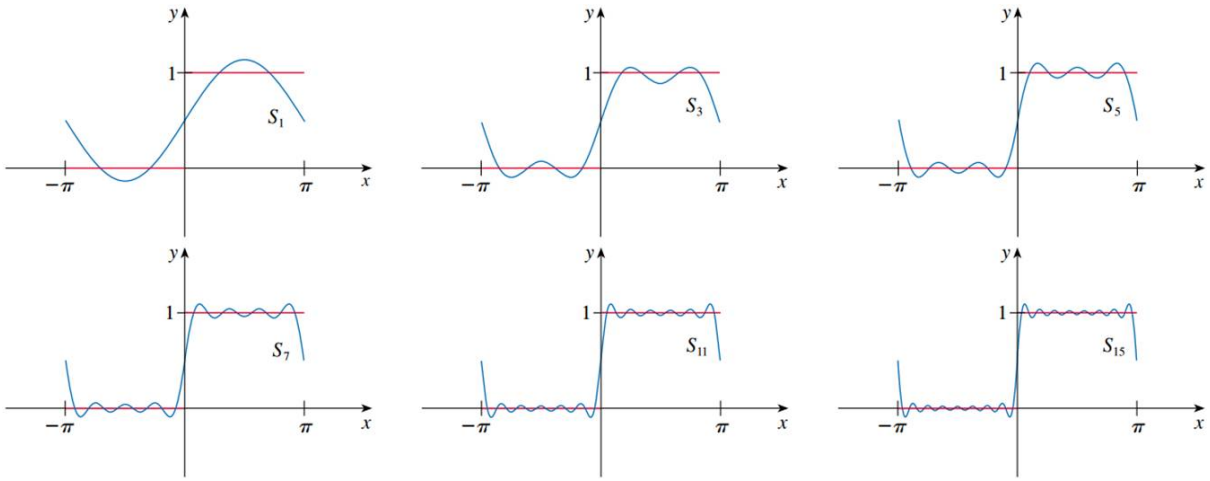
$$\begin{aligned}a_n&=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos nxdx\\&=\frac{1}{\pi}\int_{-\pi}^{0}f(x)\cos nxdx+\frac{1}{\pi}\int_{0}^{\pi}f(x)\cos nxdx\\&=\frac{1}{\pi}\int_{-\pi}^{0}0dx+\frac{1}{\pi}\int_{0}^{\pi}\cos nxdx\\&=\frac{\sin nx}{n\pi}\Big|_{0}^{\pi}=0\end{aligned}$$

$$\begin{aligned}b_n&=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\sin nxdx\\&=\frac{1}{\pi}\int_{-\pi}^{0}f(x)\sin nxdx+\frac{1}{\pi}\int_{0}^{\pi}f(x)\sin nxdx\\&=\frac{1}{\pi}\int_{-\pi}^{0}0dx+\frac{1}{\pi}\int_{0}^{\pi}\sin nxdx\\&=-\frac{\cos nx}{n\pi}\Big|_{0}^{\pi}\end{aligned}$$

$$= \begin{cases} 0 & \text{if } n \text{ is even} \\ \dfrac{2}{n\pi} & \text{if } n \text{ is odd} \end{cases}$$

Fourier series of f is

$$\frac{1}{2} + \frac{2}{\pi}\sin x + \frac{2}{3\pi}\sin 3x + \frac{2}{5\pi}\sin 5x + \cdots$$



Definition 5.5.12

An $n \times n$ matrix Q is said to be an **orthogonal matrix** if the column vectors of Q form an orthonormal set in $\boldsymbol{R}^n$.

Theorem 5.5.13

An $n \times n$ matrix Q is orthogonal if and only if $Q^T Q = I$.

Proof

Q is orthogonal iff $Q = \begin{pmatrix} | & | & & | \\ \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_n \\ | & | & & | \end{pmatrix}_{n\times n}$ where $\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n$ are orthonormal vectors.

$$\text{iff } Q^T Q = \begin{pmatrix} - & \boldsymbol{u}_1^T & - \\ - & \boldsymbol{u}_2^T & - \\ & \vdots & \\ - & \boldsymbol{u}_n^T & - \end{pmatrix} \begin{pmatrix} | & | & & | \\ \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_n \\ | & | & & | \end{pmatrix} = \begin{pmatrix} \boldsymbol{u}_1^T\boldsymbol{u}_1 & \boldsymbol{u}_1^T\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_1^T\boldsymbol{u}_n \\ \boldsymbol{u}_2^T\boldsymbol{u}_1 & \boldsymbol{u}_2^T\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_2^T\boldsymbol{u}_n \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{u}_n^T\boldsymbol{u}_1 & \boldsymbol{u}_n^T\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_n^T\boldsymbol{u}_n \end{pmatrix} = I_n$$

It follows that from Theorem 5.5.13 that if Q is orthogonal, then Q is invertible and $Q^{-1} = Q^T$.

Theorem 5.5.14

If Q is an $n \times n$ orthogonal matrix, then

$$Q\boldsymbol{x} \cdot Q\boldsymbol{y} = \boldsymbol{x} \cdot \boldsymbol{y} \qquad \text{and} \qquad \|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$$

Proof

$Q\boldsymbol{x} \cdot Q\boldsymbol{y} = (Q\boldsymbol{x})^T Q\boldsymbol{y} = \boldsymbol{x}^T Q^T Q\boldsymbol{y} = \boldsymbol{x}^T \boldsymbol{y} = \boldsymbol{x} \cdot \boldsymbol{y}$. If $\boldsymbol{x} = \boldsymbol{y}$, then $\|Q\boldsymbol{x}\|^2 = Q\boldsymbol{x} \cdot Q\boldsymbol{x} = \boldsymbol{x} \cdot \boldsymbol{x} = \|\boldsymbol{x}\|^2$.

Hence $\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$.

Example 5.5.15

Rotation matrix

$$Q = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

is orthogonal and

$$Q^{-1} = Q^T = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

Rotation preserves length of vector and angle between 2 vectors (Example 4.1.16).

Example 5.5.16

Permutation matrix is a matrix formed from the identity by reordering its column.

$$\begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}, \begin{pmatrix} 1 & & \\ & & 1 \\ & 1 & \end{pmatrix}, \begin{pmatrix} & 1 & \\ 1 & & \\ & & 1 \end{pmatrix}, \begin{pmatrix} & 1 & \\ & & 1 \\ 1 & & \end{pmatrix}, \begin{pmatrix} & & 1 \\ 1 & & \\ & 1 & \end{pmatrix}, \begin{pmatrix} & & 1 \\ & 1 & \\ 1 & & \end{pmatrix}$$

3×3 permutation matrices

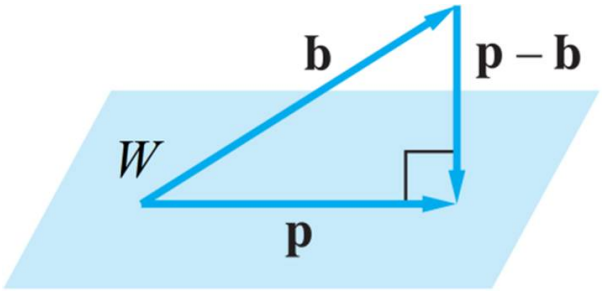
Permutation matrices are orthogonal.

Definition 5.5.17

Let W be a subspace of an inner product space V. Then V can be written as $V = W \oplus W^{\perp}$. For any $\boldsymbol{b} \in V$,

$$\boldsymbol{b} = \underbrace{\boldsymbol{p}}_{\in W} + \underbrace{\boldsymbol{b} - \boldsymbol{p}}_{\in W^{\perp}}$$

where $\boldsymbol{p} \in W$ and $\boldsymbol{b} - \boldsymbol{p} \in W^{\perp}$. $\boldsymbol{p}$ is called orthogonal projection of $\boldsymbol{b}$ onto W, denoted by $\text{proj}_W \boldsymbol{b}$. Similarly, $\boldsymbol{b} - \boldsymbol{p}$ is called orthogonal projection of $\boldsymbol{b}$ onto $W^{\perp}$, denoted by $\text{proj}_{W^{\perp}} \boldsymbol{b}$.



Theorem 5.5.18

Under the hypothesis of Definition 5.5.17, $\boldsymbol{p}$ is the element of W that is closest to $\boldsymbol{x}$; that is,

$$\|\boldsymbol{y} - \boldsymbol{b}\| > \|\boldsymbol{p} - \boldsymbol{b}\|$$

for any $\boldsymbol{y} \neq \boldsymbol{p}$ in W.

The proof is similar to the proof of Theorem 5.3.2.

Example 5.5.19 (Projection)

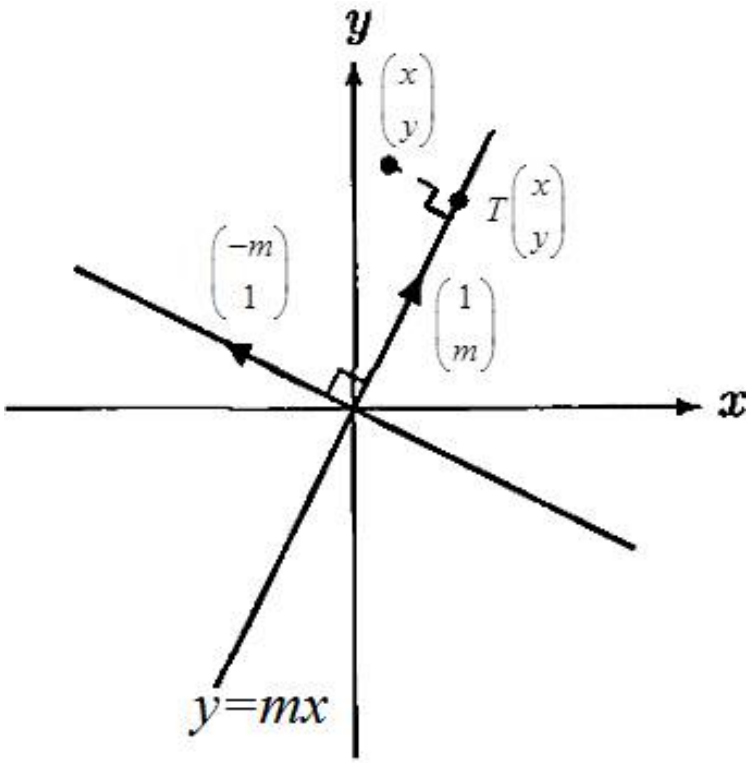
Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be linear transformation defined by $T\begin{pmatrix} 1 \\ m \end{pmatrix} = \begin{pmatrix} 1 \\ m \end{pmatrix}$ and $T\begin{pmatrix} -m \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Then $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{1+m^2} & \frac{m}{1+m^2} \\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$. Let $W = span\left\{\begin{pmatrix} 1 \\ m \end{pmatrix}\right\}$. Then $W^{\perp} = span\left\{\begin{pmatrix} -m \\ 1 \end{pmatrix}\right\}$. Let

$\boldsymbol{b}=\begin{pmatrix} x \\ y \end{pmatrix}$. In Example 5.1.11,

$$\boldsymbol{p}=\frac{(x,y)\begin{pmatrix} 1 \\ m \end{pmatrix}}{(1,m)\begin{pmatrix} 1 \\ m \end{pmatrix}}\begin{pmatrix} 1 \\ m \end{pmatrix}=\frac{x+my}{1+m^2}\begin{pmatrix} 1 \\ m \end{pmatrix}=\begin{pmatrix} \frac{1}{1+m^2} & \frac{m}{1+m^2} \\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\text{proj}_W\begin{pmatrix} x \\ y \end{pmatrix}.$$

$$\begin{aligned}
\boldsymbol{b}-\boldsymbol{p} &= \begin{pmatrix} x \\ y \end{pmatrix}-\frac{x+my}{1+m^2}\begin{pmatrix} 1 \\ m \end{pmatrix} \\
&= \begin{pmatrix} x-\frac{x+my}{1+m^2} \\ y-m\frac{x+my}{1+m^2} \end{pmatrix} \\
&= \begin{pmatrix} \frac{x(1+m^2)-(x+my)}{1+m^2} \\ \frac{y(1+m^2)-(mx+m^2y)}{1+m^2} \end{pmatrix} \\
&= \begin{pmatrix} \frac{xm^2-my}{1+m^2} \\ \frac{y-mx}{1+m^2} \end{pmatrix} \\
&= \frac{y-mx}{1+m^2}\begin{pmatrix} -m \\ 1 \end{pmatrix} \\
&= \frac{(x,y)\begin{pmatrix} -m \\ 1 \end{pmatrix}}{(-m,1)\begin{pmatrix} -m \\ 1 \end{pmatrix}}\begin{pmatrix} -m \\ 1 \end{pmatrix}\in W^{\perp}
\end{aligned}$$



Indeed, $\boldsymbol{b}-\boldsymbol{p}=\text{proj}_{W^{\perp}}\begin{pmatrix} x \\ y \end{pmatrix}$ and $\boldsymbol{x}=\text{proj}_W\begin{pmatrix} x \\ y \end{pmatrix}+\text{proj}_{W^{\perp}}\begin{pmatrix} x \\ y \end{pmatrix}$.

Theorem 5.5.20

Let W be a subspace of a finite dimensional inner product space V and let $\boldsymbol{b}\in V$. Let $\{\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_n\}$ and $\{\boldsymbol{v}_{n+1},\boldsymbol{v}_{n+2},\cdots,\boldsymbol{v}_m\}$ be orthogonal base of W and $W^{\perp}$ respectively. Then

$$\boldsymbol{p}=\text{proj}_W\boldsymbol{b}=\frac{\langle \boldsymbol{b},\boldsymbol{v}_1\rangle}{\langle \boldsymbol{v}_1,\boldsymbol{v}_1\rangle}\boldsymbol{v}_1+\frac{\langle \boldsymbol{b},\boldsymbol{v}_2\rangle}{\langle \boldsymbol{v}_2,\boldsymbol{v}_2\rangle}\boldsymbol{v}_2+\cdots+\frac{\langle \boldsymbol{b},\boldsymbol{v}_n\rangle}{\langle \boldsymbol{v}_n,\boldsymbol{v}_n\rangle}\boldsymbol{v}_n.$$

Furthermore, $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n, \cdots, \boldsymbol{v}_m\}$ is an orthogonal basis of V and

$$\boldsymbol{b} - \boldsymbol{p} = \text{proj}_{W^\perp}\boldsymbol{b} = \frac{\langle \boldsymbol{b}, \boldsymbol{v}_{n+1}\rangle}{\langle \boldsymbol{v}_{n+1}, \boldsymbol{v}_{n+1}\rangle}\boldsymbol{v}_{n+1} + \frac{\langle \boldsymbol{b}, \boldsymbol{v}_{n+2}\rangle}{\langle \boldsymbol{v}_{n+2}, \boldsymbol{v}_{n+2}\rangle}\boldsymbol{v}_{n+2} + \cdots + \frac{\langle \boldsymbol{b}, \boldsymbol{v}_m\rangle}{\langle \boldsymbol{v}_m, \boldsymbol{v}_m\rangle}\boldsymbol{v}_m.$$

Proof

Normalize the orthogonal basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_m\}$ of V, $\left\{\frac{\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|}, \frac{\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|}, \cdots, \frac{\boldsymbol{v}_m}{\|\boldsymbol{v}_m\|}\right\}$ is an orthonormal basis of V. Since $\boldsymbol{b} \in V$, by Theorem 5.5.7,

$$\begin{aligned}\boldsymbol{b} &= \left\langle \boldsymbol{b}, \frac{\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|}\right\rangle \frac{\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|} + \cdots + \left\langle \boldsymbol{b}, \frac{\boldsymbol{v}_n}{\|\boldsymbol{v}_n\|}\right\rangle \frac{\boldsymbol{v}_n}{\|\boldsymbol{v}_n\|} + \left\langle \boldsymbol{b}, \frac{\boldsymbol{v}_{n+1}}{\|\boldsymbol{v}_{n+1}\|}\right\rangle \frac{\boldsymbol{v}_{n+1}}{\|\boldsymbol{v}_{n+1}\|} + \cdots + \left\langle \boldsymbol{b}, \frac{\boldsymbol{v}_m}{\|\boldsymbol{v}_m\|}\right\rangle \frac{\boldsymbol{v}_m}{\|\boldsymbol{v}_m\|} \\ &= \underbrace{\frac{\langle \boldsymbol{b}, \boldsymbol{v}_1\rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1\rangle}\boldsymbol{v}_1 + + \cdots \frac{\langle \boldsymbol{b}, \boldsymbol{v}_n\rangle}{\langle \boldsymbol{v}_n, \boldsymbol{v}_n\rangle}\boldsymbol{v}_n}_{\in W} + \underbrace{\frac{\langle \boldsymbol{b}, \boldsymbol{v}_{n+1}\rangle}{\langle \boldsymbol{v}_{n+1}, \boldsymbol{v}_{n+1}\rangle}\boldsymbol{v}_{n+1} + \cdots + \frac{\langle \boldsymbol{b}, \boldsymbol{v}_m\rangle}{\langle \boldsymbol{v}_m, \boldsymbol{v}_m\rangle}\boldsymbol{v}_m}_{\in W^\perp}\end{aligned}$$

Furthermore, $V = W \oplus W^\perp$. $\boldsymbol{b}$ can be written uniquely as sum of a vector in W and a vector in $W^\perp$. We have

$$\boldsymbol{p} = \text{proj}_W\boldsymbol{b} = \frac{\langle \boldsymbol{b}, \boldsymbol{v}_1\rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1\rangle}\boldsymbol{v}_1 + \frac{\langle \boldsymbol{b}, \boldsymbol{v}_2\rangle}{\langle \boldsymbol{v}_2, \boldsymbol{v}_2\rangle}\boldsymbol{v}_2 + \cdots + \frac{\langle \boldsymbol{b}, \boldsymbol{v}_n\rangle}{\langle \boldsymbol{v}_n, \boldsymbol{v}_n\rangle}\boldsymbol{v}_n \quad \text{and}$$

$$\boldsymbol{b} - \boldsymbol{p} = \text{proj}_{W^\perp}\boldsymbol{b} = \frac{\langle \boldsymbol{b}, \boldsymbol{v}_{n+1}\rangle}{\langle \boldsymbol{v}_{n+1}, \boldsymbol{v}_{n+1}\rangle}\boldsymbol{v}_{n+1} + \frac{\langle \boldsymbol{b}, \boldsymbol{v}_{n+2}\rangle}{\langle \boldsymbol{v}_{n+2}, \boldsymbol{v}_{n+2}\rangle}\boldsymbol{v}_{n+2} + \cdots + \frac{\langle \boldsymbol{b}, \boldsymbol{v}_m\rangle}{\langle \boldsymbol{v}_m, \boldsymbol{v}_m\rangle}\boldsymbol{v}_m.$$

Theorem 5.5.21

Let $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n\}$ be orthonormal vectors in $\boldsymbol{R}^m$ and $W = \text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \cdots, \boldsymbol{u}_n\}$. Let

$$A = \begin{pmatrix} | & | & & | \\ \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_n \\ | & | & & | \end{pmatrix}_{m\times n}$$ be an $m \times n$ matrix with $m > n$. For each $\boldsymbol{b} \in \boldsymbol{R}^m$, the unique vector $\boldsymbol{p} = A\hat{\boldsymbol{x}} \in \text{Col}(A) = W$ in Theorem 5.3.2 is equal to

$$\boldsymbol{p} = \text{proj}_{\text{Col}(A)}\boldsymbol{b}$$

and

$$\hat{\boldsymbol{x}} = A^T\boldsymbol{b} = (\boldsymbol{u}_1 \cdot \boldsymbol{b}, \boldsymbol{u}_2 \cdot \boldsymbol{b}, \cdots, \boldsymbol{u}_n \cdot \boldsymbol{b})^T$$

Proof

By Theorem 5.3.3,

$$\begin{aligned}\hat{\boldsymbol{x}} &= (A^TA)^{-1}A^T\boldsymbol{b}\\ &= \left(\begin{pmatrix} - & \boldsymbol{u}_1^T & - \\ - & \boldsymbol{u}_2^T & - \\ & \vdots & \\ - & \boldsymbol{u}_n^T & - \end{pmatrix}\begin{pmatrix} | & | & & | \\ \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_n \\ | & | & & | \end{pmatrix}\right)^{-1}\begin{pmatrix} - & \boldsymbol{u}_1^T & - \\ - & \boldsymbol{u}_2^T & - \\ & \vdots & \\ - & \boldsymbol{u}_n^T & - \end{pmatrix}\boldsymbol{b}\\ &= \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{pmatrix}_{n\times n}^{-1}\begin{pmatrix} - & \boldsymbol{u}_1^T & - \\ - & \boldsymbol{u}_2^T & - \\ & \vdots & \\ - & \boldsymbol{u}_2^T & - \end{pmatrix}\boldsymbol{b}\\ &= (\boldsymbol{u}_1\cdot\boldsymbol{b}, \boldsymbol{u}_2\cdot\boldsymbol{b}, \cdots, \boldsymbol{u}_n\cdot\boldsymbol{b})^T\end{aligned}$$

$$\begin{aligned}\boldsymbol{p} &= A\hat{\boldsymbol{x}}\\ &= \begin{pmatrix} | & | & & | \\ \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_n \\ | & | & & | \end{pmatrix}\begin{pmatrix} \boldsymbol{u}_1\cdot\boldsymbol{b} \\ \boldsymbol{u}_2\cdot\boldsymbol{b} \\ \vdots \\ \boldsymbol{u}_n\cdot\boldsymbol{b} \end{pmatrix}\\ &= (\boldsymbol{u}_1\cdot\boldsymbol{b})\boldsymbol{u}_1 + (\boldsymbol{u}_2\cdot\boldsymbol{b})\boldsymbol{u}_2 + \cdots + (\boldsymbol{u}_n\cdot\boldsymbol{b})\boldsymbol{u}_n\\ &= \text{proj}_{\text{Col}(A)}\boldsymbol{b}\end{aligned}$$

Example 5.5.22

Let $A = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix}$. Clearly, the 3 columns of A are orthonormal (Check it).

The least-squares solution of the inconsistent system $A\boldsymbol{x} = \boldsymbol{b}$ is $\hat{\boldsymbol{x}} = (\frac{2+3+5+7}{2}, \frac{-2+3-5+7}{2}, \frac{-2-3+5+7}{2})^T$

$= (\frac{17}{2}, \frac{3}{2}, \frac{7}{2})^T$. The orthogonal projection $\boldsymbol{p} = \text{proj}_{\text{Col}(A)}\boldsymbol{b} = \frac{17}{2}\begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} + \frac{3}{2}\begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix} + \frac{7}{2}\begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{7}{4} \\ \frac{13}{4} \\ \frac{21}{4} \\ \frac{27}{4} \end{pmatrix}$.

Section 5.6 Gram-Schmidt Process

In Theorem 5.5.21, the least-squares solution $\hat{\boldsymbol{x}}$ of the inconsistent system $A\boldsymbol{x}=\boldsymbol{b}$ has a simple expression if the column vectors of A are orthonormal. How about if not? We need Gram-Schmidt Process.

Theorem 5.6.1 (Gram-Schmidt Process)

Given a basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n\}$ for a nonzero subspace W of an inner product space V, define

$$\boldsymbol{v}_1 = \boldsymbol{x}_1$$

$$\boldsymbol{v}_2 = \boldsymbol{x}_2 - \text{proj}_{\text{span}\{\boldsymbol{v}_1\}}\boldsymbol{x}_2 = \boldsymbol{x}_2 - \frac{\langle \boldsymbol{x}_2, \boldsymbol{v}_1 \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle}\boldsymbol{v}_1 \in \text{span}\{\boldsymbol{v}_1\}^{\perp}$$

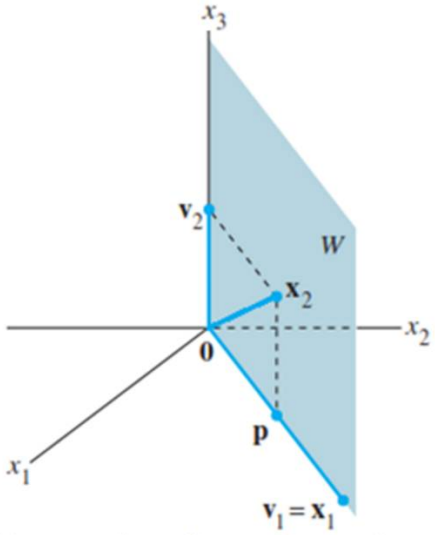
$$\boldsymbol{v}_3 = \boldsymbol{x}_3 - \text{proj}_{\text{span}\{\boldsymbol{v}_1,\boldsymbol{v}_2\}}\boldsymbol{x}_3 = \boldsymbol{x}_3 - \frac{\langle \boldsymbol{x}_3, \boldsymbol{v}_1 \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle}\boldsymbol{v}_1 - \frac{\langle \boldsymbol{x}_3, \boldsymbol{v}_2 \rangle}{\langle \boldsymbol{v}_2, \boldsymbol{v}_2 \rangle}\boldsymbol{v}_2 \in \text{span}\{\boldsymbol{v}_1,\boldsymbol{v}_2\}^{\perp}$$

$$\vdots$$

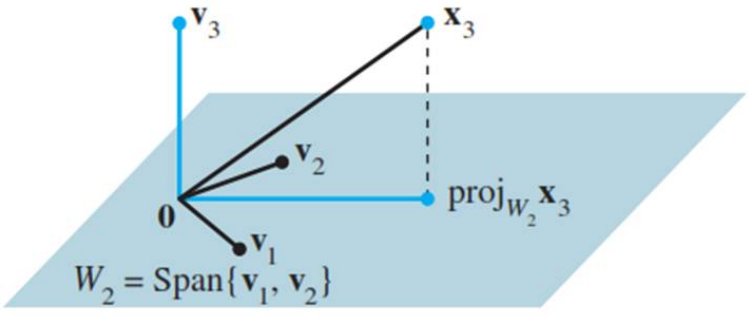
$$\boldsymbol{v}_n = \boldsymbol{x}_n - \text{proj}_{\text{span}\{\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_{n-1}\}}\boldsymbol{x}_n = \boldsymbol{x}_n - \frac{\langle \boldsymbol{x}_n, \boldsymbol{v}_1 \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle}\boldsymbol{v}_1 - \cdots - \frac{\langle \boldsymbol{x}_n, \boldsymbol{v}_{n-1} \rangle}{\langle \boldsymbol{v}_{n-1}, \boldsymbol{v}_{n-1} \rangle}\boldsymbol{v}_{n-1} \in \text{span}\{\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_{n-1}\}^{\perp}$$

Then $\left\{\frac{\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|}, \frac{\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|}, \cdots, \frac{\boldsymbol{v}_n}{\|\boldsymbol{v}_n\|}\right\}$ is an orthonormal basis for W. In addition

$$\text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_k\} = \text{span}\{\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_k\} \quad \text{for } 1 \le k \le n.$$



Construction of an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$.



The construction of $\mathbf{v}_3$ from $\mathbf{x}_3$ and W_2.

Example 5.6.2

Consider the column vectors of A in Example 5.3.7. Let $\boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, $\boldsymbol{x}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$ and $\boldsymbol{x}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$.

Construct an orthonormal basis for Col(A) of $\boldsymbol{R}^4$.

Solution

Solution

Let $\boldsymbol{v}_1 = \boldsymbol{x}_1$ and $W_1 = \text{span}\{\boldsymbol{v}_1\} = \text{span}\{\boldsymbol{x}_1\}$.

Let $\boldsymbol{v}_2 = \boldsymbol{x}_2 - \text{proj}_{W_1}\boldsymbol{x}_2 = \boldsymbol{x}_2 - \underbrace{\frac{\langle \boldsymbol{x}_2, \boldsymbol{v}_1 \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle}\boldsymbol{v}_1}_{\substack{\text{projection of} \\ \boldsymbol{x}_2 \text{ onto } \boldsymbol{v}_1}} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} - \frac{2}{4}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$.

$\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ is an orthogonal basis for the subspace $W_2 = \text{span}\{\boldsymbol{x}_1, \boldsymbol{x}_2\}$.

Let $\boldsymbol{v}_3 = \boldsymbol{x}_3 - \text{proj}_{W_2}\boldsymbol{x}_3 = \boldsymbol{x}_3 - \underbrace{\frac{\langle \boldsymbol{x}_3, \boldsymbol{v}_1 \rangle}{\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle}\boldsymbol{v}_1}_{\substack{\text{projection of} \\ \boldsymbol{x}_3 \text{ onto } \boldsymbol{v}_1}} - \underbrace{\frac{\langle \boldsymbol{x}_3, \boldsymbol{v}_2 \rangle}{\langle \boldsymbol{v}_2, \boldsymbol{v}_2 \rangle}\boldsymbol{v}_2}_{\substack{\text{projection of} \\ \boldsymbol{x}_3 \text{ onto } \boldsymbol{v}_2}} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{4}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$

$\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ is an orthogonal basis for Col(A).

Normalizing $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$, $\left\{ \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \right\}$ is an orthonormal basis for Col(A).

Theorem 5.6.3 (QR factorization)

Use the same notation as in Theorem 5.6.1 where $V = \boldsymbol{R}^m$. If $A = \begin{pmatrix} \boldsymbol{x}_1 \mid \boldsymbol{x}_2 \mid \cdots \mid \boldsymbol{x}_n \end{pmatrix}$ is an $m \times n$ matrix of rank n, then A can be factored into a product QR, where

$$Q=\left(\frac{\boldsymbol{v}_1}{\|\boldsymbol{v}_1\|}\;\middle|\;\frac{\boldsymbol{v}_2}{\|\boldsymbol{v}_2\|}\;\middle|\;\cdots\;\middle|\;\frac{\boldsymbol{v}_n}{\|\boldsymbol{v}_n\|}\right)\quad\text{and}\quad R=\begin{pmatrix}\frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1\cdot\boldsymbol{x}_1 & \frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1\cdot\boldsymbol{x}_2 & \cdots & \frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1\cdot\boldsymbol{x}_n\\ 0 & \frac{1}{\|\boldsymbol{v}_2\|}\boldsymbol{v}_2\cdot\boldsymbol{x}_2 & \cdots & \frac{1}{\|\boldsymbol{v}_2\|}\boldsymbol{v}_2\cdot\boldsymbol{x}_n\\ \vdots & \ddots & \ddots & \vdots\\ 0 & \cdots & 0 & \frac{1}{\|\boldsymbol{v}_n\|}\boldsymbol{v}_n\cdot\boldsymbol{x}_n\end{pmatrix}$$

Proof

Since $A = QR$, we have

$$R=Q^TQR=Q^TA=\begin{pmatrix}\frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1^T\\ \frac{1}{\|\boldsymbol{v}_2\|}\boldsymbol{v}_2^T\\ \vdots\\ \frac{1}{\|\boldsymbol{v}_n\|}\boldsymbol{v}_n^T\end{pmatrix}\begin{pmatrix}\boldsymbol{x}_1 & \boldsymbol{x}_2 & \cdots & \boldsymbol{x}_n\end{pmatrix}=\begin{pmatrix}\frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1\cdot\boldsymbol{x}_1 & \frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1\cdot\boldsymbol{x}_2 & \cdots & \frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1\cdot\boldsymbol{x}_n\\ 0 & \frac{1}{\|\boldsymbol{v}_2\|}\boldsymbol{v}_2\cdot\boldsymbol{x}_2 & \cdots & \frac{1}{\|\boldsymbol{v}_2\|}\boldsymbol{v}_2\cdot\boldsymbol{x}_n\\ \vdots & \ddots & \ddots & \vdots\\ 0 & \cdots & 0 & \frac{1}{\|\boldsymbol{v}_n\|}\boldsymbol{v}_n\cdot\boldsymbol{x}_n\end{pmatrix}.$$

R is an upper triangular matrix, that is $\frac{1}{\|\boldsymbol{v}_i\|}\boldsymbol{v}_i\cdot\boldsymbol{x}_j=0$ for $j < i$, because $\boldsymbol{v}_i\in\operatorname{span}\{\boldsymbol{v}_1,\cdots,\boldsymbol{v}_{i-1}\}^\perp$

$=\operatorname{span}\{\boldsymbol{x}_1,\cdots,\boldsymbol{x}_{i-1}\}^\perp$ for $i=1,2,\cdots,n.$

Example 5.6.4

Let $A=\begin{pmatrix}1&0&0\\1&1&0\\1&0&1\\1&1&1\end{pmatrix}$ and $Q=\begin{pmatrix}\frac12&-\frac12&-\frac12\\\frac12&\frac12&-\frac12\\\frac12&-\frac12&\frac12\\\frac12&\frac12&\frac12\end{pmatrix}$ as in Example 5.6.2. We have

$$R=Q^TA=\begin{pmatrix}\frac12&\frac12&\frac12&\frac12\\-\frac12&\frac12&-\frac12&\frac12\\-\frac12&-\frac12&\frac12&\frac12\end{pmatrix}\begin{pmatrix}1&0&0\\1&1&0\\1&0&1\\1&1&1\end{pmatrix}=\begin{pmatrix}2&1&1\\0&1&0\\0&0&1\end{pmatrix}.$$

Theorem 5.6.5

If A is an $m\times n$ matrix of rank n, then the least squares solution of $A\boldsymbol{x} = \boldsymbol{b}$ is given by $\hat{\boldsymbol{x}}=R^{-1}Q^T\boldsymbol{b}$, where Q and R are the matrices obtained from QR factorization.

Proof

Clearly, the diagonal elements of R are $\frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_i \cdot \boldsymbol{x}_i = \frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_i \cdot \left(\boldsymbol{v}_i + \text{proj}_{\text{span}\{\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_{i-1}\}}\boldsymbol{x}_n\right) = \frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_i \cdot \boldsymbol{v}_i$

$= \|\boldsymbol{v}_i\|$. R is invertible because $\det R = \|\boldsymbol{v}_1\|\|\boldsymbol{v}_2\|\cdots\|\boldsymbol{v}_n\| \neq 0$.

$\boldsymbol{u} \in \text{Col}(A) \Rightarrow \boldsymbol{u} = A\boldsymbol{v} = Q(R\boldsymbol{v})$ for some $\boldsymbol{v} \Rightarrow \boldsymbol{u} \in \text{Col}(Q)$. That means $\text{Col}(A) \subset \text{Col}(Q)$. Similarly, $\text{Col}(Q) \subset \text{Col}(A)$ because R is invertible and $Q = AR^{-1}$. We have Col(A) = Col(Q).

The equation $A\boldsymbol{x} = \boldsymbol{b}$ becomes $Q\boldsymbol{y} = \boldsymbol{b}$ where $\boldsymbol{y} = R\boldsymbol{x}$. By Theorem 5.3.2, the projection of $\boldsymbol{b}$ onto Col(A) and Col(Q) must be the same because of uniqueness of $\boldsymbol{p}$ and Col(A) = Col(Q), that is

$$\begin{aligned} \boldsymbol{p} &= A\hat{\boldsymbol{x}} = Q\hat{\boldsymbol{y}} \\ QR\hat{\boldsymbol{x}} &= Q\hat{\boldsymbol{y}} \\ Q^TQR\hat{\boldsymbol{x}} &= Q^TQ\hat{\boldsymbol{y}} \\ R\hat{\boldsymbol{x}} &= \hat{\boldsymbol{y}} \qquad [Q^TQ = I \text{ proof of Theorem 5.2.21}] \end{aligned}$$

We have $\hat{\boldsymbol{x}} = R^{-1}\hat{\boldsymbol{y}}$. Since the column vectors of Q are orthonormal, $\hat{\boldsymbol{y}} = Q^T\boldsymbol{b}$ by Theorem 5.5.21. The least square solution of $A\boldsymbol{x} = \boldsymbol{b}$ is $\hat{\boldsymbol{x}} = R^{-1}Q^T\boldsymbol{b}$.

Example 5.6.6

Using QR factorization, find the least square solution in Example 5.3.7.

Solution

Since $R\hat{\boldsymbol{x}} = \hat{\boldsymbol{y}} = Q^T\boldsymbol{b}$, we have

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{17}{2} \\ \frac{3}{2} \\ \frac{7}{2} \end{pmatrix}}_{\text{Example 5.5.22}}$$

$$x_3 = \tfrac{7}{2}, \quad x_2 = \tfrac{3}{2}, \quad 2x_1 + x_2 + x_3 = \tfrac{17}{2}, \quad x_1 = \tfrac{7}{4}$$

The orthogonal projection $A\hat{\boldsymbol{x}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} \frac{7}{4} \\ \frac{3}{2} \\ \frac{7}{2} \end{pmatrix} = \begin{pmatrix} \frac{7}{4} \\ \frac{13}{4} \\ \frac{21}{4} \\ \frac{27}{4} \end{pmatrix}$ which is the same as the answer in Example 5.5.22.

Section 6.1 Eigenvectors and Eigenvalues

Definition 6.1.1

Let A be an $n \times n$ matrix. A scalar λ is said to be an **eigenvalue** of A if there exists a *nonzero* vector $\boldsymbol{v} \in \boldsymbol{R}^n$ such that $A\boldsymbol{v} = \lambda\boldsymbol{v}$. The vector $\boldsymbol{v}$ is said to be an **eigenvector** of A corresponding to λ. The pair λ, $\boldsymbol{v}$ is called an eigenpair for A.

Theorem 6.1.2

Let A be an $n \times n$ matrix and λ be a scalar. The following statements are equivalent:

(a) λ is an eigenvalue of A.

(b) the **eigenspace** of λ, $N(A - \lambda I_n) \neq \boldsymbol{0}$.

(c) $\det(A - \lambda I_n) = 0$.

In particular, 0 is an eigenvalue of A if and only if $\det(A) = 0$.

Definition 6.1.3

Let A be an $n \times n$ matrix. The polynomial $p_A(\lambda) = \det(A - \lambda I_n)$ is called the **characteristic polynomial** of A. The equation $\det(A - \lambda I_n) = 0$ is called the **characteristic equation** of A.

Remark 6.1.4

λ is an eigenvalue of A if and only if λ is a root of characteristic polynomial of A.

$\boldsymbol{v}$ is an eigenvector of A corresponding to eigenvalue λ if and only if $\boldsymbol{v} \in N(A - \lambda I_n) \setminus \{\boldsymbol{0}\}$.

If λ, $\boldsymbol{v}$ is an eigenpair for A, then λ, $c\boldsymbol{v}$ is also an eigenpair for A for any nonzero constant c. Furthermore, if A is invertible, then λ^{-1}, $\boldsymbol{v}$ is an eigenpair for A^{-1}.

Definition 6.1.5

Let $f(x) = a_k x^k + \cdots + a_1 x + a_0$ be a polynomial and A be an $n \times n$ matrix. We define

$$f(A) = a_k A^k + \cdots + a_1 A + a_0 I_n.$$

Theorem 6.1.6

Let $f(x) = a_k x^k + \cdots + a_1 x + a_0$ be a polynomial. If λ, $\boldsymbol{v}$ is an eigenpair for an $n \times n$ matrix A, then $f(\lambda)$, $\boldsymbol{v}$ is an eigenvector pair of $f(A)$.

Proof

Since $A\boldsymbol{v} = \lambda \boldsymbol{v}$, we have $A^i \boldsymbol{v} = A^{i-1} A\boldsymbol{v} = A^{i-1}(\lambda \boldsymbol{v}) = \lambda A^{i-1}\boldsymbol{v} = \cdots = \lambda^i \boldsymbol{v}$. Then

$$\begin{aligned} f(A)\boldsymbol{v} &= (a_k A^k + \cdots + a_1 A + a_0 I_n)\boldsymbol{v} \\ &= a_k A^k \boldsymbol{v} + \cdots + a_1 A\boldsymbol{v} + a_0 \boldsymbol{v} \\ &= a_k \lambda^k \boldsymbol{v} + \cdots + a_1 \lambda \boldsymbol{v} + a_0 \boldsymbol{v} \\ &= (a_k \lambda^k + \cdots + a_1 \lambda + a_0)\boldsymbol{v} \\ &= f(\lambda)\boldsymbol{v} \end{aligned}$$

Theorem 6.1.7

The eigenvalues of an upper or lower triangular matrix are the entries on its main diagonal.

Proof

We only consider an upper triangular matrix. The lower triangular case can be done similarly.

Let $A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1,n} \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix}$. The characteristic polynomial of A is

$$p_A(\lambda) = \det(A - \lambda I_n) = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} - \lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1,n} \\ 0 & \cdots & 0 & a_{nn} - \lambda \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda)\cdots(a_{nn} - \lambda)$$

Eigenvalues of A are roots of $p_A(\lambda) = 0$, which are $a_{11}, a_{22}, \cdots, a_{nn}$.

Let a_{ij} be the *i-j* entry of an $n \times n$ matrix A. Then

$$p_A(\lambda) = \det(A - \lambda I_n) = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1,n} \\ a_{n1} & \cdots & a_{n,n-1} & a_{nn} - \lambda \end{vmatrix}$$

Choosing diagonal element in every determinant expansion, we have $(a_{11} - \lambda)(a_{22} - \lambda)\cdots(a_{nn} - \lambda)$. Choosing non-diagonal element in determinant expansion, the degree of polynomial is decreased at least by 2.

$$\begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1,n} \\ a_{n1} & \cdots & a_{n,n-1} & a_{nn} - \lambda \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda)\cdots(a_{nn} - \lambda) + \text{polynomial with degree at most } n-2$$

$$= (-1)^n \lambda^n + (-1)^{n-1}(a_{11} + \cdots + a_{nn})\lambda^{n-1} + \cdots + p_A(0)$$

$$= (-1)^n \lambda^n + (-1)^{n-1}\operatorname{tr}(A)\lambda^{n-1} + \cdots + \det(A)$$

Keep in mind that the constant term of $p_A(\lambda) = \det(A - \lambda I_n)$ is $p_A(0) = \det(A)$. The sum of the diagonal elements of A is called the **trace** of A and is denoted by $\operatorname{tr}(A)$.

If $\lambda_1, \lambda_2, \cdots, \lambda_n$ are all eigenvalues of A, then $p_A(\lambda) = (-1)^n(\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n)$

$= (-1)^n \lambda^n + (-1)^{n-1}(\lambda_1 + \cdots + \lambda_n)\lambda^{n-1} + \cdots + \lambda_1\lambda_2\cdots\lambda_n$. By comparing coefficient of λ^{n-1} and constant term, we have the following Theorem.

Theorem 6.1.8

If $\lambda_1, \lambda_2, \cdots, \lambda_n$ are all eigenvalues of A, then

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = \operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn} \quad \text{and} \quad \lambda_1\lambda_2\cdots\lambda_n = \det(A).$$

Example 6.1.9

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Its characteristic polynomial is

$$\det(A-\lambda I_2)=\det\begin{pmatrix}a-\lambda & b\\ c & d-\lambda\end{pmatrix}=(a-\lambda)(d-\lambda)-bc=\lambda^2-(\underbrace{a+d}_{\mathrm{tr}(A)})\lambda+\underbrace{ad-bc}_{\det(A)}.$$

Suppose $A=\begin{pmatrix}1 & 3\\ 4 & 2\end{pmatrix}$. Its characteristic polynomial is

$$\det(\lambda I_2-A)=\det\begin{pmatrix}1-\lambda & -3\\ -4 & 2-\lambda\end{pmatrix}=(1-\lambda)(2-\lambda)-12=\lambda^2-3\lambda-10=(\lambda-5)(\lambda+2)$$

The eigenvalues of A are $\lambda_1=5$ and $\lambda_2=-2$.

Consider $\lambda_1=5$.

$$N(A-5I_2)=N\begin{pmatrix}1-5 & 3\\ 4 & 2-5\end{pmatrix}=N\begin{pmatrix}-4 & 3\\ 4 & -3\end{pmatrix}=span\left\{\begin{pmatrix}3\\ 4\end{pmatrix}\right\}.$$

Eigenvector $\boldsymbol{v}_1=\begin{pmatrix}3\\ 4\end{pmatrix}$.

Consider $\lambda_2=-2$.

$$N(A+2I_2)=N\begin{pmatrix}1+2 & 3\\ 4 & 2+2\end{pmatrix}=N\begin{pmatrix}3 & 3\\ 4 & 4\end{pmatrix}=span\left\{\begin{pmatrix}1\\ -1\end{pmatrix}\right\}.$$

Eigenvector $\boldsymbol{v}_2=\begin{pmatrix}1\\ -1\end{pmatrix}$.

Definition 6.1.10

Let A be an $n\times n$ matrix and let λ be an eigenvalue of A. The **algebraic multiplicity** of λ is the largest positive integer k for which $(x-\lambda)^k$ is a factor of $p_A(\lambda)$, i.e., there are exactly k copies of eigenvalue λ. $\dim N(A-\lambda I_n)$ is called **geometric multiplicity** of λ.

Example 6.1.11

Find all eigenvectors and eigenvalues of $A=\begin{pmatrix}1 & \cdots & 1\\ \vdots & \ddots & \vdots\\ 1 & \cdots & 1\end{pmatrix}_{n\times n}$.

Solution

Using characteristic equation to find eigenvalue is not practical if the size of matrix is large. Since rank(A) = 1, we have $\dim N(A) = n-1$ by rank-nullity Theorem. The geometric multiplicity of 0 is $n-1$. Clearly, $\begin{pmatrix}1\\-1\\0\\\vdots\\0\end{pmatrix}, \begin{pmatrix}1\\0\\-1\\0\\\vdots\\0\end{pmatrix}, \cdots, \begin{pmatrix}1\\0\\\vdots\\0\\-1\end{pmatrix} \in N(A)$. $\begin{pmatrix}1\\-1\\0\\\vdots\\0\end{pmatrix}, \begin{pmatrix}1\\0\\-1\\0\\\vdots\\0\end{pmatrix}, \cdots, \begin{pmatrix}1\\0\\\vdots\\0\\-1\end{pmatrix}$ are $n-1$ eigenvectors of A corresponding to 0.

Let λ be the last eigenvalue of A. By Theorem 6.1.8, we have $0+\cdots+0+\lambda = \text{tr}(A) = n$.

Clearly, $\begin{pmatrix}1\\1\\\vdots\\1\end{pmatrix} \in N(A-nI_n) = N\begin{pmatrix}1-n & 1 & \cdots & 1\\ 1 & 1-n & \ddots & \vdots\\ \vdots & \ddots & \ddots & 1\\ 1 & \cdots & 1 & 1-n\end{pmatrix}_{n\times n}$. The last eigenvector is $\begin{pmatrix}1\\1\\\vdots\\1\end{pmatrix}$.

$$\text{eigenvectors}: \begin{pmatrix}1\\-1\\0\\\vdots\\0\end{pmatrix}, \begin{pmatrix}1\\0\\-1\\0\\\vdots\\0\end{pmatrix}, \cdots, \begin{pmatrix}1\\0\\\vdots\\0\\-1\end{pmatrix}, \begin{pmatrix}1\\1\\\vdots\\1\\1\end{pmatrix}$$

$$\text{eigenvalues}: \quad 0, \quad 0, \quad \cdots \quad 0, \quad 1$$

Theorem 6.1.12

If two $n\times n$ matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same algebraic multiplicities). Suppose there exists a nonsingular matrix P such that $A = PBP^{-1}$. λ, $\boldsymbol{v}$ is an eigenpair for A if and only if $\lambda, P^{-1}\boldsymbol{v}$ is an eigenpair for B (with the same geometric multiplicities).

Proof

Suppose A and B are similar. There exists a nonsingular matrix P such that $A = PBP^{-1}$.

$$p_A(\lambda) = \det(A - \lambda I_n)$$

$$\begin{aligned}
&= \det(PBP^{-1} - P(\lambda I_n)P^{-1}) \\
&= \det(P(B - \lambda I_n)P^{-1}) \\
&= \det P \det(B - \lambda I_n) \det(P^{-1}) \\
&= \det P (\det P)^{-1} \det(B - \lambda I_n) \\
&= \det(B - \lambda I_n) \\
&= p_B(\lambda).
\end{aligned}$$

Clearly, $A\boldsymbol{v} = \lambda\boldsymbol{v} \quad \Leftrightarrow \quad PBP^{-1}\boldsymbol{v} = \lambda\boldsymbol{v} \quad \Leftrightarrow \quad B(P^{-1}\boldsymbol{v}) = \lambda(P^{-1}\boldsymbol{v})$.

Theorem 6.1.13

If $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \lambda_2, \cdots, \lambda_r$ of an $n \times n$ matrix A, then the set $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_r\}$ is linearly independent.

Proof

Suppose $a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_r\boldsymbol{v}_r = \boldsymbol{0}$ ------(1). Then

$$a_1\lambda_1\boldsymbol{v}_1 + a_2\lambda_2\boldsymbol{v}_2 + \cdots + a_r\lambda_r\boldsymbol{v}_r = A(a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_r\boldsymbol{v}_r) = \boldsymbol{0} \text{ ----------(2)}$$

Consider $(2) - \lambda_r(1)$, we have

$$a_1(\lambda_1 - \lambda_r)\boldsymbol{v}_1 + a_2(\lambda_2 - \lambda_r)\boldsymbol{v}_2 + \cdots + a_{r-1}(\lambda_{r-1} - \lambda_r)\boldsymbol{v}_{r-1} = \boldsymbol{0} \text{ ----------(3)}$$

That means we can reduce the number of eigenvectors of equation (1) to the number of eigenvectors of equation (3) by 1. By repeating this process, we have

$$a_1(\lambda_1 - \lambda_r)(\lambda_1 - \lambda_2)\cdots(\lambda_1 - \lambda_{r-1})\boldsymbol{v}_1 = \boldsymbol{0}$$

Since $\lambda_1, \lambda_2, \cdots, \lambda_r$ are distinct eigenvalues, we have $a_1 = 0$. Similarly, $a_1 = a_2 = \cdots = a_r = 0$.

Theorem 6.1.14 (Companion Matrix)

Let $A = \begin{pmatrix} -a_{n-1} & \cdots & -a_1 & -a_0 \\ 1 & 0 & \cdots & 0 \\ & \ddots & \ddots & \vdots \\ \mathbf{0} & & 1 & 0 \end{pmatrix}_{n\times n}$.

(a) The characteristic polynomial of A is $p_A(\lambda) = (-1)^n(\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0)$.

(b) If λ_i is an eigenvalue of A, then its corresponding eigenvector is $(\lambda_i^{n-1}, \lambda_i^{n-2}, \cdots, \lambda_i, 1)^T$.

Solution

(a) We will prove it by induction. When $n = 1$, we have

$$p_A(\lambda) = \det(A - \lambda I_n) = \det(-a_0 - \lambda) = (-1)(\lambda + a_0).$$

Suppose (a) is true for $n = k$, that means

$$\det \begin{pmatrix} -c_{k-1} - \lambda & \cdots & -c_1 & -c_0 \\ 1 & -\lambda & & 0 \\ & \ddots & \ddots & \\ 0 & & 1 & -\lambda \end{pmatrix}_{k \times k} = (-1)^k (\lambda^k + c_{k-1}\lambda^{k-1} + \cdots + c_1\lambda + c_0).$$

$$\det \begin{pmatrix} -a_k - \lambda & -a_{k-1} & \cdots & -a_1 & -a_0 \\ 1 & -\lambda & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & 1 & -\lambda & 0 \\ 0 & \cdots & 0 & 1 & -\lambda \end{pmatrix}_{(k+1)\times(k+1)}$$ [expand along the last column]

$$= (-1)^{k+1+1}(-a_0) \det \begin{pmatrix} 1 & -\lambda & & 0 \\ 0 & 1 & \ddots & \\ \vdots & \ddots & \ddots & -\lambda \\ 0 & \cdots & 0 & 1 \end{pmatrix}_{k \times k} + (-1)^{k+1+k+1}(-\lambda) \begin{pmatrix} -a_k - \lambda & -a_{k-1} & \cdots & -a_1 \\ 1 & -\lambda & & 0 \\ & \ddots & \ddots & \\ 0 & & 1 & -\lambda \end{pmatrix}_{k \times k}$$

$$= (-1)^{k+1} a_0 + (-\lambda)(-1)^k (\lambda^k + a_k \lambda^{k-1} + \cdots + a_1)$$

$$= (-1)^{k+1} (\lambda^{k+1} + a_k \lambda^k + \cdots + a_1 \lambda + a_0)$$

By induction, (a) is true for all positive integer n.

(b) If λ_i is an eigenvalue of A, then $\lambda_i^n + a_{n-1}\lambda_i^{n-1} + \cdots + a_1\lambda_i + a_0 = 0$.

$$\begin{pmatrix} -a_{n-1} & \cdots & -a_1 & -a_0 \\ 1 & 0 & \cdots & 0 \\ & \ddots & \ddots & \vdots \\ 0 & & 1 & 0 \end{pmatrix} \begin{pmatrix} \lambda_i^{n-1} \\ \lambda_i^{n-2} \\ \vdots \\ \lambda_i \\ 1 \end{pmatrix} = \begin{pmatrix} -(a_{n-1}\lambda_i^{n-1} + \cdots + a_1\lambda_i + a_0) \\ \lambda_i^{n-1} \\ \vdots \\ \lambda_i^2 \\ \lambda_i \end{pmatrix} = \begin{pmatrix} \lambda_i^n \\ \lambda_i^{n-1} \\ \vdots \\ \lambda_i^2 \\ \lambda_i \end{pmatrix} = \lambda_i \begin{pmatrix} \lambda_i^{n-1} \\ \lambda_i^{n-2} \\ \vdots \\ \lambda_i \\ 1 \end{pmatrix}$$

$(\lambda_i^{n-1}, \lambda_i^{n-2}, \cdots, \lambda_i, 1)^T$ is an eigenvector of A corresponding to eigenvalue λ_i.

Section 6.2 Diagonalization

Definition 6.2.1

An $n \times n$ matrix A is said to be **diagonalizable** if there exists a nonsingular matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

That means, A is similar to a diagonal matrix D. The k-th diagonal entry of D is an eigenvalue of A corresponding to the eigenvector, the k-th column vector of P.

Theorem 6.2.2

Let A be an $n \times n$ matrix and $\lambda_1, \lambda_2, \cdots, \lambda_k$ the distinct eigenvalues of A. Then

$$\text{geometric multiplicity of } \lambda_i \leq \text{ algebraic multiplicity of } \lambda_i \text{ for all } i.$$

Proof

Let m be geometric multiplicity of λ_1 and $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_m$ the eigenvectors corresponding to λ_1.

Extend $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_m$ to $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_m, \underbrace{\boldsymbol{v}_{m+1}, \cdots, \boldsymbol{v}_n}_{\substack{\text{not necessary} \\ \text{eigenvector of } A}}$ and form a basis of $\boldsymbol{R}^n$.

$$P^{-1}AP = \begin{pmatrix} | & & | & & | \\ \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_m & \cdots & \boldsymbol{v}_n \\ | & & | & & | \end{pmatrix}^{-1} \begin{pmatrix} | & & | & & | \\ A\boldsymbol{v}_1 & \cdots & A\boldsymbol{v}_m & \cdots & A\boldsymbol{v}_n \\ | & & | & & | \end{pmatrix}$$

$$= \begin{pmatrix} | & & | & & | \\ \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_m & \cdots & \boldsymbol{v}_n \\ | & & | & & | \end{pmatrix}^{-1} \left(\begin{array}{ccc|ccc} | & & | & | & & | \\ \lambda_1\boldsymbol{v}_1 & \cdots & \lambda_1\boldsymbol{v}_m & A\boldsymbol{v}_{m+1} & \cdots & A\boldsymbol{v}_n \\ | & & | & | & & | \end{array}\right) = \left(\begin{array}{c|c} \lambda_1 I_m & C_{m\times(n-m)} \\ \hline \boldsymbol{0} & D_{(n-m)\times(n-m)} \end{array}\right)$$

That means A is similar to $\left(\begin{array}{c|c} \lambda_1 I_m & C_{m\times(n-m)} \\ \hline \boldsymbol{0} & D_{(n-m)\times(n-m)} \end{array}\right)$. By Theorem 6.1.12, we have

$$p_A(\lambda) = \det\left(\begin{array}{c|c} (\lambda_1 - \lambda)I_m & C_{m\times(n-m)} \\ \hline \boldsymbol{0} & D_{(n-m)\times(n-m)} - \lambda I_m \end{array}\right) = (\lambda_1 - \lambda)^m \det(D - \lambda I_m) = (\lambda_1 - \lambda)^m p_D(\lambda).$$

Geometric multiplicity of $\lambda_1 \leq$ algebraic multiplicity of λ_1, hence for all eigenvalues.

Theorem 6.2.3

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. In other words, A is diagonalizable if and only if geometric multiplicity is equal to algebraic multiplicity for all eigenvalues.

Proof

Suppose that the matrix A has n linearly independent eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$. Let λ_i be the eigenvalue of A corresponding to $\boldsymbol{v}_i$ for each i. (Some of the λ_i's may be equal.) Let P be the matrix whose i-th column vector is $\boldsymbol{v}_i$ for $i = 1, 2, \cdots, n$. It follows that $A\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$ is the i-th column vector of AP. Thus,

$$\begin{aligned} AP &= \begin{pmatrix} | & | & & | \\ A\boldsymbol{v}_1 & A\boldsymbol{v}_2 & \cdots & A\boldsymbol{v}_n \\ | & | & & | \end{pmatrix} \\ &= \begin{pmatrix} | & | & & | \\ \lambda_1\boldsymbol{v}_1 & \lambda_2\boldsymbol{v}_2 & \cdots & \lambda_n\boldsymbol{v}_n \\ | & | & & | \end{pmatrix} \\ &= \begin{pmatrix} | & | & & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix} \\ &= PD \end{aligned}$$

Since P has n linearly independent column vectors, it follows that P is invertible and hence $A = PDP^{-1}$.

Conversely, suppose that A is diagonalizable. Then there exists a nonsingular matrix P such that $AP = PD$. If $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ are the column vectors of P, then

$$A\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$$

for each i. Thus, for each i, λ_i is an eigenvalue of A and $\boldsymbol{v}_i$ is an eigenvector belonging to λ_i. Since the column vectors of P are linearly independent, it follows that A has n linearly independent eigenvectors.

Corollary 6.2.4

If A is $n \times n$ matrix and A has n distinct eigenvalues, then A is diagonalizable.

Proof

By Theorem 6.1.13, all eigenvectors of A are linear independent. Hence A is diagonalizable by Theorem 6.2.3.

If A is diagonalizable, then A can be expressed as $A = PDP^{-1}$ where D is a diagonal matrix. Then $A^k = \underbrace{(PDP^{-1})(PDP^{-1})\cdots(PDP^{-1})}_{k \text{ times}} = PD^kP^{-1}$ where k is any positive integer.

Consider $A = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}$. In Example 6.1.9, we have

$$\begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 4 & -1 \end{pmatrix}\begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ 4 & -1 \end{pmatrix}^{-1}.$$

Hence

$$\begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}^k = \begin{pmatrix} 3 & 1 \\ 4 & -1 \end{pmatrix}\begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}^k\begin{pmatrix} 3 & 1 \\ 4 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & 1 \\ 4 & -1 \end{pmatrix}\begin{pmatrix} 5^k & 0 \\ 0 & (-2)^k \end{pmatrix}\begin{pmatrix} 3 & 1 \\ 4 & -1 \end{pmatrix}^{-1}$$

where k is any positive integer.

Are all matrices diagonalizable? No.

It is because algebraic multiplicities of an eigenvalue is larger than its geometric multiplicities.

Example 6.2.5

Consider rotation matrix $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. Geometrically, for any nonzero vector $\boldsymbol{v}$, $\boldsymbol{v}$ and $A\boldsymbol{v}$ are not collinear (if $\theta \neq n\pi$); hence $A\boldsymbol{v}$ is not a multiple of $\boldsymbol{v}$. Therefore A has no eigenvectors and, consequently, no eigenvalues.

A is not diagonalizable in $\boldsymbol{R}$ but diagonalizable in $\boldsymbol{C}$.

Characteristic polynomial of $A = \det\begin{pmatrix} \cos\theta - \lambda & -\sin\theta \\ \sin\theta & \cos\theta - \lambda \end{pmatrix} = (\lambda - \cos\theta)^2 + \sin^2\theta.$

There are 2 eigenvalues $\cos\theta + i\sin\theta$ and $\cos\theta - i\sin\theta$ corresponding to eigenvectors $\begin{pmatrix} 1 \\ -i \end{pmatrix}$ and $\begin{pmatrix} 1 \\ i \end{pmatrix}$ respectively. Then

Consider $\lambda_1 = \cos\theta + i\sin\theta.$

$$N\left(A - (\cos\theta + i\sin\theta)I_2\right) = N\begin{pmatrix} -i\sin\theta & -\sin\theta \\ \sin\theta & -i\sin\theta \end{pmatrix} = N\begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix} = span\left\{\begin{pmatrix} 1 \\ -i \end{pmatrix}\right\}.$$

Eigenvector $\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix}.$

Consider $\lambda_2 = \cos\theta - i\sin\theta.$

$$N\left(A - (\cos\theta - i\sin\theta)I_2\right) = N\begin{pmatrix} i\sin\theta & -\sin\theta \\ \sin\theta & i\sin\theta \end{pmatrix} = N\begin{pmatrix} i & -1 \\ 1 & i \end{pmatrix} = span\left\{\begin{pmatrix} 1 \\ i \end{pmatrix}\right\}.$$

Eigenvector $\boldsymbol{v}_2 = \begin{pmatrix} 1 \\ i \end{pmatrix}.$ We have

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}\begin{pmatrix} \cos\theta + i\sin\theta & 0 \\ 0 & \cos\theta - i\sin\theta \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}^{-1}.$$

Example 6.2.6

Consider $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$ Its characteristic polynomial is $p_A(\lambda) = \lambda^2.$ If A is diagonalizable, then there exists an invertible matrix P such that

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = A = P\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}P^{-1} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Contradiction.

Indeed, 0 is the only eigenvalue with algebraic multiplicity 2. However, dim $N(A)$ = 1 and $(0,1)^T$ is the only independent eigenvector. Number of independent eigenvector corresponding to eigenvalue 0 is not enough. In other words, the geometric multiplicities of 0 is less than its algebraic multiplicities.

Section 6.3 Hermitian Matrices

In Example 6.2.5, the eigenpair of a matrix with real entries can be in complex number. In the rest of this course, we consider vector spaces and matrices over $\boldsymbol{C}$.

Definition 6.3.1

The conjugate transpose of a matrix A with complex entries is denoted by A^H and defined by $A^H = \overline{A}^T$, which $\overline{A}$ is the entrywise conjugate. A matrix A is said to be **Hermitian** if $A^H = A$.

Example 6.3.2

$A = \begin{pmatrix} 3 & 2-i \\ 2+i & 4 \end{pmatrix}$ is Hermitian since $A^H = \begin{pmatrix} \overline{3} & \overline{2-i} \\ \overline{2+i} & \overline{4} \end{pmatrix}^T = \begin{pmatrix} \overline{3} & \overline{2-i} \\ \overline{2+i} & \overline{4} \end{pmatrix}^T = \begin{pmatrix} 3 & \overline{2+i} \\ \overline{2-i} & 4 \end{pmatrix} = A.$

However, $A = \begin{pmatrix} 3 & 2+i \\ 2+i & 4 \end{pmatrix}$ is symmetric but not Hermitian.

If A is a matrix with real entries, then $A^H = A^T$. In particular, if A is a real symmetric matrix, then A is Hermitian. Thus we may view Hermitian matrices as the complex analogue of real symmetric matrices.

Theorem 6.3.3

For any matrices A, B with complex entries and any complex scalars α and β, we have

(i) $(A^H)^H = A$,

(ii) $(\alpha A + \beta B)^H = \overline{\alpha} A^H + \overline{\beta} B^H$,

(iii) $(AB)^H = B^H A^H$.

Definition 6.3.4 (complex dot product)

We can define a dot product on $\boldsymbol{C}^n$ by $\boldsymbol{u}\cdot\boldsymbol{v}=\boldsymbol{v}^H\boldsymbol{u}$ for all $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\boldsymbol{C}^n$. $\boldsymbol{u}$ and $\boldsymbol{v}$ are said to be **orthogonal** (to each other) if $\boldsymbol{u}\cdot\boldsymbol{v}=0$.

The length of a vector $\boldsymbol{z}=(z_1,z_2,\cdots,z_n)^T$ in $\boldsymbol{C}^n$ is given by

$$||\boldsymbol{z}||=\sqrt{|z_1|^2+|z_2|^2+\cdots+|z_n|^2}.$$

Theorem 6.3.5

Suppose $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{w}$ are vectors in $\boldsymbol{C}^n$ and let α and β be scalars in $\boldsymbol{C}$. Then

(i) $\boldsymbol{u}\cdot\boldsymbol{u}\geq 0$, with equality if and only if $\boldsymbol{u}=\boldsymbol{0}$.

(ii) $\boldsymbol{u}\cdot\boldsymbol{v}=\overline{\boldsymbol{v}\cdot\boldsymbol{u}}$ for all $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\boldsymbol{C}^n$.

(iii) $(\alpha\boldsymbol{u}+\beta\boldsymbol{v})\cdot\boldsymbol{w}=\alpha(\boldsymbol{u}\cdot\boldsymbol{w})+(\beta\boldsymbol{v}\cdot\boldsymbol{w})$.

The proofs are easy.

If A is an $n\times n$ Hermitian matrix, then $A\boldsymbol{u}\cdot\boldsymbol{v}=\boldsymbol{v}^HA\boldsymbol{u}=\boldsymbol{v}^HA^H\boldsymbol{u}=(A\boldsymbol{v})^H\boldsymbol{u}=\boldsymbol{u}\cdot A\boldsymbol{v}$

Theorem 6.3.6

All eigenvalues of a Hermitian matrix are real. Furthermore, eigenvectors of a Hermitian matrix belonging to distinct eigenvalues are orthogonal.

Proof

Let A be a Hermitian matrix. Let λ, $\boldsymbol{v}$ be an eigenpair for A.

$$\begin{aligned}&\boldsymbol{v}^HA\boldsymbol{v}=\lambda\boldsymbol{v}^H\boldsymbol{v}=\lambda||\boldsymbol{v}||^2\\&\overline{\lambda}\,\|\boldsymbol{v}\|^2=\left(\lambda||\boldsymbol{v}||^2\right)^H=\left(\boldsymbol{v}^HA\boldsymbol{v}\right)^H=\boldsymbol{v}^HA^H\boldsymbol{v}=\boldsymbol{v}^HA\boldsymbol{v}=\lambda||\boldsymbol{v}||^2\\&\overline{\lambda}=\lambda\quad\because\boldsymbol{v}\neq\boldsymbol{0},\therefore||\boldsymbol{v}||\neq 0\end{aligned}$$

Hence λ is real. If $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are eigenvectors belonging to distinct eigenvalues λ_1 and

λ_2, respectively, then $\lambda_1(\boldsymbol{v}_1 \cdot \boldsymbol{v}_2) = A\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = \boldsymbol{v}_1 \cdot A\boldsymbol{v}_2 = \boldsymbol{v}_1 \cdot \lambda_2 \boldsymbol{v}_2 = \overline{(\lambda_2 \boldsymbol{v}_2) \cdot \boldsymbol{v}_1} = \overline{\lambda_2}(\overline{\boldsymbol{v}_2 \cdot \boldsymbol{v}_1}) = \lambda_2 \boldsymbol{v}_1 \cdot \boldsymbol{v}_2$.

Since $\lambda_1 \neq \lambda_2$, it follows that $\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = 0$.

Definition 6.3.7

An $n \times n$ matrix U is said to be **unitary** if its column vectors form an orthonormal set in $\boldsymbol{C}^n$.

That means $U^H U = I_n$ or $U^{-1} = U^H$.

A real unitary matrix is an orthogonal matrix.

Theorem 6.3.8 (Schur's Theorem)

For each $n \times n$ matrix A, there exists a unitary matrix U such that $U^H AU$ is upper triangular.

Proof

The proof is by induction on n. The result is obvious if $n = 1$. Assume that the hypothesis holds for $k \times k$ matrices, and let A be a $(k+1) \times (k+1)$ matrix. Let $\boldsymbol{w}_1$ be a unit eigenvector of A corresponding to eigenvalue λ_1, that is, $\boldsymbol{w}_1^H \boldsymbol{w}_1 = 1$ and $A\boldsymbol{w}_1 = \lambda_1 \boldsymbol{w}_1$. Using the Gram–Schmidt process, construct $\boldsymbol{w}_2, \cdots, \boldsymbol{w}_{k+1}$ such that $\{\boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_{k+1}\}$ is an orthonormal basis for $\boldsymbol{C}^{k+1}$. Let $U_1 = \left(\begin{array}{c|c|c|c|c} | & | & | & & | \\ \boldsymbol{w}_1 & \boldsymbol{w}_2 & \cdots & & \boldsymbol{w}_{k+1} \\ | & | & | & & | \end{array}\right)$. Then, by construction, U_1 is unitary.

$$U_1^H A U_1 = U_1^H \left(\begin{array}{c|c|c|c|c} | & | & | & & | \\ A\boldsymbol{w}_1 & A\boldsymbol{w}_2 & \cdots & & A\boldsymbol{w}_{k+1} \\ | & | & | & & | \end{array}\right)$$

$$= \left(\begin{array}{c} - \ \boldsymbol{w}_1^H \ - \\ \hline - \ \boldsymbol{w}_2^H \ - \\ \hline \vdots \\ \hline - \ \boldsymbol{w}_{k+1}^H \ - \end{array}\right) \left(\begin{array}{c|c|c|c|c} | & | & | & & | \\ \lambda_1 \boldsymbol{w}_1 & A\boldsymbol{w}_2 & \cdots & & A\boldsymbol{w}_{k+1} \\ | & | & | & & | \end{array}\right)$$

$$= \left(\begin{array}{c|c} \lambda_1 \boldsymbol{w}_1^H \boldsymbol{w}_1 & \boldsymbol{w}_1^H A\boldsymbol{w}_2 \ \cdots \ \boldsymbol{w}_1^H A\boldsymbol{w}_{k+1} \\ \hline \lambda_1 \boldsymbol{w}_2^H \boldsymbol{w}_1 & \\ \vdots & \quad B_{k \times k} \\ \lambda_1 \boldsymbol{w}_{k+1}^H \boldsymbol{w}_1 & \end{array}\right) = \left(\begin{array}{c|c} \lambda_1 & \boldsymbol{w}_1^H A\boldsymbol{w}_2 \ \cdots \ \boldsymbol{w}_1^H A\boldsymbol{w}_{k+1} \\ \hline 0 & \\ \vdots & \quad B_{k \times k} \\ 0 & \end{array}\right)$$

where B is a $k \times k$ matrix. By the induction hypothesis, there exists a $k \times k$ unitary matrix V such that $V^H BV = T_1$, where T_1 is upper triangular. Let

$$V_1 = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & V_{k\times k} & \\ 0 & & & \end{array}\right)$$

Here V_1 is unitary and

$$V_1^H U_1^H A U_1 V_1 = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & V^H_{k\times k} & \\ 0 & & & \end{array}\right)\left(\begin{array}{c|ccc} \lambda_1 & \boldsymbol{w}_1^H A\boldsymbol{w}_2 & \cdots & \boldsymbol{w}_1^H A\boldsymbol{w}_{k+1} \\ \hline 0 & & & \\ \vdots & & B_{k\times k} & \\ 0 & & & \end{array}\right)\left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & V_{k\times k} & \\ 0 & & & \end{array}\right)$$

$$= \left(\begin{array}{c|ccc} \lambda_1 & *_{1,2} & \cdots & *_{1,k+1} \\ \hline 0 & & & \\ \vdots & & V^H_{k\times k} B_{k\times k} V_{k\times k} & \\ 0 & & & \end{array}\right)$$

$$= \left(\begin{array}{c|ccc} \lambda_1 & *_{1,2} & \cdots & *_{1,k+1} \\ \hline 0 & & & \\ \vdots & & T_1 & \\ 0 & & & \end{array}\right) = T, \text{ an upper triangular matrix.}$$

Let $U = U_1 V_1$. Since $U^H U = (U_1 V_1)^H U_1 V_1 = V_1^H U_1^H U_1 V_1 = I_{k+1}$, U is unitary and $U^H AU$ $= (U_1 V_1)^H AU_1 V_1 = V_1^H U_1^H AU_1 V_1 = T -$ an upper triangular matrix.

Theorem 6.3.9

Let A be an $n \times n$ matrix with real entries. Suppose that λ is a real eigenvalue of A corresponding to complex eigenvector $\boldsymbol{x} = \boldsymbol{u} + i\boldsymbol{v} \in \boldsymbol{C}^n$, where $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\boldsymbol{R}^n$ are real and imaginary parts of $\boldsymbol{x}$ respectively. Then either $\boldsymbol{u}$ or $\boldsymbol{v}$ (can be both) is an eigenvector of A corresponding to eigenvalue λ.

Proof

Since A is a real matrix and λ is a real eigenvalue, we have

$$A\boldsymbol{x}=\lambda\boldsymbol{x} \quad\Rightarrow\quad A(\boldsymbol{u}+i\boldsymbol{v})=\lambda(\boldsymbol{u}+i\boldsymbol{v}) \quad\Rightarrow\quad A\boldsymbol{u}+iA\boldsymbol{v}=\lambda\boldsymbol{u}+i\lambda\boldsymbol{v} \quad\Rightarrow\quad A\boldsymbol{u}=\lambda\boldsymbol{u},\quad A\boldsymbol{v}=\lambda\boldsymbol{v}$$

Either $\boldsymbol{u}$ or $\boldsymbol{v}$ is not $\boldsymbol{0}$. Otherwise $\boldsymbol{x}=\boldsymbol{0}$, which contradicts to the definition of eigenvector. That means, either $\boldsymbol{u}$ or $\boldsymbol{v}$ (can be both) is an eigenvector of A corresponding to eigenvalue λ.

Theorem 6.3.10

If A is an $n\times n$ matrix with real entries and all eigenvalues are real, then A can be factored into a product QTQ^T, where Q is orthogonal and T is upper triangular.

Proof

If all the eigenvalues of real matrix A are real, then all of the eigenvectors and unitary matrices in the algorithm of Theorem 6.3.8 can be chosen to be real by Theorem 6.3.9.

Example 6.3.11

Consider $A=\begin{pmatrix}0.6 & 0 & -0.75\\ 0.56 & -0.2 & -0.55\\ 0.08 & 0.4 & -0.4\end{pmatrix}$. Find an orthogonal matrix Q and an upper triangular T such that $A=QTQ^T$.

Solution

Its characteristic polynomial is

$$\begin{aligned}\det(A-\lambda I_3)&=\det\begin{pmatrix}0.6-\lambda & 0 & -0.75\\ 0.56 & -0.2-\lambda & -0.55\\ 0.08 & 0.4 & -0.4-\lambda\end{pmatrix}\\ &=(0.6-\lambda)\det\begin{pmatrix}-0.2-\lambda & -0.55\\ 0.4 & -0.4-\lambda\end{pmatrix}-0.75\det\begin{pmatrix}0.56 & -0.2-\lambda\\ 0.08 & 0.4\end{pmatrix}\\ &=(0.6-\lambda)\left(\lambda^2+0.6\lambda+0.08+0.22\right)-0.75\left(0.224+0.08(0.2+\lambda)\right)\\ &=(0.6-\lambda)\left(\lambda^2+0.6\lambda+0.3\right)-0.75\left(0.08\lambda+0.24\right)\\ &=-\lambda^3+0.36\lambda-0.3\lambda+0.18-0.06\lambda-0.18\\ &=-\lambda^3\end{aligned}$$

Clearly, rank(A) = 2 and dim $N(A)$ = 1. Geometric multiplicity of eigenvalue 0 = 1 and algebraic multiplicity of eigenvalue 0 = 3. The eigenvector of A corresponding to eigenvalue 0 is $\boldsymbol{v}_1 = \boldsymbol{x}_1 = (5,3,4)^T$. Choose any 2 more vectors $\boldsymbol{x}_2$ and $\boldsymbol{x}_3$ such that $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3$ forms a basis in $\boldsymbol{R}^3$. Let $\boldsymbol{x}_2 = (1,0,0)^T$ and $\boldsymbol{x}_3 = (0,1,0)^T$. By Gram-Schmidt Process,

$$\boldsymbol{v}_2 = \boldsymbol{x}_2 - \frac{\boldsymbol{v}_1 \cdot \boldsymbol{x}_2}{\boldsymbol{v}_1 \cdot \boldsymbol{v}_1}\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{5}{50}\begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 0.5 \\ -0.3 \\ -0.4 \end{pmatrix}$$

$$\boldsymbol{v}_3 = \boldsymbol{x}_3 - \frac{\boldsymbol{v}_1 \cdot \boldsymbol{x}_3}{\boldsymbol{v}_1 \cdot \boldsymbol{v}_1}\boldsymbol{v}_1 - \frac{\boldsymbol{v}_2 \cdot \boldsymbol{x}_3}{\boldsymbol{v}_2 \cdot \boldsymbol{v}_2}\boldsymbol{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \frac{3}{50}\begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix} - \frac{-0.3}{0.5}\begin{pmatrix} 0.5 \\ -0.3 \\ -0.4 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0.3 \\ 0.18 \\ 0.24 \end{pmatrix} + \begin{pmatrix} 0.3 \\ -0.18 \\ -0.24 \end{pmatrix} = \begin{pmatrix} 0 \\ 0.64 \\ -0.48 \end{pmatrix}$$

Normalizing $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ to form an orthogonal matrix U_1, we have

$$U_1 = \begin{pmatrix} 0.5\sqrt{2} & 0.5\sqrt{2} & 0 \\ 0.3\sqrt{2} & -0.3\sqrt{2} & 0.8 \\ 0.4\sqrt{2} & -0.4\sqrt{2} & -0.6 \end{pmatrix}.$$

$$U_1^T A U_1 = \begin{pmatrix} 0.5\sqrt{2} & 0.3\sqrt{2} & 0.4\sqrt{2} \\ 0.5\sqrt{2} & -0.3\sqrt{2} & -0.4\sqrt{2} \\ 0 & 0.8 & -0.6 \end{pmatrix}\begin{pmatrix} 0.6 & 0 & -0.75 \\ 0.56 & -0.2 & -0.55 \\ 0.08 & 0.4 & -0.4 \end{pmatrix}\begin{pmatrix} 0.5\sqrt{2} & 0.5\sqrt{2} & 0 \\ 0.3\sqrt{2} & -0.3\sqrt{2} & 0.8 \\ 0.4\sqrt{2} & -0.4\sqrt{2} & -0.6 \end{pmatrix}$$

$$= \begin{pmatrix} 0.5\sqrt{2} & 0.3\sqrt{2} & 0.4\sqrt{2} \\ 0.5\sqrt{2} & -0.3\sqrt{2} & -0.4\sqrt{2} \\ 0 & 0.8 & -0.6 \end{pmatrix}\begin{pmatrix} 0 & 0.6\sqrt{2} & 0.45 \\ 0 & 0.56\sqrt{2} & 0.17 \\ 0 & 0.08\sqrt{2} & 0.56 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & 0.5\sqrt{2} \\ 0 & 0.2 & -0.05\sqrt{2} \\ 0 & 0.4\sqrt{2} & -0.2 \end{pmatrix}$$

Let $B = \begin{pmatrix} 0.2 & -0.05\sqrt{2} \\ 0.4\sqrt{2} & -0.2 \end{pmatrix}$. Its characteristic polynomial is

$$\det(B - \lambda I) = \begin{pmatrix} 0.2 - \lambda & -0.05\sqrt{2} \\ 0.4\sqrt{2} & -0.2 - \lambda \end{pmatrix} = \lambda^2 - 0.04 + \left(0.4\sqrt{2}\right)\left(0.05\sqrt{2}\right) = \lambda^2$$

Clearly, rank(B) = 1 and dim $N(B)$ = 1. Geometric multiplicity of eigenvalue 0 = 1 and algebraic multiplicity of eigenvalue 0 = 2. The eigenvector of B corresponding to eigenvalue 0 is $\boldsymbol{y}_1 = (1, 2\sqrt{2})^T$. Then choose $\boldsymbol{y}_2 = (2\sqrt{2}, -1)^T$ such that $\boldsymbol{y}_1, \boldsymbol{y}_2$ forms an orthogonal basis in $\boldsymbol{R}^2$. Normalizing $\boldsymbol{y}_1, \boldsymbol{y}_2$ to form a 2×2 orthogonal matrix V, we have

$$V = \begin{pmatrix} \frac{1}{3} & \frac{2\sqrt{2}}{3} \\ \frac{2\sqrt{2}}{3} & -\frac{1}{3} \end{pmatrix}.$$

Let $V_1 = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & & \\ 0 & & V \end{array}\right) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{2\sqrt{2}}{3} \\ 0 & \frac{2\sqrt{2}}{3} & -\frac{1}{3} \end{pmatrix}$.

$$V_1^T U_1^T A U_1 V_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{2\sqrt{2}}{3} \\ 0 & \frac{2\sqrt{2}}{3} & -\frac{1}{3} \end{pmatrix} \underbrace{\begin{pmatrix} 0 & 1 & 0.5\sqrt{2} \\ 0 & 0.2 & -0.05\sqrt{2} \\ 0 & 0.4\sqrt{2} & -0.2 \end{pmatrix}}_{U_1^H A U_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{2\sqrt{2}}{3} \\ 0 & \frac{2\sqrt{2}}{3} & -\frac{1}{3} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{2\sqrt{2}}{3} \\ 0 & \frac{2\sqrt{2}}{3} & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0.5\sqrt{2} \\ 0 & 0 & 0.15\sqrt{2} \\ 0 & 0 & 0.6 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & 0.5\sqrt{2} \\ 0 & 0 & 0.45\sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} = T - \text{an upper triangular matrix}$$

$A = U_1 V_1 T V_1^T U_1^T = (U_1 V_1) T (U_1 V_1)^T = QTQ^T$ where

$$Q = U_1 V_1 = \begin{pmatrix} 0.5\sqrt{2} & 0.5\sqrt{2} & 0 \\ 0.3\sqrt{2} & -0.3\sqrt{2} & 0.8 \\ 0.4\sqrt{2} & -0.4\sqrt{2} & -0.6 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{2\sqrt{2}}{3} \\ 0 & \frac{2\sqrt{2}}{3} & -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} 0.5\sqrt{2} & \frac{0.5\sqrt{2}}{3} & \frac{2}{3} \\ 0.3\sqrt{2} & \frac{1.3\sqrt{2}}{3} & -\frac{2}{3} \\ 0.4\sqrt{2} & -\frac{1.6\sqrt{2}}{3} & -\frac{1}{3} \end{pmatrix}$$ is an orthogonal matrix.

Theorem 6.3.12

If A is a Hermitian matrix, then there exists an unitary matrix U that diagonalizes A; that is, $U^H A U = D$, where D is diagonal. If A is a real symmetric matrix, then A is **orthogonally diagonalizable**; that is, $Q^T A Q = D$, where Q is orthogonal and D is real diagonal.

Proof

By Theorem 6.3.8, $A = UTU^H$ where T is upper triangular and U is unitary. Suppose A is an $n \times n$ Hermitian matrix.

$$A = A^H \quad \Rightarrow \quad UTU^H = (UTU^H)^H = (U^H)^H T^H U^H = UT^H U^H$$

Since U is unitary, U^{-1} exists ($= U^H$) and therefore $T = T^H$. Hence T is diagonal.

Suppose A is an $n \times n$ real symmetric matrix. Then A is Hermitian. By Theorem 6.3.6, all eigenvalues of A are real. By Theorem 6.3.10, $A = QTQ^T$, where Q is orthogonal and T is upper triangular. Since A is symmetric, we have

$$A = A^T \quad \Rightarrow \quad QTQ^T = (QTQ^T)^T = (Q^T)^T T^T Q^T = QT^T Q^T.$$

Since Q is orthogonal, Q^{-1} exists ($= Q^T$) and $T = T^T$. Hence T is real diagonal.

Theorem 6.3.13

Let $R = \left(\begin{array}{c|c} 0_{k\times k} & R_{k\times(n-k)} \\ \hline 0_{(n-k)\times k} & R_{(n-k)\times(n-k)} \end{array}\right)$ and $T = \left(\begin{array}{c|c} T_{k\times k} & T_{k\times(n-k)} \\ \hline 0_{(n-k)\times k} & T_{(n-k)\times(n-k)} \end{array}\right)$ be two $n \times n$ matrices. The subscript of each block matrix of R and T are their sizes. Suppose the first column of the $(n-k)\times(n-k)$ matrix $T_{(n-k)\times(n-k)}$ is zero. That is $T_{(n-k)\times(n-k)} = \left(\begin{array}{c|c} 0 & \\ \vdots & U_{(n-k)\times(n-k-1)} \\ 0 & \end{array}\right)$. Then

$$RT = \left(\begin{array}{c|c|c} & 0 & \\ 0_{k\times k} & \vdots & *_{k\times(n-k-1)} \\ & 0 & \\ \hline & 0 & \\ 0_{(n-k)\times k} & \vdots & *_{(n-k)\times(n-k-1)} \\ & 0 & \end{array}\right).$$

Proof

$$RT = \left(\begin{array}{c|c} 0_{k\times k} & R_{k\times(n-k)} \\ \hline 0_{(n-k)\times k} & R_{(n-k)\times(n-k)} \end{array}\right)\left(\begin{array}{c|c} T_{k\times k} & T_{k\times(n-k)} \\ \hline 0_{(n-k)\times k} & \begin{array}{c} 0 \\ \vdots \\ 0 \end{array} \quad U_{(n-k)\times(n-k-1)} \end{array}\right)$$

$$= \left(\begin{array}{c|c} 0_{k\times k}T_{k\times k} + R_{k\times(n-k)}0_{(n-k)\times k} & 0_{k\times k}T_{k\times(n-k)} + R_{k\times(n-k)}\left(\begin{array}{c|c} 0 & \\ \vdots & U_{(n-k)\times(n-k-1)} \\ 0 & \end{array}\right) \\ \hline 0_{(n-k)\times k}T_{k\times k} + R_{(n-k)\times(n-k)}0_{(n-k)\times k} & 0_{(n-k)\times k}T_{k\times(n-k)} + R_{(n-k)\times(n-k)}\left(\begin{array}{c|c} 0 & \\ \vdots & U_{(n-k)\times(n-k-1)} \\ 0 & \end{array}\right) \end{array}\right)$$

$$= \left(\begin{array}{c|c|c} 0_{k\times k} & \begin{array}{c} 0 \\ \vdots \\ 0 \end{array} & R_{k\times(n-k)}U_{(n-k)\times(n-k-1)} \\ \hline 0_{(n-k)\times k} & \begin{array}{c} 0 \\ \vdots \\ 0 \end{array} & R_{(n-k)\times(n-k)}U_{(n-k)\times(n-k-1)} \end{array}\right).$$

Theorem 6.3.14 (Cayley-Hamilton)

Let A be an $n\times n$ matrix (not necessary diagonalizable). Then $p_A(A) = 0_{n\times n}$, an $n\times n$ zero matrix.

Proof

By Theorem 6.3.8, $A = UTU^H$, in which U is unitary, T is upper triangular, and the main diagonal entries of T are $\lambda_1, \lambda_2, \cdots, \lambda_n$. We can write $p_A(t) = p_T(t) = (\lambda_1 - t)(\lambda_2 - t)\cdots(\lambda_n - t)$ because A is similar to T. We have

$$\begin{aligned} p_A(A) &= p_A(UTU^H) \\ &= (\lambda_1 I - UTU^H)(\lambda_2 I - UTU^H)\cdots(\lambda_n I - UTU^H) \\ &= (U\lambda_1 U^H - UTU^H)(U\lambda_2 U^H - UTU^H)\cdots(U\lambda_n U^H - UTU^H) \\ &= U(\lambda_1 I - T)U^H U(\lambda_2 I - T)U^H \cdots U(\lambda_n I - T)U^H \\ &= U(\lambda_1 I - T)(\lambda_2 I - T)\cdots(\lambda_n I - T)U^H. \end{aligned}$$

We can show by induction on m that the first m-th column of $(\lambda_1 I - T)\cdots(\lambda_m I - T) = 0_{n\times m}$.

Clearly, $m = 1$ is true. Suppose it is true for $m = k$. By Theorem 6.3.13,

$$(\lambda_1 I - T)\cdots(\lambda_k I - T)(\lambda_{k+1} I - T) = \underbrace{\left(\begin{array}{c|c} 0_{k\times k} & R_{k\times(n-k)} \\ \hline 0_{(n-k)\times k} & R_{(n-k)\times(n-k)} \end{array}\right)}_{(\lambda_1 I - T)\cdots(\lambda_k I - T)} \underbrace{\left(\begin{array}{c|c} T_{k\times k} & T_{k\times(n-k)} \\ \hline 0_{(n-k)\times k} & \begin{array}{c|c} 0 & \\ \vdots & U_{(n-k)\times(n-k-1)} \\ 0 & \end{array} \end{array}\right)}_{\lambda_{k+1} I - T}$$

$$= \left(\begin{array}{c|c|c} 0_{k\times k} & \begin{array}{c} 0 \\ \vdots \\ 0 \end{array} & R_{k\times(n-k)}U_{(n-k)\times(n-k-1)} \\ \hline 0_{(n-k)\times k} & \begin{array}{c} 0 \\ \vdots \\ 0 \end{array} & R_{(n-k)\times(n-k)}U_{(n-k)\times(n-k-1)} \end{array}\right).$$

It is true for $m = k + 1$. By induction, it is true for $m = n$. That is $p_A(A) = 0_{n\times n}$.

Remark 6.3.15

$p_A(A) = \det(A - AI_n) = \det(0_{n\times n}) = 0$. Is it correct? No.

λ is a scalar but A is a matrix. We cannot substitute A into λ. On the other hand, $p_A(A) = 0_{n\times n}$ is a zero matrix but $\det(0_{n\times n}) = 0$, a zero number.

Example 6.3.16

Consider $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 2 \\ 2 & 1 & 0 \end{pmatrix}$. Verify Cayley-Hamilton Theorem. Hence calculate A^{-1}.

Solution

$$p_A(\lambda) = \det\begin{pmatrix} 2-\lambda & 0 & 0 \\ 0 & 1-\lambda & 2 \\ 2 & 1 & -\lambda \end{pmatrix} = (2-\lambda)\det\begin{pmatrix} 1-\lambda & 2 \\ 1 & -\lambda \end{pmatrix} = (2-\lambda)(\lambda^2-\lambda-2) = -(\lambda-2)^2(\lambda+1)$$

$$A-2I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 2 \\ 2 & 1 & -2 \end{pmatrix}, \quad (A-2I)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 2 \\ 2 & 1 & -2 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 & 0 \\ 4 & 3 & -6 \\ -4 & -3 & 6 \end{pmatrix}, \quad A+I = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 2 \\ 2 & 1 & 1 \end{pmatrix}.$$

$$(A-2I)^2(A+I) = \begin{pmatrix} 0 & 0 & 0 \\ 4 & 3 & -6 \\ -4 & -3 & 6 \end{pmatrix}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 2 \\ 2 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Clearly, $\det(A) = p_A(0) = -4 \neq 0$. A^{-1} exists

$$A^3 - 3A^2 + 4I_3 = 0_{3\times 3} \quad \Rightarrow \quad A(3A - A^2) = (3A - A^2)A = 4I_3$$
$$\Rightarrow \quad A\left(\tfrac{1}{4}(3A - A^2)\right) = \left(\tfrac{1}{4}(3A - A^2)\right)A = I_3$$

$$A^{-1} = \frac{1}{4}(3A - A^2) = \frac{1}{4}\left(3\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 2 \\ 2 & 1 & 0 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 4 & 3 & 2 \\ 4 & 1 & 2 \end{pmatrix}\right) = \frac{1}{4}\begin{pmatrix} 2 & 0 & 0 \\ -4 & 0 & 4 \\ 2 & 2 & -2 \end{pmatrix} = \begin{pmatrix} 1/2 & 0 & 0 \\ -1 & 0 & 1 \\ 1/2 & 1/2 & -1/2 \end{pmatrix}.$$

Section 6.4 Quadratic Forms

Definition 6.4.1

A **quadratic equation** in two variables x and y is an equation of the form

$$ax^2 + 2bxy + cy^2 + dx + ey + f = 0 \qquad (1)$$

Equation (1) may be rewritten in the form

$$(x \quad y)\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + (d \quad e)\begin{pmatrix} x \\ y \end{pmatrix} + f = 0$$

Let $\boldsymbol{x} = \begin{pmatrix} x \\ y \end{pmatrix}$ and $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$. The term

$$\boldsymbol{x}^T A\boldsymbol{x} = ax^2 + 2bxy + cy^2$$

is called the **quadratic form** associated with (1).

The graph of an equation of the form (1) is called a **conic section**. If the graph of (1) consists of a single point, a double line, or a pair of lines, (1) is called a **degenerate conic**. Of more interest are the **nondegenerate conics**. Graphs of nondegenerate conics are circles, ellipses, parabolas, or hyperbolas. Their equations can be put into one of the following standard forms:

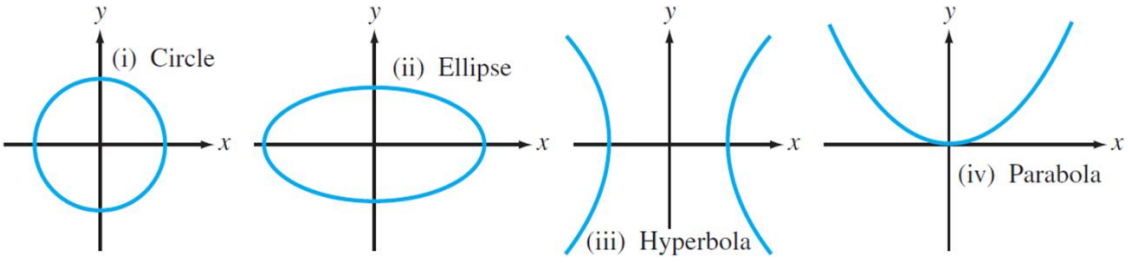
(i) $x^2 + y^2 = r^2$ (circle)

(ii) $\dfrac{x^2}{\alpha^2} + \dfrac{y^2}{\beta^2} = 1$ (ellipse)

(iii) $\dfrac{x^2}{\alpha^2} - \dfrac{y^2}{\beta^2} = 1$ or $\dfrac{y^2}{\alpha^2} - \dfrac{x^2}{\beta^2} = 1$ (hyperbola)

(iv) $x^2 = \alpha y$ or $y^2 = \alpha x$ (parabola)

Here α and β are nonzero real numbers; and $r > 0$.



Example 6.4.2

Given the quadratic equation

$$3x^2+2xy+3y^2+8\sqrt{2}y-4=0$$

find a change of coordinates so that the resulting equation represents a conic in standard position.

Solution

This equation can be written in the form

$$(x \quad y)\begin{pmatrix}3 & 1\\ 1 & 3\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}+(0 \quad 8\sqrt{2})\begin{pmatrix}x\\ y\end{pmatrix}-4=0$$

The matrix $\begin{pmatrix}3 & 1\\ 1 & 3\end{pmatrix}$ has eigenvalues $\lambda=2$ and $\lambda=4$ with corresponding unit eigenvectors

$$\left(\tfrac{1}{\sqrt{2}} \quad -\tfrac{1}{\sqrt{2}}\right)^T \quad \text{and} \quad \left(\tfrac{1}{\sqrt{2}} \quad \tfrac{1}{\sqrt{2}}\right)^T$$

Let $Q=\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}=\begin{pmatrix}\cos 45^\circ & \sin 45^\circ\\ -\sin 45^\circ & \cos 45^\circ\end{pmatrix}$ and set $\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}x'\\ y'\end{pmatrix}$. Thus,

$$Q^TAQ=\begin{pmatrix}2 & 0\\ 0 & 4\end{pmatrix}$$

and the equation of the conic becomes

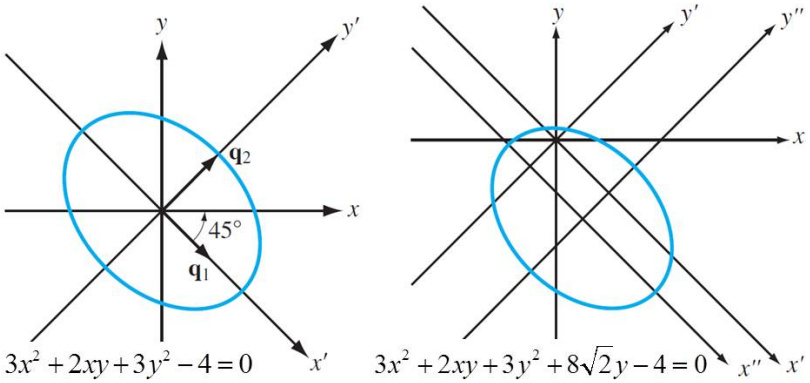
$$(x' \quad y')\begin{pmatrix}2 & 0\\ 0 & 4\end{pmatrix}\begin{pmatrix}x'\\ y'\end{pmatrix}+(0 \quad 8\sqrt{2})\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}x'\\ y'\end{pmatrix}-4=0 \quad \text{or} \quad 2x'^2+4y'^2-8x'+8y'-4=0$$

If we complete the square, we get $(x'-2)^2+2(y'+1)=8.$ If we set $x''=x'-2$ and $y''=y'+1,$ the equation simplifies to

$$\frac{(x'')^2}{8}+\frac{(y'')^2}{4}=1$$

where

$$\begin{pmatrix}x\\ y\end{pmatrix}=\overbrace{\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}}^{\text{rotation}}\begin{pmatrix}x'\\ y'\end{pmatrix}=\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\overbrace{\begin{pmatrix}x''+2\\ y''-1\end{pmatrix}}^{\text{translation}} \quad \text{or} \quad \begin{pmatrix}x''\\ y''\end{pmatrix}=\begin{pmatrix}x'\\ y'\end{pmatrix}+\begin{pmatrix}-2\\ 1\end{pmatrix}=\begin{pmatrix}\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}+\begin{pmatrix}-2\\ 1\end{pmatrix}$$



In general, a quadratic equation in n variables $x_1, \cdots, x_n$ is one of the form

$$\boldsymbol{x}^T A\boldsymbol{x} + B\boldsymbol{x} + \alpha = 0$$

where $\boldsymbol{x} = (x_1, \cdots, x_n)^T$, A is an $n \times n$ symmetric matrix, B is a $1 \times n$ matrix, and α is a scalar.

The vector function

$$f(\boldsymbol{x}) = \boldsymbol{x}^T A\boldsymbol{x} = \sum_{i,j=1}^{n} a_{ij} x_i x_j$$

is the **quadratic form in n variables** associated with the quadratic equation. The quadratic equation in three variables is of the form

$$ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz + gx + hy + iz + \alpha = 0.$$

Its graph is called a **quadric surface**. There are four basic types of nondegenerate quadric surfaces:

1. Ellipsoids
2. Hyperboloids (of one or two sheets)
3. Cones
4. Paraboloids (either elliptic or hyperbolic)

As in the two-dimensional case, we can use translations and rotations to transform the equation into the standard form

$$\lambda_1 (x')^2 + \lambda_2 (y')^2 + \lambda_3 (z')^2 + \alpha = 0$$

where $\lambda_1, \lambda_2, \lambda_3$ are the eigenvalues of A. For the general n-dimensional case, the quadratic form can always be translated to a simpler diagonal form.

Theorem 6.4.3 (Principal Axes Theorem)

Let A be an $n \times n$ real symmetric matrix. Then there is an orthogonal change of variable, $\boldsymbol{x} = Q\boldsymbol{y}$, that transforms the quadratic form $\boldsymbol{x}^T A\boldsymbol{x}$ into a quadratic form $\boldsymbol{y}^T D\boldsymbol{y}$ with no cross-product term.

Proof

If A is a real symmetric matrix, then by Theorem 6.3.12 there is an orthogonal matrix Q that diagonalizes A; that is, $Q^T AQ = D$ (a diagonal matrix). If we set $\boldsymbol{y} = Q^T \boldsymbol{x}$, then $\boldsymbol{x} = Q\boldsymbol{y}$ and

$$\boldsymbol{x}^T A\boldsymbol{x} = (Q\boldsymbol{y})^T AQ\boldsymbol{y} = \boldsymbol{y}^T Q^T AQ\boldsymbol{y} = \boldsymbol{y}^T D\boldsymbol{y}.$$

The columns of Q in Theorem 6.4.3 are called the **principal axes** of the quadratic form $\boldsymbol{x}^T A\boldsymbol{x}$.

Let us consider the problem of maximizing and minimizing functions of several variables.

Definition 6.4.4

Let $f(\boldsymbol{x})$ be a real-valued vector function on $\boldsymbol{R}^n$. A point $\boldsymbol{x}_0$ in $\boldsymbol{R}^n$ is said to be a **stationary point** of f if all the first partial derivatives of f at $\boldsymbol{x}_0$ exist and are zero.

If $f(\boldsymbol{x})$ has either a local maximum or a local minimum at a point $\boldsymbol{x}_0$ and the first partial derivatives of f exist at $\boldsymbol{x}_0$, they will all be zero. Thus, if $f(\boldsymbol{x})$ has first partial derivatives everywhere, its local maxima and minima will occur at stationary points.

Consider the quadratic form $f(x, y) = ax^2 + 2bxy + cy^2$. The first partial derivatives of f are

$$f_x = 2ax + 2by \quad \text{and} \quad f_y = 2bx + 2cy$$

Setting these equal to zero, we see that (0, 0) is a stationary point. Moreover, if the matrix

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

is nonsingular, we have

$$\left.\begin{aligned} f_x &= 2ax + 2by = 0 \\ f_y &= 2bx + 2cy = 0 \end{aligned}\right\} \Rightarrow \begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

(0, 0) will be the only critical point. f will have either a global minimum, a global maximum, or a saddle point at (0, 0).

Let us write f in matrix form

$$f(\boldsymbol{x}) = f(x, y) = ax^2 + 2bxy + cy^2 = (x \quad y)\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \boldsymbol{x}^T A\boldsymbol{x} \quad \text{where} \quad \boldsymbol{x} = \begin{pmatrix} x \\ y \end{pmatrix}.$$

Since $f(0, 0) = 0$, it follows that

(a) f will have a global minimum at (0, 0) if and only if $\boldsymbol{x}^T A\boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x}$.

(b) f will have a global maximum at (0, 0) if and only if $\boldsymbol{x}^T A\boldsymbol{x} < 0$ for all nonzero $\boldsymbol{x}$.

(c) f will have a saddle point at (0, 0) if and only if $\boldsymbol{x}^T A\boldsymbol{x}$ changes sign.

Definition 6.4.5

A quadratic form $f(\boldsymbol{x}) = \boldsymbol{x}^T A\boldsymbol{x}$ is said to be **definite** if it takes on only one sign as $\boldsymbol{x}$ varies over all nonzero vectors in $\boldsymbol{R}^n$. The form is **positive definite** if $\boldsymbol{x}^T A\boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x}$ in $\boldsymbol{R}^n$ and **negative definite** if $\boldsymbol{x}^T A\boldsymbol{x} < 0$ for all nonzero $\boldsymbol{x}$ in $\boldsymbol{R}^n$. A quadratic form is said to be **indefinite** if it takes on values that differ in sign. If $f(\boldsymbol{x}) = \boldsymbol{x}^T A\boldsymbol{x} \geq 0$ and assumes the value 0 for some $\boldsymbol{x} \neq \boldsymbol{0}$, then $f(\boldsymbol{x})$ is said to be **positive semidefinite**. If $f(\boldsymbol{x}) \leq 0$ and assumes the value 0 for some $\boldsymbol{x} \neq \boldsymbol{0}$, then $f(\boldsymbol{x})$ is said to be **negative semidefinite**.

Whether the quadratic form is positive definite or negative definite depends on the matrix A. If the quadratic form is positive definite, we say simply that A is positive definite. The preceding definition can then be restated as follows.

Definition 6.4.6

A real symmetric $n \times n$ matrix A is said to be

I. **positive definite** if $\boldsymbol{x}^T A\boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x}$ in $\boldsymbol{R}^n$.

II. **negative definite** if $\boldsymbol{x}^T A\boldsymbol{x} < 0$ for all nonzero $\boldsymbol{x}$ in $\boldsymbol{R}^n$.

III. **positive semidefinite** if $\boldsymbol{x}^T A\boldsymbol{x} \geq 0$ for all nonzero $\boldsymbol{x}$ in $\boldsymbol{R}^n$.

IV. **negative semidefinite** if $\boldsymbol{x}^T A\boldsymbol{x} \leq 0$ for all nonzero $\boldsymbol{x}$ in $\boldsymbol{R}^n$.

V. **indefinite** if $\boldsymbol{x}^T A\boldsymbol{x}$ takes on values that differ in sign.

Theorem 6.4.7

Let A be a real symmetric $n \times n$ matrix. Then A is

(a) positive definite if and only if eigenvalues of A are all positive,

(b) negative definite if and only if eigenvalues of A are all negative, or

(c) indefinite if and only if A has both positive and negative eigenvalues.

Proof

(a) If A is positive definite and λ is an eigenvalue of A, then, for any real eigenvalue λ (Theorem 6.3.6) belonging to real eigenvector $\boldsymbol{x}$ (Theorem 6.3.9),

$$\boldsymbol{x}^T A\boldsymbol{x} = \lambda \boldsymbol{x}^T \boldsymbol{x} = \lambda \| \boldsymbol{x} \|^2 .$$

Hence,

$$\lambda = \frac{\boldsymbol{x}^T A\boldsymbol{x}}{\| \boldsymbol{x} \|^2} > 0.$$

Conversely, suppose that all the eigenvalues of A are positive. Let $\boldsymbol{u}_1, \cdots, \boldsymbol{u}_n$ be an orthonormal basis of eigenvectors of A corresponding to eigenvalues $\lambda_1, \cdots, \lambda_n$ (by Theorem 6.3.12). If $\boldsymbol{x}$ is any nonzero vector in $\boldsymbol{R}^n$, then $\boldsymbol{x}$ can be written in the form

$$\boldsymbol{x} = c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n$$

where some $c_i \neq 0$ (otherwise if all $c_i = 0$, then $\boldsymbol{x} = \boldsymbol{0}$. Contradiction). It follows that

$$\begin{aligned}
\boldsymbol{x}^T A\boldsymbol{x} &= (c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n)^T A(c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_n\boldsymbol{u}_n) \\
&= (c_1\boldsymbol{u}_1^T + c_2\boldsymbol{u}_2^T + \cdots + c_n\boldsymbol{u}_n^T)(c_1\lambda_1\boldsymbol{u}_1 + c_2\lambda_2\boldsymbol{u}_2 + \cdots + c_n\lambda_n\boldsymbol{u}_n) \\
&= c_1^2\lambda_1 + c_2^2\lambda_2 + \cdots + c_n^2\lambda_n \\
&> 0
\end{aligned}$$

and hence A is positive definite.

(b) eigenvalues of A are all negative $\Leftrightarrow$ eigenvalues of $-A$ are all positive

$\Leftrightarrow$ $-A$ is positive definite

$\Leftrightarrow$ A is negative definite

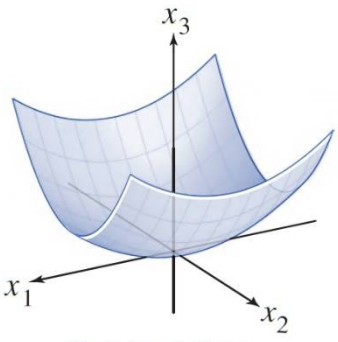
(c) If A is indefinite, then A is neither positive definite nor negative definite. Hence eigenvalues of A are neither all positive nor all negative.
A has both positive and negative eigenvalues $\lambda_{\max} > 0$ and $\lambda_{\min} < 0$. If $\boldsymbol{x}$ is an eigenvector corresponding to $\lambda_{\max}$, then

$$\boldsymbol{x}^T A\boldsymbol{x} = \lambda_{\max}\boldsymbol{x}^T\boldsymbol{x} = \lambda_{\max} \| \boldsymbol{x} \|^2 > 0.$$

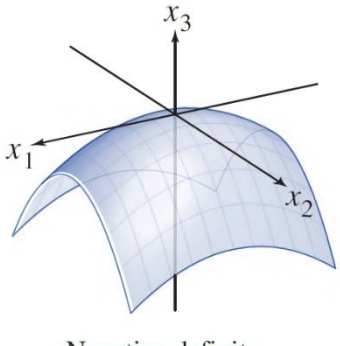
If $\boldsymbol{y}$ is an eigenvector corresponding to $\lambda_{\min}$, then

$$\boldsymbol{y}^T A\boldsymbol{y} = \lambda_{\min}\boldsymbol{y}^T\boldsymbol{y} = \lambda_{\min} \| \boldsymbol{y} \|^2 < 0.$$

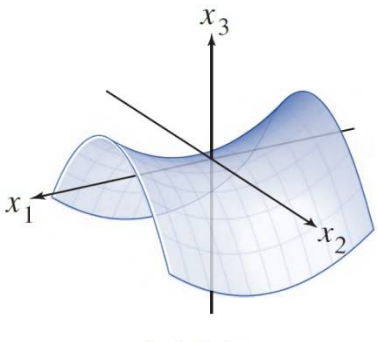
The first partial derivatives of quadratic form $f(\boldsymbol{x}) = \boldsymbol{x}^T A\boldsymbol{x}$ is $\nabla f(\boldsymbol{x}) = 2A\boldsymbol{x}$. f has the only critical point $\mathbf{0}$ if A is nonsingular. $\mathbf{0}$ will be a global minimum of f if A is positive definite and a global maximum of f if A is negative definite. If A is indefinite, then $\mathbf{0}$ is a saddle point of f.



Positive definite



Negative definite



Indefinite

Suppose now that we have a function $f(x, y)$ with a stationary point (x_0, y_0). If f has continuous third partial derivatives in a neighborhood of (x_0, y_0), it can be expanded in a Taylor series about that point.

$$f(x_0+h,y_0+k)=f(x_0,y_0)+\left(hf_x(x_0,y_0)+kf_y(x_0,y_0)\right)$$
$$+\frac{1}{2}\left(h^2f_{xx}(x_0,y_0)+2hkf_{xy}(x_0,y_0)+k^2f_{yy}(x_0,y_0)\right)+R$$
$$=f(x_0,y_0)+\frac{1}{2}\left(h^2f_{xx}(x_0,y_0)+2hkf_{xy}(x_0,y_0)+k^2f_{yy}(x_0,y_0)\right)+R$$

and the remainder R is given by

$$R=\frac{1}{6}\left(h^3f_{xxx}(z)+3h^2kf_{xxy}(z)+3hk^2f_{xyy}(z)+k^3f_{yyy}(z)\right),\quad z=(x_0+\theta h,y_0+\theta k),\quad 0<\theta<1$$

If h and k are sufficiently small, $|R|<\frac{1}{2}\left|h^2f_{xx}(x_0,y_0)+2hkf_{xy}(x_0,y_0)+k^2f_{yy}(x_0,y_0)\right|$, and hence $f(x_0+h,y_0+k)-f(x_0,y_0)$ will have the same sign as the quadratic form $h^2f_{xx}(x_0,y_0)+2hkf_{xy}(x_0,y_0)+k^2f_{yy}(x_0,y_0)$ in the variables h and k. Thus, $f(x, y)$ will have a local minimum (maximum) at (x_0,y_0) if and only if $f(h, k)$ has a minimum (maximum) at (0, 0). Let

$$H=\begin{pmatrix}f_{xx}(x_0,y_0) & f_{xy}(x_0,y_0)\\ f_{xy}(x_0,y_0) & f_{yy}(x_0,y_0)\end{pmatrix}$$

and let λ_1 and λ_2 be the eigenvalues of H. If H is nonsingular, then λ_1 and λ_2 are nonzero and we can classify the stationary points as follows:

(i) f has a minimum at (x_0,y_0) if $\lambda_1,\lambda_2>0$.

(ii) f has a maximum at (x_0,y_0) if $\lambda_1,\lambda_2<0$.

(iii) f has a saddle point at (x_0,y_0) if λ_1 and λ_2 differ in sign.

Example 6.4.8

Classify each stationary point of the function $f(x,y)=\frac{1}{3}x^3+xy^2-4xy$.

Solution

The first partials of f are $f_x=x^2+y^2+4y$ and $f_y=2xy+4x=2x(y+2)$. Setting $f_y=0$, we get $x=0$ or $y=-2$. Setting $f_x=0$ we see that if $x=0$, then y must either be 0 or -4,

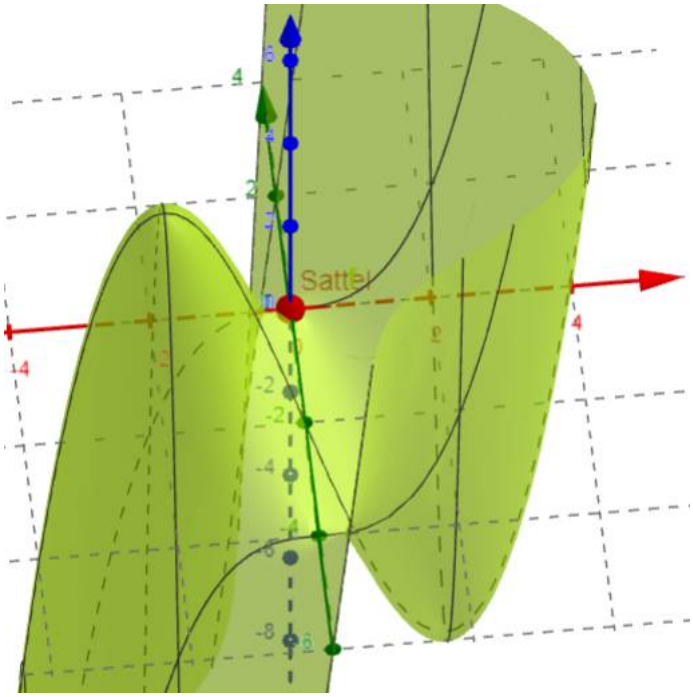
and if $y=-2$, then $x=\pm 2$. Thus, (0, 0), (0, −4), (2, −2) and (−2, −2) are the stationary points of f. To classify the stationary points, we compute the second partial derivatives:

$$f_{xx}=2x,\quad f_{xy}=2y+4,\quad f_{yy}=2x.$$

For each stationary point (x_0,y_0), we determine the eigenvalues of $\begin{pmatrix} 2x_0 & 2y_0+4 \\ 2y_0+4 & 2x_0 \end{pmatrix}$.

These values are summarized in below table:

Stationary Point (x_0, y_0)	λ_1	λ_2	Description
(0, 0)	4	−4	Saddle point
(0,−4)	4	−4	Saddle point
(2,−2)	4	4	Local minimum
(−2,−2)	−4	−4	Local maximum



We can now generalize our method of classifying stationary points to functions of more than two variables. Let $f(\boldsymbol{x})=f(x_1,\cdots,x_n)$ be a real-valued function whose third partial derivatives are all continuous. Let $\boldsymbol{x}_0$ be a stationary point of f and define the matrix $H=H(\boldsymbol{x}_0)$ by

$$H_{ij}=f_{x_ix_j}(\boldsymbol{x}_0).$$

$H(\boldsymbol{x}_0)$ is called the *Hessian* of f at $\boldsymbol{x}_0$.

The stationary point can be classified as follows:

(i) $\boldsymbol{x}_0$ is a local minimum of f if $H(\boldsymbol{x}_0)$ is positive definite.

(ii) $\boldsymbol{x}_0$ is a local maximum of f if $H(\boldsymbol{x}_0)$ is negative definite.

(iii) $\boldsymbol{x}_0$ is a saddle point of f if $H(\boldsymbol{x}_0)$ is indefinite.

Example 6.4.9

Classify all stationary points of the function $f(x,y,z) = x^2 + xz - 3\cos y + z^2$.

Solution

The first partial derivatives of f are

$$f_x = 2x + z, \quad f_y = 3\sin y, \quad f_z = x + 2z.$$

It follows that (x, y, z) is a stationary point of f if and only if $x = z = 0$ and $y = n\pi$, where n is an integer. Let $\boldsymbol{x}_0 = (0, 2k\pi, 0)^T$. The Hessian of f at $\boldsymbol{x}_0$ is given by

$$H(\boldsymbol{x}_0) = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

The eigenvalues of $H(\boldsymbol{x}_0)$ are 3, 3, and 1. Since the eigenvalues are all positive, it follows that $H(\boldsymbol{x}_0)$ is positive definite and hence F has a local minimum at $\boldsymbol{x}_0$. At a stationary point of the form $\boldsymbol{x}_1 = (0, (2k-1)\pi, 0)^T$, the Hessian will be

$$H(\boldsymbol{x}_0) = \begin{pmatrix} 2 & 0 & 1 \\ 0 & -3 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

The eigenvalues of $H(\boldsymbol{x}_1)$ are -3, 3, and 1. It follows that $H(\boldsymbol{x}_1)$ is indefinite and hence $\boldsymbol{x}_1$ is a saddle point of f.

Section 6.5 Positive Definite Matrices

Since the eigenvalues of an $n \times n$ real symmetric matrix A are real, we may (and do) adopt the convention that they are always arranged in algebraically ascending order:

$$\lambda_{\min}(A) = \lambda_1(A) \le \lambda_2(A) \le \cdots \le \lambda_n(A) = \lambda_{\max}(A)$$

Theorem 6.5.1

Let $\boldsymbol{u}_1, \cdots, \boldsymbol{u}_k$ be an orthonormal set of eigenvectors of an $n \times n$ real symmetric matrix A. Then $\lambda_{\min} \le \boldsymbol{x}^T A\boldsymbol{x} \le \lambda_{\max}$ for all unit vector $\boldsymbol{x}$ in $\text{span}\{\boldsymbol{u}_1, \cdots, \boldsymbol{u}_k\}$. ($\lambda_{\min}$, $\lambda_{\max}$ is min/max eigenvalues corresponding to $\boldsymbol{u}_1, \cdots, \boldsymbol{u}_k$)

Proof

Let $\boldsymbol{x} = c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_k\boldsymbol{u}_k$ be an unit vector as in the proof of Theorem 6.4.7. Then by Theorem 5.5.7, we have $c_1^2 + c_2^2 + \cdots + c_k^2 = 1$.

$$\left(c_1^2 + c_2^2 + \cdots + c_k^2\right)\lambda_{\min} \le c_1^2\lambda_1 + c_2^2\lambda_2 + \cdots + c_k^2\lambda_k \le \left(c_1^2 + c_2^2 + \cdots + c_k^2\right)\lambda_{\max}$$
$$\lambda_{\min} \le \boldsymbol{x}^T A\boldsymbol{x} \le \lambda_{\max}$$

Theorem 6.5.2

Let A be an $n \times n$ real symmetric matrix. Then

(i) $\lambda_i(-A) = -\lambda_{n-i+1}(A)$ and $\lambda_1(-A) \ge \lambda_2(-A) \ge \cdots \ge \lambda_n(-A)$.

(ii) $\lambda_i(A + cI_n) = \lambda_i(A) + c$ and $\lambda_1(A + cI_n) \le \lambda_2(A + cI_n) \le \cdots \le \lambda_n(A + cI_n)$ for any real constant c.

Theorem 6.5.3

Let $\boldsymbol{y} \in \boldsymbol{R}^{n-1}$ and $a \in \boldsymbol{R}$ be given, and let $K = \left(\begin{array}{c|c} 0_{(n-1)\times(n-1)} & \boldsymbol{y} \\ \hline -\boldsymbol{y}^T - & a \end{array}\right)_{n\times n}$. Then the eigenvalues of K

are $\frac{a-\sqrt{a^2+4\|\boldsymbol{y}\|^2}}{2}, \underbrace{0,\cdots,0}_{n-2}, \frac{a+\sqrt{a^2+4\|\boldsymbol{y}\|^2}}{2}$ in ascending order.

Proof

The case for $\boldsymbol{y}=\boldsymbol{0}$ is easy.

Suppose $\boldsymbol{y}\neq\boldsymbol{0}$. Let $\boldsymbol{v}_1,\cdots,\boldsymbol{v}_{n-2}$ be an orthonormal basis of span$\{\boldsymbol{y}\}$. That is $\boldsymbol{y}^T\boldsymbol{v}_i=0$ for $1,\cdots,n-2$. Then $\left(\begin{array}{ccc|c} | & & | & | \\ \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_{n-2} & \frac{1}{\|y\|}\boldsymbol{y} \\ | & & | & | \\ \hline 0 & \cdots & 0 & 0 \end{array}\begin{array}{c|} 0 \\ \vdots \\ 0 \\ \hline 1 \end{array}\right)$ is an orthogonal matrix. Since

$$\left(\begin{array}{ccc|c} | & & | & | \\ \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_{n-2} & \frac{1}{\|y\|}\boldsymbol{y} \\ | & & | & | \\ \hline 0 & \cdots & 0 & 0 \end{array}\begin{array}{c|} 0 \\ \vdots \\ 0 \\ \hline 1 \end{array}\right)^T \left(\begin{array}{c|c} \boldsymbol{0}_{n\times n} & \begin{array}{c} | \\ \boldsymbol{y} \\ | \end{array} \\ \hline -\boldsymbol{y}^H- & a \end{array}\right)\left(\begin{array}{ccc|c} | & & | & | \\ \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_{n-2} & \frac{1}{\|y\|}\boldsymbol{y} \\ | & & | & | \\ \hline 0 & \cdots & 0 & 0 \end{array}\begin{array}{c|} 0 \\ \vdots \\ 0 \\ \hline 1 \end{array}\right)$$

$$=\left(\begin{array}{ccc|c} - & \boldsymbol{v}_1^T & - & 0 \\ & \vdots & & \vdots \\ - & \boldsymbol{v}_{n-2}^T & - & 0 \\ - & \frac{1}{\|y\|}\boldsymbol{y}^T & - & 0 \\ \hline 0 & \cdots & 0 & 1 \end{array}\right)\left(\begin{array}{c|c} \boldsymbol{0}_{n\times(n-2)} & \begin{array}{c} 0 \\ \vdots \\ 0 \\ \hline \|\boldsymbol{y}\| \end{array}\begin{array}{c|} \boldsymbol{y} \\ \\ \hline a \end{array} \end{array}\right)$$

$$=\left(\begin{array}{c|cc} \boldsymbol{0}_{n\times(n-2)} & \begin{array}{c} 0 \\ \vdots \\ 0 \\ 0 \\ \hline \|\boldsymbol{y}\| \end{array} & \begin{array}{c} 0 \\ \vdots \\ 0 \\ \|\boldsymbol{y}\| \\ \hline a \end{array} \end{array}\right),$$

K is similar to $\left(\begin{array}{c|cc} \boldsymbol{0}_{n\times(n-2)} & \begin{array}{c} 0 \\ \vdots \\ 0 \\ 0 \\ \hline \|\boldsymbol{y}\| \end{array} & \begin{array}{c} 0 \\ \vdots \\ 0 \\ \|\boldsymbol{y}\| \\ \hline a \end{array} \end{array}\right)$. Hence the characteristic polynomial of K is

$$p_K(\lambda)=(-\lambda)^{n-2}\det\begin{pmatrix}-\lambda & \|y\| \\ \|y\| & a-\lambda\end{pmatrix}=(-\lambda)^{n-2}\left(\lambda^2-a\lambda-\|y\|^2\right)$$

The eigenvalues of K are $\frac{a-\sqrt{a^2+4\|\boldsymbol{y}\|^2}}{2}, \underbrace{0,\cdots,0}_{n-2}, \frac{a+\sqrt{a^2+4\|\boldsymbol{y}\|^2}}{2}$ in ascending order.

Theorem 6.5.4

Let $H=\left(\begin{array}{c|c} B_{(n-1)\times(n-1)} & \begin{matrix}0\\ \vdots\\ 0\end{matrix} \\ \hline 0\cdots 0 & 0 \end{array}\right)_{n\times n}$, $K=\left(\begin{array}{c|c} 0_{(n-1)\times(n-1)} & \begin{matrix}|\\ \boldsymbol{y}\\ |\end{matrix} \\ \hline -\boldsymbol{y}^T- & a \end{array}\right)_{n\times n}$ be two $n\times n$ real symmetric matrices.

Then for each $i=1,\cdots,n$,

$$\lambda_i(H+K)\le\lambda_{i+1}(H)$$

Proof

Let $x_1,\cdots,x_n;\ y_1,\cdots,y_n,$ and $z_1,\cdots,z_n$ be orthonormal sets of eigenvectors of H, K, and $H+K$, respectively such that $Hx_i=\lambda_i(H)x_i,\ Ky_i=\lambda_i(K)y_i,$ and $(H+K)z_i=\lambda_i(H+K)z_i$ for each $i=1,\cdots,n.$ For a given $i\in\{1,\cdots,n\}$ and any $j\in\{1,\cdots,n-i\},$ let $S_1=\text{span}\{x_1,\cdots,x_{i+1}\}$, $S_2=\text{span}\{y_1,\cdots,y_{n-1}\},$ and $S_3=\text{span}\{z_i,\cdots,z_n\}.$ Then

$$\dim S_1+\dim S_2+\dim S_3=(i+1)+(n-1)+(n-i+1)=2n+1.$$

On the other hand,

$$\dim(S_1\cap S_2)=\dim S_1+\dim S_2-\dim(S_1+S_2)\ge\dim S_1+\dim S_2-n$$
$$\dim(S_1\cap S_2\cap S_3)\ge\dim S_1+\dim(S_2\cap S_3)-n\ge\dim S_1+\dim S_2+\dim S_3-2n=1$$

There is a unit vector $\boldsymbol{x}\in S_1\cap S_2\cap S_3.$ We obtain

$$\lambda_i(H+K)\le\boldsymbol{x}^T(H+K)\boldsymbol{x}=\boldsymbol{x}^TH\boldsymbol{x}+\boldsymbol{x}^TK\boldsymbol{x}\le\lambda_{i+1}(H)+\lambda_{n-1}(K)=\lambda_{i+1}(H)$$

By Theorem 6.5.1, the first inequality follows from $\boldsymbol{x}\in S_3$ and the second inequality follows from $\boldsymbol{x}\in S_1$ and $\boldsymbol{x}\in S_2,$ respectively. The equality follows from Theorem 6.5.3.

Theorem 6.5.5

Let B be an $(n-1)\times(n-1)$ real symmetric matrix, let $\boldsymbol{y}\in\boldsymbol{R}^{n-1}$ and $a\in\boldsymbol{R}$ be given, let

$$A=\left(\begin{array}{c|c} B_{(n-1)\times(n-1)} & \begin{matrix}|\\ \boldsymbol{y}\\ |\end{matrix} \\ \hline -\boldsymbol{y}^T- & a \end{array}\right)_{n\times n}.$$

Then $\lambda_1(A)\le\lambda_1(B)\le\lambda_2(A)\le\cdots\le\lambda_{n-1}(A)\le\lambda_{n-1}(B)\le\lambda_n(A).$

Proof

Choose a large real number c such that $\lambda_i(A+cI_n)>0,\ i=1,\cdots,n$ and $\lambda_j(B+cI_{n-1})>0,$

$j = 1, \cdots, n-1$. Let $H = \left(\begin{array}{c|c} B + cI_{n-1} & \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} \\ \hline 0 \cdots 0 & 0 \end{array}\right)_{n \times n}$, $K = \left(\begin{array}{c|c} 0_{(n-1)\times(n-1)} & \begin{matrix} y \\ | \end{matrix} \\ \hline -y^T- & a+c \end{array}\right)_{n \times n}$.

Clearly, $A + cI_n = H + K$ and $0 = \lambda_1(H) \le \underbrace{\lambda_2(H)}_{\lambda_1(B+cI_{n-1})} \le \cdots \le \underbrace{\lambda_{n-1}(H)}_{\lambda_{n-2}(B+cI_{n-1})} \le \underbrace{\lambda_n(H)}_{\lambda_{n-1}(B+cI_{n-1})}$.

By Theorem 6.5.4, we have $\lambda_i(A + cI_n) = \lambda_i(H + K) \le \lambda_{i+1}(H) = \lambda_i(B + cI_{n-1})$.

By Theorem 6.5.2 (b), we have $\lambda_i(A) \le \lambda_i(B)$.

Apply the above inequality to $-A$ and $-B$, we have

$$-\lambda_{n-i+1}(A) = \lambda_i(-A) \le \lambda_i(-B) = -\lambda_{n-i}(B)$$
$$\lambda_{n-i}(B) \le \lambda_{n-i+1}(A)$$

Then $\lambda_1(A) \le \lambda_1(B) \le \lambda_2(A) \le \cdots \le \lambda_{n-1}(A) \le \lambda_{n-1}(B) \le \lambda_n(A)$.

Definition 6.5.6

Given an $n \times n$ matrix A, let A_r denote the matrix formed by deleting the last $n-r$ rows and columns of A. A_r is called the **leading principal submatrix** of A of order r.

$$A = \left(\begin{array}{cc|c|ccc} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \hline a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{array}\right)_{n \times n}$$

$$A_1 = (a_{11}), \quad A_2 = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad A_3 = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{23} & a_{33} \end{pmatrix}, \quad A_n = A$$

In order to determine whether a real symmetric matrix is positive definite or not, we need to show all eigenvalues are positive (Theorem 6.4.7). However, this method is complicated. The following theorem give us a much simpler method.

Theorem 6.5.7

Let A be a real symmetric $n \times n$ matrix. A is a positive definite matrix if and only if all leading principal submatrices have positive determinant, that is, $\det(A_r) > 0$, $r = 1, \cdots, n$.

Proof

Suppose A is symmetric positive definite. To show that A_r has positive determinant, we need to show A_r is positive definite. Let $\boldsymbol{x}_r = (x_1, \cdots, x_r)^T$ be any nonzero vector in $\boldsymbol{R}^r$ and set $\boldsymbol{x} = (x_1, \cdots, x_r, 0, \cdots, 0)^T$. Since $\boldsymbol{x}_r^T A_r \boldsymbol{x}_r = \boldsymbol{x}^T A \boldsymbol{x} > 0,$ it follows that A_r is positive definite. By Theorem 6.4.7, all eigenvalues of A_r are positive. Hence $\det(A_r) > 0.$

Suppose $\det(A_r) > 0,\ r = 1, \cdots, n.$ We will show that "all eigenvalues of A_k are positive, $k = 1, \cdots, n.$" by induction.

Clearly, $n = 1$ is true. Suppose it is true for $n = k$. Let $A_{k+1} = \left(\begin{array}{c|c} A_k & \boldsymbol{y} \\ \hline -\boldsymbol{y}^T & a \end{array}\right)_{(k+1)\times(k+1)}$ be a real symmetric $(k + 1) \times (k + 1)$ matrix such that $\det(A_r) > 0,\ r = 1, \cdots, k+1.$ By induction hypothesis, all eigenvalues of A_k are positive. By Theorem 4.3.17, we have

$$\lambda_1(A_{k+1}) \le \underset{0<}{\lambda_1(A_k)} \le \lambda_2(A_{k+1}) \le \lambda_2(A_k) \le \cdots \le \lambda_k(A_{k+1}) \le \lambda_k(A_k) \le \lambda_{k+1}(A_{k+1}).$$

Since $\det(A_{k+1}) = \lambda_1(A_{k+1}) \underbrace{\lambda_2(A_{k+1}) \cdots \lambda_{k+1}(A_{k+1})}_{>0} > 0,$ we have $\lambda_1(A_{k+1}) > 0.$ All eigenvalues of A_{k+1} are positive.

Remark 6.5.8

Let A be a real symmetric $n \times n$ matrix. A is a negative definite matrix <u>does not</u> means all leading principal submatrices have negative determinant, $\det(A_r) < 0,\ r = 1, \cdots, n.$

Indeed, A is a negative definite matrix if and only if $(-1)^r \det(A_r) > 0,\ r = 1, \cdots, n.$

Proof

A is a negative definite matrix if and only if $-A$ is a positive definite matrix

if and only if $(-1)^r \det(A_r) = \det(-A_r) > 0,\ r = 1, \cdots, n.$

Section 6.6 Singular Value Decomposition

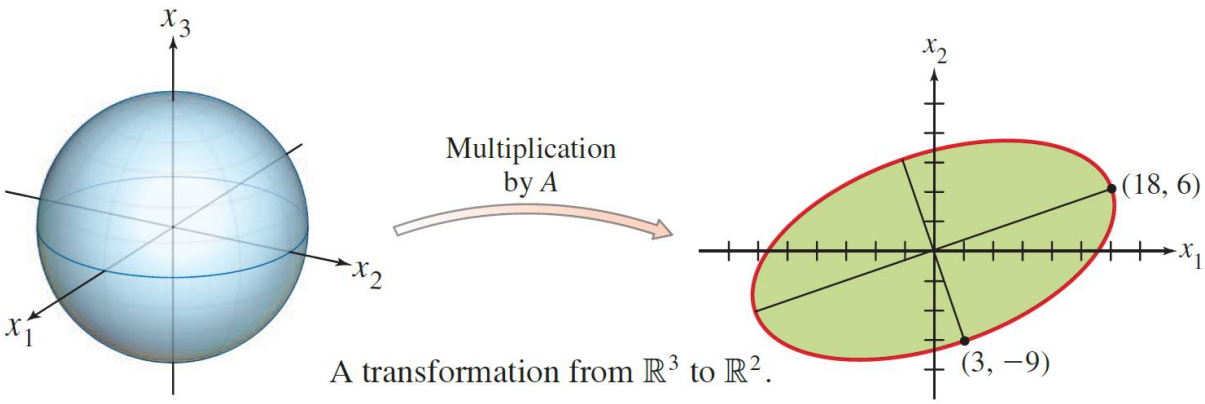
The singular value decomposition is based on the following property of the ordinary diagonalization that can be imitated for rectangular matrices: The absolute values of the eigenvalues of a symmetric matrix A measure the amounts that A stretches or shrinks certain vectors (the eigenvectors). If $A\boldsymbol{x} = \lambda \boldsymbol{x}$ and $\| \boldsymbol{x} \| = 1$, then

$$\| A\boldsymbol{x} \| = \| \lambda \boldsymbol{x} \| = |\lambda| \cdot \| \boldsymbol{x} \| = |\lambda|$$

If λ_1 is the eigenvalue with the greatest magnitude, then a corresponding unit eigenvector $\boldsymbol{v}_1$ identifies a direction in which the stretching effect of A is greatest. That is, the length of $A\boldsymbol{x}$ is maximized when $\boldsymbol{x} = \boldsymbol{v}_1$, and $\| A\boldsymbol{v}_1 \| = |\lambda_1|$.

Example 6.6.1

If $A = \begin{pmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{pmatrix}$, then the linear transformation $\boldsymbol{x} \mapsto A\boldsymbol{x}$ maps the unit sphere $\{\boldsymbol{x} : \| \boldsymbol{x} \| = 1\}$ in $\boldsymbol{R}^3$ onto an ellipse in $\boldsymbol{R}^2$, shown in below figure. Find a unit vector $\boldsymbol{x}$ at which the length $\| A\boldsymbol{x} \|$ is maximized, and compute this maximum length.



A transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

Solution

The quantity $\| A\boldsymbol{x} \|^2$ is maximized at the same $\boldsymbol{x}$ that maximizes $\| A\boldsymbol{x} \|$, and $\| A\boldsymbol{x} \|^2$ is easier to study. Observe that

$$\| A\boldsymbol{x} \|^2 = (A\boldsymbol{x})^T (A\boldsymbol{x}) = \boldsymbol{x}^T A^T A\boldsymbol{x} = \boldsymbol{x}^T (A^T A)\boldsymbol{x}.$$

Also, $A^T A$ is a symmetric matrix, since $(A^T A)^T = A^T (A^T)^T = A^T A$. So the problem now is to

maximize the quadratic form $\boldsymbol{x}^T(A^TA)\boldsymbol{x}$ subject to the constraint $\|\boldsymbol{x}\|=1$. By Theorem 6.5.1, the maximum value is the greatest eigenvalue λ_1 of A^TA. Also, the maximum value is attained at a unit eigenvector of A^TA corresponding to λ_1.

For the matrix A in this example,

$$A^TA=\begin{pmatrix}4&8\\11&7\\14&-2\end{pmatrix}\begin{pmatrix}4&11&14\\8&7&-2\end{pmatrix}=\begin{pmatrix}80&100&40\\100&170&140\\40&140&200\end{pmatrix}.$$

The eigenvalues of A^TA are $\lambda_1=360, \lambda_2=90,$ and $\lambda_3=0$. Corresponding unit eigenvectors are, respectively,

$$\boldsymbol{v}_1=\begin{pmatrix}1/3\\2/3\\2/3\end{pmatrix},\quad \boldsymbol{v}_2=\begin{pmatrix}-2/3\\-1/3\\2/3\end{pmatrix},\quad \boldsymbol{v}_3=\begin{pmatrix}2/3\\-2/3\\1/3\end{pmatrix}$$

The maximum value of $\|A\boldsymbol{x}\|^2$ is 360, attained when $\boldsymbol{x}$ is the unit vector $\boldsymbol{v}_1$. The vector $A\boldsymbol{v}_1$ is a point on the ellipse in above figure farthest from the origin, namely,

$$A\boldsymbol{v}_1=\begin{pmatrix}4&11&14\\8&7&-2\end{pmatrix}\begin{pmatrix}1/3\\2/3\\2/3\end{pmatrix}=\begin{pmatrix}18\\6\end{pmatrix}$$

For $\|\boldsymbol{x}\|=1,$ the maximum value of $\|A\boldsymbol{x}\|$ is $\|A\boldsymbol{v}_1\|=\sqrt{360}=6\sqrt{10}.$

Let A be an $m\times n$ matrix. Then A^TA is symmetric and can be orthogonally diagonalized. Let $\{\boldsymbol{v}_1,\cdots,\boldsymbol{v}_n\}$ be an orthonormal basis for $\boldsymbol{R}^n$ consisting of eigenvectors of $A^TA,$ consisting of eigenvectors of $A^TA,$ and let $\lambda_1,\cdots,\lambda_n$ be the associated eigenvalues of A^TA. Then, for $1\le i\le n,$

$$0\le\|A\boldsymbol{v}_i\|^2=(A\boldsymbol{v}_i)^T(A\boldsymbol{v}_i)=\boldsymbol{v}_i^TA^TA\boldsymbol{v}_i=\boldsymbol{v}_i^T(\lambda_i\boldsymbol{v}_i)=\lambda_i \tag{1}$$

So the eigenvalues of A^TA are all nonnegative. By renumbering, if necessary, we may assume that the eigenvalues are arranged so that

$$\lambda_1\ge\cdots\ge\lambda_n\ge 0.$$

The **singular values** of A are the square roots of the eigenvalues of $A^TA,$ denoted by $\sigma_1,\cdots,\sigma_n,$ and they are arranged in decreasing order. That is, $\sigma_i=\sqrt{\lambda_i}$ for $1\le i\le n.$ By equation (1), the singular values of A are the lengths of the vectors $A\boldsymbol{v}_1,\cdots,A\boldsymbol{v}_n.$

Example 6.6.2

Let A be the matrix in Example 6.6.1. Since the eigenvalues of A^TA are 360, 90, and 0, the singular values of A are

$$\sigma_1 = \sqrt{360} = 6\sqrt{10}, \quad \sigma_1 = \sqrt{90} = 3\sqrt{10}, \quad \sigma_3 = 0.$$

From Example 6.6.1, the first singular value of A is the maximum of $\|A\boldsymbol{x}\|$ over all unit vectors, and the maximum is attained at the unit eigenvector $\boldsymbol{v}_1$. Theorem 6.5.1 and Theorem 6.3.6 shows that the second singular value of A is the maximum of $\|A\boldsymbol{x}\|$ over all unit vectors that are *orthogonal to* $\boldsymbol{v}_1$, and this maximum is attained at the second unit eigenvector, $\boldsymbol{v}_2$. For the $\boldsymbol{v}_2$ in Example 6.6.1,

$$A\boldsymbol{v}_2 = \begin{pmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{pmatrix} \begin{pmatrix} -2/3 \\ -1/3 \\ 2/3 \end{pmatrix} = \begin{pmatrix} 3 \\ -9 \end{pmatrix}.$$

This point is on the minor axis of the ellipse in above figure, just as $A\boldsymbol{v}_1$ is on the major axis. (See above figure on the left). The first two singular values of A are the lengths of the major and minor semiaxes of the ellipse.

Theorem 6.6.3

Let A be an $m \times n$ matrix. Suppose $\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n\}$ is an orthonormal basis of $\boldsymbol{R}^n$ consisting of eigenvectors of A^TA, arranged so that the corresponding eigenvalues of A^TA, satisfy $\lambda_1 \geq \cdots \geq \lambda_n$, and suppose A has r nonzero singular values. Then $\{A\boldsymbol{v}_1, \cdots, A\boldsymbol{v}_r\}$ is an orthogonal basis for Col(A), and rank(A) = r.

Proof

Because $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$ are orthogonal for $i \neq j$,

$$(A\boldsymbol{v}_i)^T(A\boldsymbol{v}_j) = \boldsymbol{v}_i^T A^T A\boldsymbol{v}_j = \boldsymbol{v}_i^T(\lambda_j \boldsymbol{v}_j) = 0$$

Thus $\{A\boldsymbol{v}_1, \cdots, A\boldsymbol{v}_n\}$ is an orthogonal set. Furthermore, since the lengths of the vectors $A\boldsymbol{v}_1, \cdots, A\boldsymbol{v}_n$ are the singular values of A, and since there are r nonzero singular values, $A\boldsymbol{v}_1 \neq \boldsymbol{0}$ if and only if $1 \leq i \leq r$. So $A\boldsymbol{v}_1, \cdots, A\boldsymbol{v}_r$ are linearly independent vectors by Theorem 5.5.3, and they are in Col(A). Finally, for any $\boldsymbol{y}$ in Col(A)—say, $\boldsymbol{y} = A\boldsymbol{x}$—we can write $\boldsymbol{x} = c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n$, and

$$\boldsymbol{y} = A\boldsymbol{x} = c_1 A\boldsymbol{v}_1 + \cdots + c_r A\boldsymbol{v}_r + c_{r+1} A\boldsymbol{v}_{r+1} + \cdots + c_n A\boldsymbol{v}_n = c_1 A\boldsymbol{v}_1 + \cdots + c_r A\boldsymbol{v}_r$$

Thus $\boldsymbol{y}$ is in $\text{span}\{A\boldsymbol{v}_1, \cdots, A\boldsymbol{v}_r\}$, which shows that $\{A\boldsymbol{v}_1, \cdots, A\boldsymbol{v}_r\}$ is an (orthogonal) basis for Col(A). Hence rank(A) = dim Col(A) = r.

Theorem 6.6.4

Let A be an $m \times n$ matrix with rank r. Then there exists an $m \times n$ matrix $\Sigma = \left(\begin{array}{c|c} D_{r\times r} & \mathbf{0}_{r\times(n-r)} \\ \hline \mathbf{0}_{(m-r)\times r} & \mathbf{0}_{(m-r)\times(n-r)} \end{array}\right)$ for which the diagonal entries in D are the first r singular values of A, $\sigma_1 \geq \cdots \geq \sigma_r > 0$, and there exist an $m \times m$ orthogonal matrix U and an $n \times n$ orthogonal matrix V such that

$$A = U\Sigma V^T$$

Proof

Let λ_i and $\boldsymbol{v}_i$ be as in Theorem 6.6.3, so that $\{A\boldsymbol{v}_1, \cdots, A\boldsymbol{v}_r\}$ is an orthogonal basis for Col(A). Normalize each $A\boldsymbol{v}_i$ to obtain an orthonormal basis $\{\boldsymbol{u}_1, \cdots, \boldsymbol{u}_r\}$, where

$$\boldsymbol{u}_i = \frac{1}{\| A\boldsymbol{v}_i \|} A\boldsymbol{v}_i = \frac{1}{\sigma_i} A\boldsymbol{v}_i \quad \text{and} \quad A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i \quad (1 \leq i \leq r) \tag{2}$$

Now extend $\{\boldsymbol{u}_1, \cdots, \boldsymbol{u}_r\}$ to an orthonormal basis $\{\boldsymbol{u}_1, \cdots, \boldsymbol{u}_m\}$ of $\boldsymbol{R}^n$, and let

$$U = \begin{pmatrix} | & & | \\ \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_m \\ | & & | \end{pmatrix} \quad \text{and} \quad V = \begin{pmatrix} | & & | \\ \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_n \\ | & & | \end{pmatrix}.$$

By construction, U and V are orthogonal matrices. Also, from (2)

$$AV = \left(\begin{array}{ccc|c} | & & | & \\ A\boldsymbol{v}_1 & \cdots & A\boldsymbol{v}_r & \mathbf{0}_{m\times(n-r)} \\ | & & | & \end{array}\right) = \left(\begin{array}{ccc|c} | & & | & \\ \sigma_1\boldsymbol{u}_1 & \cdots & \sigma_r\boldsymbol{u}_r & \mathbf{0}_{m\times(n-r)} \\ | & & | & \end{array}\right)$$

Let D be the diagonal matrix with diagonal entries $\sigma_1, \cdots, \sigma_r$. Then

$$U\Sigma = \begin{pmatrix} | & & | \\ \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_m \\ | & & | \end{pmatrix} \left(\begin{array}{ccc|c} \sigma_1 & & \mathbf{0} & \\ & \ddots & & \mathbf{0}_{r\times(n-r)} \\ \mathbf{0} & & \sigma_r & \\ \hline \multicolumn{3}{c|}{\mathbf{0}_{(m-r)\times r}} & \mathbf{0}_{(m-r)\times(n-r)} \end{array}\right) = \left(\begin{array}{ccc|c} | & & | & \\ \sigma_1\boldsymbol{u}_1 & \cdots & \sigma_r\boldsymbol{u}_r & \mathbf{0}_{m\times(n-r)} \\ | & & | & \end{array}\right) = AV$$

Since V is an orthogonal matrix, $U\Sigma V^T = AVV^T = A$.

Any factorization $A = U\Sigma V^T$, with U and V orthogonal, Σ as in Theorem 6.6.4, and positive

diagonal entries in D, is called a **singular value decomposition** (or **SVD**) of A. The matrices U and V are not uniquely determined by A, but the diagonal entries of $\sum$ are necessarily the singular values of A.

The columns of U, $\boldsymbol{u}_i$'s, are called the **left singular vectors** of A, and the columns of V, $\boldsymbol{v}_j$'s, are called the **right singular vectors** of A.

Since V diagonalizes A^TA, it follows that the $\boldsymbol{v}_j$'s are eigenvectors of A^TA.

Since $AA^T = U\sum\sum^T U^T$, it follows that U diagonalizes AA^T and that the $\boldsymbol{u}_i$'s are eigenvectors of AA^T.

If A has rank r, then

(i) $\boldsymbol{v}_1,\cdots,\boldsymbol{v}_r$ form an orthonormal basis for $\text{Col}(A^T)$.

(ii) $\boldsymbol{v}_{r+1},\cdots,\boldsymbol{v}_n$ form an orthonormal basis for $N(A)$.

(iii) $\boldsymbol{u}_1,\cdots,\boldsymbol{u}_r$ form an orthonormal basis for $\text{Col}(A)$.

(iv) $\boldsymbol{u}_{r+1},\cdots,\boldsymbol{u}_m$ form an orthonormal basis for $N(A^T)$.

$\boldsymbol{v}_{r+1},\cdots,\boldsymbol{v}_n$ is an orthonormal set of eigenvectors corresponding to eigenvalue 0 and $\text{rank}(A) = r$. (ii) holds. Theorem 6.6.3 and equation (2) implies (iii). (i) and (iv) follows from (iii) and (ii) by considering A^T.

Example 6.6.5

Use the results of Examples 6.6.1 and 6.6.2 to construct a singular value decomposition of $A = \begin{pmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{pmatrix}$.

Solution

Step 1 Find an orthogonal diagonalization of A^TA.

See Example 6.6.1

Step 2 Set up V and Σ.

Using Example 6.6.1, construct

$$V=\begin{pmatrix} | & | & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \\ | & | & | \end{pmatrix}=\begin{pmatrix} 1/3 & -2/3 & 2/3 \\ 2/3 & -1/3 & -2/3 \\ 2/3 & 2/3 & 1/3 \end{pmatrix}$$

The square roots of the eigenvalues are the singular values:

$$\sigma_1=\sqrt{360}=6\sqrt{10},\quad \sigma_1=\sqrt{90}=3\sqrt{10},\quad \sigma_3=0.$$

The nonzero singular values are the diagonal entries of D. The matrix Σ is the same size as A, with D in its upper left corner and with 0's elsewhere.

$$D=\begin{pmatrix} 6\sqrt{10} & 0 \\ 0 & 3\sqrt{10} \end{pmatrix},\quad \Sigma=\left(D \mid \mathbf{0}\right)=\begin{pmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{pmatrix}$$

Step 3 Construct U

When A has rank r, the first r columns of U are the normalized vectors obtained from $A\boldsymbol{v}_1,\cdots,A\boldsymbol{v}_r$. In this example, A has two nonzero singular values, so rank(A) = 2. Recall from equation (2) and the paragraph before Example 6.6.2 that $\|A\boldsymbol{v}_1\|=\sigma_1$ and $\|A\boldsymbol{v}_2\|=\sigma_2$. Thus

$$\boldsymbol{u}_1=\frac{1}{\sigma_1}A\boldsymbol{v}_1=\frac{1}{6\sqrt{10}}\begin{pmatrix}18\\6\end{pmatrix}=\begin{pmatrix}3/\sqrt{10}\\1/\sqrt{10}\end{pmatrix},\quad \boldsymbol{u}_2=\frac{1}{\sigma_2}A\boldsymbol{v}_2=\frac{1}{3\sqrt{10}}\begin{pmatrix}3\\-9\end{pmatrix}=\begin{pmatrix}1/\sqrt{10}\\-3/\sqrt{10}\end{pmatrix}$$

Note that $\{\boldsymbol{u}_1,\boldsymbol{u}_2\}$ is already a basis for $\boldsymbol{R}^2$. Thus no additional vectors are needed for U, and $U=\left(\boldsymbol{u}_1 \mid \boldsymbol{u}_2\right)$. The singular value decomposition of A is

$$A=\underbrace{\begin{pmatrix}3/\sqrt{10} & 1/\sqrt{10}\\ 1/\sqrt{10} & -3/\sqrt{10}\end{pmatrix}}_{U}\underbrace{\begin{pmatrix}6\sqrt{10} & 0 & 0\\ 0 & 3\sqrt{10} & 0\end{pmatrix}}_{\Sigma}\underbrace{\begin{pmatrix}1/3 & 2/3 & 2/3\\ -2/3 & -1/3 & 2/3\\ 2/3 & -2/3 & 1/3\end{pmatrix}}_{V^T}$$

Example 6.6.6

Find a singular value decomposition of $A=\begin{pmatrix}1 & -1\\ -2 & 2\\ 2 & -2\end{pmatrix}$.

Solution

First, compute $A^TA=\begin{pmatrix}9 & -9\\ -9 & 9\end{pmatrix}$. The eigenvalues of A^TA are 18 and 0, with corresponding unit eigenvectors $\boldsymbol{v}_1=\begin{pmatrix}1/\sqrt{2}\\ -1/\sqrt{2}\end{pmatrix}$, $\boldsymbol{v}_2=\begin{pmatrix}1/\sqrt{2}\\ 1/\sqrt{2}\end{pmatrix}$.

These unit vectors form the columns of $V=\begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ -1/\sqrt{2} & 1/\sqrt{2}\end{pmatrix}$.

The singular values are $\sigma_1=\sqrt{18}=3\sqrt{2}$ and $\sigma_2=0$. Then $D=\sigma_1=3\sqrt{2}$ and

$$\Sigma=\begin{pmatrix}D & 0\\ 0 & 0\\ 0 & 0\end{pmatrix}=\begin{pmatrix}3\sqrt{2} & 0\\ 0 & 0\\ 0 & 0\end{pmatrix}.$$

To construct U, first construct $A\boldsymbol{v}_1$ and $A\boldsymbol{v}_2$:

$$A\boldsymbol{v}_1=\begin{pmatrix}1 & -1\\ -2 & 2\\ 2 & -2\end{pmatrix}\begin{pmatrix}1/\sqrt{2}\\ -1/\sqrt{2}\end{pmatrix}=\begin{pmatrix}2/\sqrt{2}\\ -4/\sqrt{2}\\ 4/\sqrt{2}\end{pmatrix},\quad A\boldsymbol{v}_2=\begin{pmatrix}1 & -1\\ -2 & 2\\ 2 & -2\end{pmatrix}\begin{pmatrix}1/\sqrt{2}\\ 1/\sqrt{2}\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}.$$

The only column found for U so far is $\boldsymbol{u}_1=\frac{1}{\sigma_1}A\boldsymbol{v}_1=\frac{1}{3\sqrt{2}}\begin{pmatrix}2/\sqrt{2}\\ -4/\sqrt{2}\\ 4/\sqrt{2}\end{pmatrix}=\begin{pmatrix}1/3\\ -2/3\\ 2/3\end{pmatrix}$

Apply the Gram-Schmidt process (with normalizations) that are orthogonal to $\boldsymbol{u}_1$, and obtain

$$\boldsymbol{u}_2=\begin{pmatrix}2/\sqrt{5}\\ 1/\sqrt{5}\\ 0\end{pmatrix},\quad \boldsymbol{u}_3=\begin{pmatrix}-2/\sqrt{45}\\ 4/\sqrt{45}\\ 5/\sqrt{45}\end{pmatrix}$$

Finally, set $\left(\boldsymbol{u}_1 \mid \boldsymbol{u}_2 \mid \boldsymbol{u}_3\right)$, take Σ and V^T from above, and write

$$A=\begin{pmatrix}1 & -1\\ -2 & 2\\ 2 & -2\end{pmatrix}=\begin{pmatrix}1/3 & 2/\sqrt{5} & -2/\sqrt{45}\\ -2/3 & 1/\sqrt{5} & 4/\sqrt{45}\\ 2/3 & 0 & 5/\sqrt{45}\end{pmatrix}\begin{pmatrix}3\sqrt{2} & 0\\ 0 & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ -1/\sqrt{2} & 1/\sqrt{2}\end{pmatrix}^T.$$